

MATHEMATICS GRADE 10: SIMILARITY AND ENLARGEMENT

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.1 (12 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Mathematics
Strand	Strand 2.0: Geometry
Sub-Strand	Sub-Strand 2.1: Similarity and Enlargement
Total Duration	12 lessons × 40 minutes = 480 minutes total
Sub-Strand Content	– Similar figures: identifying and describing similar shapes in the environment – Properties of similar figures: corresponding angles and ratios of corresponding sides – Linear Scale Factor (LSF): calculating and applying LSF between similar figures – Areas of similar figures: relationship between LSF and Area Scale Factor (ASF) – Volumes of similar solids: relationship between LSF and Volume Scale Factor (VSF) – Enlargement on a grid: constructing enlargements using a centre of enlargement and positive scale factor – Enlargement with negative scale factors: constructing and interpreting enlargements with negative LSF – Centre of enlargement: locating the centre of enlargement given an object and its image – Real-life applications: maps, architectural plans, passport photos, and scale models
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - identify similar figures and describe their properties in terms of corresponding angles and ratios of corresponding sides, - determine the linear scale factor (LSF) between similar figures, - apply the linear scale factor to calculate lengths of corresponding sides in similar figures, - determine the relationship between the linear scale factor and the area scale factor (ASF) of similar figures, - determine the relationship between the linear scale factor and the volume scale factor (VSF) of similar solids, - construct enlargements of plane figures on a grid using a given centre and scale factor, - locate the centre of enlargement given an object and its image, - apply similarity and enlargement concepts to solve real-life problems such as map reading, passport photo resizing, and architectural drawing, - appreciate the role of similarity and enlargement in day-to-day activities and careers.
Core Competencies	– Communication and Collaboration: Learners discuss in groups to identify similar figures in Kenyan contexts (e.g., matatu route maps, Nairobi city plans, passport photos), share reasoning about scale factors, and co-construct enlargements on grids, articulating their mathematical thinking to peers. – Critical Thinking and Problem Solving: Learners analyse why a passport photo must be enlarged or reduced by a specific LSF to fit official dimensions, derive the ASF and VSF from first principles, and apply these relationships to solve multi-step real-life problems. – Creativity and Imagination: Learners design scale models of familiar structures (e.g., a Kenyan homestead or a school

	building) by selecting appropriate linear scale factors and justifying their choices. – Digital Literacy: Learners use calculators, geometry software (e.g., GeoGebra), and digital resources to construct enlargements, verify scale factor relationships, and explore similarity in images and maps. – Learning to Learn: Learners reflect on errors in scale factor calculations, revise their constructed models, and develop self-monitoring strategies for checking ratio consistency across corresponding sides.
Core Values	– Responsibility: The learner takes care of geometric instruments (rulers, protractors, graph paper) when constructing enlargements and scale drawings, and handles shared digital devices with care. – Respect: The learner listens to peers' explanations of scale factor reasoning during group discussions without interruption, valuing diverse approaches to solving similarity problems. – Unity: The learner works collaboratively in groups to construct enlargements on grids and to derive the relationship between LSF, ASF, and VSF, contributing equitably to shared tasks. – Integrity: The learner accurately measures and records corresponding lengths in similar figures without manipulating data, ensuring honest reporting of scale factor calculations. – Social Justice: The learner appreciates how affordable passport photo services and accessible map-reading skills (linked to similarity and enlargement) support equal opportunity for all Kenyans, including those in rural areas.
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners ask why a passport photo printed at different sizes still looks the same, and define what mathematical relationship must hold between the original and the resized image. – Developing and Using Models: Learners build and iteratively revise diagrammatic models showing an object, its centre of enlargement, and its image under various scale factors, linking visual representations to algebraic relationships. – Planning and Carrying Out Investigations: Learners measure corresponding sides of similar figures cut from Kenyan newspapers and magazines, record data in tables, and investigate the ratio of corresponding sides to determine the LSF. – Analysing and Interpreting Data: Learners tabulate LSF, ASF, and VSF values for a series of similar figures and solids, identify the pattern ($ASF = LSF^2$, $VSF = LSF^3$), and interpret what these relationships mean in real contexts such as comparing sizes of githeri portions or water tank capacities. – Using Mathematics and Computational Thinking: Learners apply ratio and proportion, index laws, and coordinate geometry to calculate scale factors, locate centres of enlargement algebraically, and solve problems involving maps and architectural plans. – Constructing Explanations: Learners write and present explanations of why the ASF equals the square of the LSF, using evidence from their investigations of similar rectangles representing Kenyan farm plots. – Obtaining, Evaluating, and Communicating Information: Learners research real-world uses of enlargement (passport photo studios in Nairobi, survey maps of the Rift Valley) and communicate findings to the class, evaluating the accuracy of scale representations.
Pertinent & Contemporary Issues (PCIs)	– Life Skills (Critical Thinking): The learner applies critical thinking to determine whether a printed map or passport photo has been scaled correctly, questioning assumptions about size and proportion in everyday documents. – Life Skills (Self-Esteem): The learner gains confidence by successfully deriving and applying LSF, ASF, and VSF relationships independently, reinforcing belief in their own mathematical ability. – Citizenship (National Identity): The learner uses Kenyan survey maps, county boundary diagrams, and Nairobi city plans as contexts for scale factor work, fostering pride in and understanding of Kenya's geography. – Financial Literacy: The learner explores how businesses such as passport photo studios, printing shops, and architectural firms in Kenya price services based on the scale of reproduction, connecting mathematics to entrepreneurial contexts. – Environmental Education: The learner investigates how similarity and scale are used in environmental mapping (e.g., mapping deforestation in the Mau Forest or water levels in Lake Victoria) to make informed conservation decisions.

Career Connections	– Architecture and Engineering: Architects and civil engineers in Kenya (e.g., those designing buildings in Nairobi's Westlands or road infrastructure in the Rift Valley) use scale drawings and enlargement principles daily to translate designs to real structures. – Cartography and Surveying: Land surveyors and cartographers at the Kenya National Mapping Authority use similarity and linear scale factors to create accurate maps of Kenya's counties, national parks, and coastlines. – Graphic Design and Photography: Graphic designers and passport photo studio operators apply enlargement and reduction scale factors to resize images for official documents, posters, and digital media. – Medicine and Radiology: Medical imaging professionals use scale factors when interpreting X-rays and scans — enlarging images of tissues or organs for diagnosis in hospitals such as Kenyatta National Hospital. – Fashion and Textile Design: Fashion designers in Nairobi's Gikomba and Eastleigh markets scale up pattern pieces using LSF to produce garments in different sizes, applying similarity concepts directly. – Urban Planning: Urban planners at county governments use scale models and similarity principles to design housing estates, school campuses, and market centres across Kenyan towns.
Focus for Lessons	This unit develops learners' understanding of similarity and enlargement as powerful mathematical tools for describing and creating proportional relationships between figures and solids. Anchored in the puzzling observation that a passport photo looks identical whether printed small or large — yet fits entirely different frames — learners investigate the concept of linear scale factor (LSF) and discover how it governs corresponding lengths, areas ($ASF = LSF^2$), and volumes ($VSF = LSF^3$). Throughout the 12 lessons, learners move from recognising similar figures in familiar Kenyan contexts (maps, architectural plans, everyday photographs) to constructing enlargements on grids, locating centres of enlargement, and solving complex real-life problems, building both procedural fluency and deep conceptual understanding of proportional reasoning.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why does resizing a passport photo change its size but never its shape — and what is the mathematics that makes this possible? KICD KEY INQUIRY QUESTIONS: 1. How do we identify and use similar figures in day-to-day situations? 2. How does the linear scale factor relate to the area and volume of similar figures and solids?
Anchoring Phenomenon	ANCHORING PHENOMENON — The Passport Photo Puzzle: A Form 4 student in Nairobi visits a passport photo studio along Kimathi Street before travelling to a regional athletics championship. The photographer produces two prints from the same digital image: a standard passport print (35 mm × 45 mm) and an enlarged print for the school noticeboard (105 mm × 135 mm). The student notices — with puzzlement — that both photos show exactly the same image: same facial proportions, same background, same clarity. Yet one is clearly three times the size of the other. How can two objects look identical but have completely different sizes? What mathematical rule did the photographer's software follow to resize the image so perfectly? And if the photographer accidentally resizes only the width but not the height, why does the face suddenly look distorted and unrecognisable? These questions are puzzling because students intuitively know that "making something bigger" should change it, yet here the shape is perfectly preserved. This phenomenon anchors the entire unit: the answer lies in the Linear Scale Factor — a single number that scales every length in the image by the same ratio, preserving all angles and all length ratios, and whose square gives the ratio of areas.
Storyline Thread	Lesson 1 — PREDICT: The Passport Photo Puzzle Launch Students observe two passport-sized prints of the same photo (35 mm × 45 mm and 105 mm × 135 mm). They predict in writing: "What changed? What stayed the same? What single number, if any, connects the two sizes?" Cold-call sharing of predictions; class records initial ideas on the Driving Question Board (DQB). Introduction of the unit driving question.

Lesson 2 — OBSERVE & EXPLORE: Properties of Similar Figures

Students measure corresponding sides and angles of pairs of similar figures cut from Kenyan newspapers and maps. They tabulate ratios of corresponding sides and record corresponding angles. Class discovers: angles are equal; ratios of corresponding sides are constant. Formal definition of similar figures introduced. Link back: "Is this what happened to our passport photo?"

Lesson 3 — EXPLAIN: Linear Scale Factor (LSF) — Lengths

Students formalise the concept of LSF from lesson 2 data. Worked examples using Kenyan contexts: resizing a school badge, scaling up a recipe card. Learners calculate LSF and use it to find unknown lengths in similar figures. Students revisit the passport photo: $LSF = 105/35 = 3$. DQB updated: "LSF tells us how much every length is multiplied."

Lesson 4 — OBSERVE & INVESTIGATE: Area Scale Factor (ASF)

Students draw pairs of similar rectangles (representing Kenyan farm plots) with $LSF = 2, 3$, and 4 on graph paper, count squares to find areas, and tabulate LSF vs. area ratio. They discover $ASF = LSF^2$. Teacher formalises the relationship. Students apply ASF to the passport photo: the larger photo has $3^2 = 9$ times the area of the smaller photo.

Lesson 5 — EXPLAIN: Volume Scale Factor (VSF)

Students use linking cubes or diagrams to build similar cuboids with $LSF = 2$ and $LSF = 3$, count unit cubes, and discover $VSF = LSF^3$. Worked examples use the Eldoret sufuria supporting phenomenon ($VSF = 27$). Learners solve problems comparing volumes of similar containers — water tanks, grain stores, cooking pots — in Kenyan farming contexts.

Lesson 6 — EXPLAIN & APPLY: Mixed Problems — LSF, ASF, VSF

Consolidation lesson. Learners solve a structured set of problems weaving all three relationships: a map distance problem (Nairobi to Nakuru), a roofing area problem (Mombasa architect model), and a water tank volume problem (Kericho tea farm). Peer-marking and error analysis. DQB updated with " $ASF = LSF^2$, $VSF = LSF^3$."

Lesson 7 — OBSERVE & CONSTRUCT: Enlargement on a Grid (Positive LSF)

Students are introduced to the centre of enlargement. Using graph paper, they construct enlargements of simple polygons (a Kenyan house floor plan outline) with positive $LSF = 2$ and $LSF = 3$, using a given centre. They observe that distances from the centre of enlargement also scale by the LSF. Connection to passport photo: the "centre" is the optical centre of the camera lens.

Lesson 8 — EXPLAIN & CONSTRUCT: Enlargement with Negative Scale Factor

Students construct enlargements using $LSF = -2$, discovering that the image appears on the opposite side of the centre and is inverted. Worked examples using coordinate grids. Learners discuss: "If a camera inverts an image (as pinhole cameras do), which scale factor describes this?" DQB updated: "Negative LSF → image is inverted and on the opposite side."

Lesson 9 — EXPLAIN: Locating the Centre of Enlargement

Given an object and its image, students use the property that corresponding points and the centre of enlargement are collinear to locate the centre by drawing or using coordinate methods. Worked examples with Kenyan map grids. Learners reverse-engineer: "Where was the photographer's camera positioned relative to the subject?"

Lesson 10 — MODEL BUILDING: Constructing a Scale Drawing (Real-Life Application)

Students work in groups to create a scale drawing of their classroom or school compound, selecting an appropriate LSF,

	measuring real dimensions, and producing a labelled floor plan on graph paper. Each group presents their choice of LSF and justifies it. This phase makes the passport photo phenomenon concrete at architectural scale. Lesson 11 — APPLICATION & PROBLEM SOLVING: Maps, Photos, and Plans Students solve a rich set of real-life problems: (a) using a 1:50,000 Kenya survey map to find actual distances between towns; (b) determining the correct dimensions for a passport photo enlargement for a Huduma Centre application; (c) comparing storage volumes of two geometrically similar grain silos on a Rift Valley farm. DQB is reviewed and outstanding questions are resolved. Lesson 12 — REVIEW, MODEL REVISION & ASSESSMENT: Pulling the Storyline Together Students return to the original passport photo phenomenon and write a full explanation of why resizing preserves shape but changes size, citing LSF, ASF, and VSF. They present revised cumulative models (diagram + annotated equations). A summative task (mix of short-answer and problem-solving questions in Kenyan contexts) assesses all learning outcomes. DQB closed: the driving question is answered collaboratively.
Supporting Phenomena	– The Shrinking Map: A matatu driver in Kisumu uses a road map of Western Kenya. The entire Lake Victoria basin — hundreds of kilometres wide — fits on a single A4 page. How can a 350 km stretch of road be represented by just 7 cm on paper? What number connects the map distance to the real distance, and how is this the same mathematical idea as the passport photo? – The Giant Ugali Pot: A caterer preparing for a school prize-giving in Eldoret normally uses a sufuria of diameter 30 cm to cook ugali for 20 people. For a function of 180 people, they buy a sufuria of diameter 90 cm. The diameter tripled — so does the ugali cook three times as much, or nine times as much, or twenty-seven times as much? The puzzling answer (the volume scales by $LSF^3 = 27$) connects to the VSF relationship. – The Architect's Model: An architect in Mombasa builds a scale model of a new coastal resort hotel. The model is 60 cm long; the real building will be 30 m long. The model's roof has an area of 450 cm² — what will the real roof area be? Learners are puzzled that multiplying the length by 50 does not give the roof area; instead the area scales by $50^2 = 2500$. – The Shadow and the Tree: Near a farm in Kericho, a farmer notices that a young tea plant casts a shadow of 0.4 m at the same time that a nearby fence post of known height 1.2 m casts a shadow of 0.6 m. How can shadows on flat ground be used to find the height of the tea plant without climbing it? This supporting phenomenon introduces indirect measurement through similar triangles.

LESSON 1 (40 minutes): The Passport Photo Puzzle: Launching the Phenomenon

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the unit phenomenon by engaging learners in noticing and wondering about two passport photos of different sizes that show the same image, building curiosity about what mathematical rule preserves shape during resizing.
Knowledge	Learners will recognise that two figures can differ in size yet retain the same shape, and will articulate initial ideas about what numerical relationship might connect the two photo sizes (35 mm x 45 mm and 105 mm x 135 mm).
Skills	Learners will practise careful observation, measurement estimation, ratio comparison, and structured prediction writing in response to a real-world visual stimulus.
Attitudes	Learners will demonstrate curiosity, open-mindedness, and willingness to share tentative ideas without fear of being wrong, appreciating that mathematics is rooted in everyday experiences such as visiting a Nairobi photo studio.
Key Inquiry Question	How do we identify and use similar figures in day-to-day situations?
Purpose in Storyline	This is the PREDICT opening lesson. It launches the anchoring phenomenon, surfaces prior knowledge and misconceptions, and opens the Driving Question Board (DQB) so that every subsequent lesson can be framed as answering a question the class itself has posed.
Safety Notes	No hazardous materials. If printed photos are laminated, ensure scissors or sharp edges are kept away from learners during any handling. Ensure printed photos are handled respectfully if they show a real student face — use a generic stock image or a drawn face to protect privacy.

B. LESSON OVERVIEW

In this 40-minute opening lesson, learners encounter the anchoring phenomenon: two passport-sized prints of the same photo (35 mm x 45 mm and 105 mm x 135 mm) that look identical in content yet differ dramatically in size. Working individually and then in pairs, learners predict in writing what changed, what stayed the same, and whether a single number connects the two sizes. Cold-call sharing surfaces initial ideas. The Driving Question Board is created and the unit Driving Question is introduced. No formal mathematics is taught yet — the lesson is designed entirely to provoke wonder, activate intuition, and generate the questions that the rest of the unit will answer.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Video: Balancing Act Source: PhET Interactive Simulations	Learners enter to find two printed photos displayed side by side at the front of the class — one measuring 35 mm x 45 mm (standard passport size) and one measuring 105 mm x 135 mm (noticeboard enlargement), both	Teacher displays the two photos and says: 'Before I say a single word about mathematics, I want you to just look. Take 30 seconds. Look at both photos carefully.' WAIT TIME: 30 seconds of complete silence. Teacher then	Think-Write-Pair-Share structured around three specific prediction prompts ensures every learner commits an idea to paper before hearing peers. Recording all predictions publicly on the board without evaluation creates a	Teacher reads sticky notes and listens during pair-share to identify: (a) learners who already notice the ratio (3:1 or 105/35) — these learners can be used as cold-call anchors; (b) learners who focus only on size difference and do not

(English)  Search ARES for similar videos  HTML: Proportions to Find Scale Ratios (Flexbook) Source: CK-12  Search ARES for similar readings	showing the same image. Teacher does not explain anything yet. Learners are given a sticky note or a quarter sheet of paper and asked to silently write three answers: (1) What is different about the two photos? (2) What is the same about the two photos? (3) Is there a single number that connects the two sizes — if so, what do you think it is and why? Learners write individually for 3 minutes in silence. They then share with a desk partner for 2 minutes. Selected pairs are cold-called to share one prediction each with the class. Predictions are not judged — all ideas are recorded on the whiteboard under the heading OUR PREDICTIONS.	says: 'Now, on your piece of paper, answer three questions I am writing on the board. Do not talk to anyone yet — this is your own thinking first. You have 3 minutes.' Teacher circulates silently, reading over shoulders, noting interesting predictions (correct and incorrect) to use in cold-calling. After 3 minutes, teacher says: 'Now turn to your neighbour and share what you wrote. You have 2 minutes.' WAIT TIME: 2 minutes. Teacher then says: 'I will cold-call three pairs. When I call you, tell us one thing from your prediction — just one. There are no wrong answers right now.' Cold-calls 3 to 4 pairs. After each response, teacher says: 'Interesting — let me write that up.' Teacher records each prediction verbatim on the board without correcting. If a learner says 'the size is three times bigger,' teacher writes it and says: 'Interesting claim — how did you get three? We will test that idea together.' If a learner says 'everything is the same,' teacher writes it and says: 'Say more — what do you mean by everything?' WAIT TIME after each cold-call: 5 seconds before moving to the next pair.	psychologically safe environment and gives the class a shared reference point to return to in later lessons when predictions are confirmed or revised.	yet notice shape preservation — these learners need the Observe phase to build that awareness; (c) learners who claim the photos are completely identical — probe with 'put a ruler next to each one.' This informs pacing and which predictions to prioritise on the DQB.
Observe Phase  VIDEO: Video: Balancing Act Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Proportions to	Teacher distributes one printed copy of both photos to each pair of learners (or projects them on the board with a ruler visible). Learners are asked to use a ruler to measure the width and height of each photo and record the measurements in a simple two-column table drawn in their exercise books. They then compute the ratio of the large photo measurement to the small photo measurement for both width	Teacher distributes photos and rulers and says: 'Measure the width and the height of each photo. Write the numbers in your book. Now divide the large measurement by the small measurement for the width. Do the same for the height. You have 3 minutes.' WAIT TIME: 3 minutes. Teacher circulates, checking measurements for accuracy. If a learner gets different ratios for width and height, teacher says:	The structured measurement task makes the abstract idea of ratio concrete and grounded in physical objects. The deliberate introduction of the distorted photo creates productive cognitive conflict — learners feel intuitively that something is wrong but cannot yet explain it mathematically. This discomfort is the engine of inquiry for the unit. The contrast between the well-resized and the distorted photo also previews the key	Teacher checks whether learners are obtaining consistent ratios (3 for both dimensions) or inconsistent ones due to measurement error. Learners who obtain different ratios for width and height are likely making ruler-reading errors — teacher addresses these individually during circulation. Learners who immediately say 'it is because one number was not multiplied by 3' are demonstrating intuitive

Find Scale Ratios (Flexbook) Source: CK-12 Search ARES for similar readings	and height. Teacher asks: 'Are the two ratios the same or different? What does that tell you?' Learners record their finding in one sentence. The class then examines the faces in both photos — teacher asks: 'Does the face look squashed or stretched in either photo, or does it look natural in both?' Learners record their observation. Teacher draws attention to the background, the proportions of the nose to the face, the width of the shoulders — asking: 'Do these proportions look the same in both photos?' Learners observe and record. Finally, teacher shows a third image — a deliberately distorted photo where only the width was changed (width tripled, height unchanged, making the face look very wide and squashed) — and asks: 'What is different about this one? Why does this face look wrong?' Learners discuss in pairs for 1 minute and share.	'Check your measurement again — use the ruler carefully from the very edge of the photo.' After 3 minutes, teacher asks: 'What ratio did you get for the width?' Cold-calls one learner. 'What about the height?' Cold-calls another. Teacher writes 105 divided by 35 equals 3 and 135 divided by 45 equals 3 on the board and says: 'Both ratios are 3. What might that mean?' WAIT TIME: 5 seconds. Teacher then displays the distorted photo and says: 'Now look at this one. Something went wrong in the photo studio. What do you notice? Talk to your partner for one minute.' WAIT TIME: 1 minute. Cold-calls one pair: 'Why does this face look wrong?' Teacher does not yet give the answer — records the pair response on the board and says: 'Keep that question in mind. By the end of this unit, you will be able to explain exactly what went wrong and why the mathematics matters.'	concept that both width and height must scale by the same factor.	proportional reasoning and can be acknowledged as supporting voices during the Explain phase. The distorted-photo response reveals whether learners are connecting the equal-ratio observation to the idea of shape preservation.
Explain Phase  VIDEO: Video: Balancing Act Source: PhET Interactive Simulations (English) Search ARES for similar videos  HTML: Proportions to Find Scale Ratios (Flexbook) Source: CK-12 Search ARES for similar readings	Teacher brings the class together for a brief whole-class discussion. Teacher writes the unit Driving Question on the board for the first time: 'Why does resizing a passport photo change its size but never its shape — and what is the mathematics that makes this possible?' Teacher says: 'You have just discovered something important from the measurements. Both ratios were 3. In this unit, we are going to explore what that number 3 means, where it comes from, what it does to lengths, to areas, and to volumes, and how photographers, architects, map-makers, and engineers use this idea every single day in Kenya and around the world.' Teacher	Teacher writes the Driving Question on the board slowly and says: 'Read this with me.' Class reads aloud together. Teacher then says: 'Every lesson in this unit is going to be one step toward fully answering this question. Today you have taken the first step — you have noticed the puzzle and you have found a number: 3. But 3 is just the beginning. Here are the questions I want you to sit with.' Teacher writes on the board: (1) Why does multiplying by 3 preserve the shape? (2) If the lengths are 3 times bigger, is the area 3 times bigger too? (3) What if we were comparing three-dimensional objects like water tanks or cooking pots? Teacher	Introducing the term Linear Scale Factor by name without full definition uses the pedagogical strategy of anticipatory set — learners have a vocabulary peg to hang their future learning on, and the gap between the name and the full understanding creates sustained motivation to learn. The question-generation task transfers intellectual ownership to the learners: the unit is now framed as answering their questions, not the teacher's agenda.	Teacher reads the questions learners write in their books. These questions serve as a pre-assessment of the class's conceptual starting point. Common productive questions include: 'Is the area also 3 times bigger?' (reveals readiness for Lesson 4), 'Does this work for circles too?' (reveals awareness of shape generality), and 'What if the scale factor is a fraction?' (reveals readiness for Lesson 3 extension). Less productive questions such as 'Why do we need passports?' indicate a learner who has not yet connected to the mathematical content — teacher makes a note to check in with this learner during Lesson 2.

	introduces the term Linear Scale Factor for the first time — writing it on the board and saying: 'This number — 3 in our case — has a name: the Linear Scale Factor. We will define it precisely in Lesson 3. For now, remember: it is the number that connects every length in the small photo to every corresponding length in the large photo.' Teacher deliberately does NOT fully define or formalise similarity or LSF at this stage — the goal is only to name the concept and create anticipation. Teacher asks: 'Based on everything you have observed so far, what questions do you still have? What do you not yet understand? What do you want to know?' Learners write 2 questions each in their books. They share with their partner, then the class pool is shared aloud.	says: 'Do not try to answer these now. Write two questions of your own that this photo puzzle has given you. You have 2 minutes.' WAIT TIME: 2 minutes of silent individual writing. Teacher cold-calls 4 learners to share one question each. For each question, teacher says: 'Excellent question. Let us put that on our board.' Teacher does not answer any question yet — the response to every question is: 'We will find out together.' If a learner asks a question that is outside the scope of the unit, teacher says: 'That is a fascinating question — let us park it here and come back to it.'		
Driving Question Board (DQB) Creation  VIDEO: Video: Balancing Act Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Proportions to Find Scale Ratios (Flexbook) Source: CK-12  Search ARES for similar readings	Teacher introduces the Driving Question Board — a large sheet of manila paper or a designated section of the whiteboard that will remain visible throughout the unit. Teacher explains: 'This board will hold our big question and all the smaller questions we generate as we investigate. At the end of each lesson, we will update it — adding new questions we have thought of, and ticking off questions we have answered.' Teacher writes the Driving Question at the top of the DQB in large letters. Below it, teacher creates three columns labelled: WHAT WE NOTICE, WHAT WE WONDER, and WHAT WE KNOW FOR SURE. Learners come forward in groups of four and stick their written predictions and questions onto the board under the appropriate column. Teacher	Teacher unveils the manila paper board (prepared before class) and says: 'This is our Driving Question Board. It belongs to all of us. Every question you have is valid enough to go on this board.' Teacher models adding the first item: 'I am going to put our Driving Question here at the top. Now, who will come and add their prediction from the Predict phase?' Invites 4 to 5 volunteers to come forward and place sticky notes. Teacher reads each one aloud as it goes up, saying: 'This one says — the face looks the same in both photos. Which column does that belong to — what we notice, what we wonder, or what we know for sure?' WAIT TIME: 3 seconds for class response. After all notes are placed, teacher steps back and says: 'Look at this board. This is a	The DQB is a physical, co-constructed artefact that externalises the class thinking and makes inquiry visible. Sorting items into three columns requires learners to metacognitively distinguish between observation (what we notice), hypothesis (what we wonder), and established fact (what we know for sure) — a foundational scientific thinking skill. Leaving most of the board in the WHAT WE WONDER column communicates that this is the beginning of the journey, not the end.	The DQB itself is a formative assessment tool. The distribution of sticky notes across columns tells the teacher how much prior knowledge the class has and how misconception-rich the starting point is. If many notes appear in WHAT WE KNOW FOR SURE that are actually incorrect (for example, 'the area is also 3 times bigger'), teacher does not correct them now but makes a mental note to return to them in Lesson 4. The act of placing notes publicly also identifies confident versus hesitant learners, informing future grouping decisions.

	reads a selection aloud. The board remains on display. Teacher points to the WHAT WE KNOW FOR SURE column and says: 'Right now, what can we put here with confidence?' Learners agree on two facts: (1) Both ratios (width and height) equal 3. (2) The shape of the face looks the same in both correctly resized photos. These two facts are written in the WHAT WE KNOW FOR SURE column. Everything else goes into WHAT WE WONDER.	map of what this unit is going to teach us. Every single thing on the WHAT WE WONDER column will be answered — some by you through investigation, some by working together, some through worked examples. Let us check this board at the start of every lesson and see how much we have learned.' Teacher ends by circling the ratio observation ($105/35 = 3$ and $135/45 = 3$) and writing: 'WHAT DOES THIS NUMBER 3 REALLY MEAN?' in the centre of the WHAT WE WONDER column in a different colour.		
Model Building Phase  VIDEO: Video: Balancing Act Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Proportions to Find Scale Ratios (Flexbook) Source: CK-12  Search ARES for similar readings	In the final 5 minutes of the lesson, each learner opens their exercise book to a fresh page headed LESSON 1 — MY INITIAL MODEL. Teacher says: 'A model is a diagram or a set of words that explains how something works. Right now, before we have learned the full mathematics, draw or write your best current explanation of this question: How can two photos look the same but have completely different sizes? Use arrows, diagrams, numbers, or words — whatever helps you express your thinking. You have 3 minutes.' Learners work individually in silence. They do not share this model with anyone yet. Teacher collects books or photographs the model pages with a phone to review after class. Teacher closes by saying: 'We will return to this model in Lesson 3 and in Lesson 12 to revise it as our understanding grows. Right now, it is just your best current thinking — and that is exactly what it should be.'	Teacher says: 'I want to capture where your thinking is right now, before we do any formal mathematics. Open your books to a new page. Write the heading: MY INITIAL MODEL — HOW CAN TWO PHOTOS LOOK THE SAME BUT HAVE DIFFERENT SIZES? Draw or write your explanation. There is no right answer yet. You have 3 minutes of silence.' WAIT TIME: 3 minutes of silence. Teacher circulates, reading models without commenting aloud. After 3 minutes, teacher says: 'Close your books. We will return to this model in later lessons. You are going to be amazed by how much your explanation grows.' Teacher previews Lesson 2: 'In our next lesson, you will measure pairs of similar figures cut from Kenyan newspapers and maps, and you will discover the properties that define what it means for two shapes to be truly similar — not just kind of similar, but mathematically similar. Bring your ruler and your curiosity.'	The initial model serves as a baseline record of each learner thinking at the start of the unit. In a storyline curriculum, model revision across the unit is a primary evidence of learning — learners must be able to articulate not just what the answer is but how their thinking has changed and why. Keeping the initial model private (not shared yet) removes performance pressure and encourages honest expression of current understanding, including uncertainty and incomplete thinking.	Teacher reviews initial models after class to identify: (a) learners who already use proportional language such as 'multiply by the same number' — these learners may benefit from extension tasks in Lessons 3 and 4; (b) learners whose models focus entirely on 'it is just a bigger copy' without any numerical or ratio reasoning — these learners need strong scaffolding in Lesson 2; (c) learners who draw diagrams showing the two photos side by side with arrows and numbers — these learners are using spatial reasoning and may be strong visual models to reference during whole-class discussions in later lessons.

D. TEACHER REFLECTION

After the lesson, the teacher should ask: (1) Did learners make genuine predictions or did they wait to copy peers? If copying was observed, the Think-Write phase needs a stricter no-talking protocol in future lessons. (2) Were cold-called learners able to articulate at least one observation about the photos? If not, the photos may need to be larger or projected for better visibility. (3) Did the distorted photo provoke genuine puzzlement? If learners shrugged it off, teacher should prepare a more dramatic distortion example for reference in Lesson 3. (4) Is the DQB rich with learner-generated questions, or is it sparse? A sparse DQB suggests learners need more structured scaffolding for question generation — consider using a sentence stem such as 'I wonder whether...' or 'I do not understand why...' in the next lesson. (5) Do the initial models show any engagement with the ratio of 3, or do they describe the phenomenon only qualitatively? This tells the teacher how much numerical scaffolding is needed at the start of Lesson 2.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Two passport photos of the same image: one measuring 35 mm x 45 mm and one measuring 105 mm x 135 mm. Both show the same face with the same proportions. A third distorted photo (width tripled, height unchanged) in which the face looks squashed and unnatural.
What did I learn?	When we divided the large photo measurements by the small photo measurements, we got the same number (3) for both width and height. The shape of the face is preserved only when both dimensions are scaled by the same number. The number that connects the two sizes is called the Linear Scale Factor.
How does this explain the phenomenon?	We cannot yet fully explain why multiplying by the same number preserves shape, or what happens to the area or volume when we scale. These questions are recorded on the Driving Question Board and will be answered in the lessons that follow.

LESSON 2 (80 minutes): The Passport Photo Puzzle — Discovering the Linear Scale Factor

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To introduce the anchoring phenomenon and establish the Linear Scale Factor as the mathematical rule that preserves shape during resizing.
Knowledge	By the end of this lesson, the learner should be able to: (a) identify corresponding lengths in two similar figures; (b) calculate the linear scale factor between two similar figures by dividing corresponding lengths; (c) verify that all pairs of corresponding lengths in similar figures share the same scale factor.
Skills	Measuring lengths with a ruler to the nearest millimetre; computing ratios; recording and organising data in a table; constructing and updating a class Driving Question Board (DQB).
Attitudes	Curiosity and openness: the learner questions why an enlarged image looks identical to the original and investigates the mathematics behind the observation. Responsibility: the learner handles printed photos and rulers carefully and records data honestly.
Key Inquiry Question	How do we identify and use similar figures in day-to-day situations?
Purpose in Storyline	In Lesson 1 (the launch lesson) students were introduced to the anchoring phenomenon — the Kimathi Street passport photo studio — and recorded initial questions on the DQB. Lesson 2 is the first evidence-gathering lesson: students physically measure both passport prints, compute length ratios across multiple pairs of corresponding dimensions, and discover that a single constant ratio (the Linear Scale Factor = 3) connects every pair of corresponding lengths. This is the first concrete, data-driven answer to the driving question and sets up all subsequent lessons on similarity, area ratios, and volume ratios.
Safety Notes	No significant hazards. Remind learners to handle printed photos and rulers carefully. Ensure rulers are passed point-end down.

B. LESSON OVERVIEW

Students receive printed cut-outs of two passport photos scaled at 35 mm × 45 mm and 105 mm × 135 mm — the same two prints from the Kimathi Street studio in the anchoring phenomenon. Working in pairs, they first predict in writing what changed and what stayed the same, then systematically measure at least five pairs of corresponding lengths (overall width, overall height, eye-to-chin distance, left-eye width, nose width) on both prints using a ruler. They record their measurements in a structured table and compute the ratio (large ÷ small) for each pair. Through guided sensemaking, the class discovers that every ratio equals 3 — the Linear Scale Factor. Cold-call sharing, a whole-class data merge on the board, and a first DQB entry consolidate the key idea. The lesson closes with a model-building sketch in which students annotate the two photos with arrows showing the ×3 relationship.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Watch Me Source: TED-Ed	Each learner receives a printed sheet containing side-by-side outlines of the two passport photos (35 mm × 45 mm and 105 mm ×	Distribute photo sheets and say: 'Before we measure anything, I want you to be a scientist — make your best prediction first. Write	Think-Pair-Share with Hypothesis Tracking: learners externalise prior knowledge and intuition before instruction, creating cognitive	Observable indicator: each learner has written responses to all three prompts in their exercise book before sharing. Teacher scans

Lessons Search ARES for similar videos HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 Search ARES for similar readings	135 mm) labelled 'Photo A' and 'Photo B'. Learners independently write answers to three prompt questions in their exercise books: (1) 'What is different about Photo A and Photo B?' (2) 'What is the same about Photo A and Photo B?' (3) 'If a single number connects the two sizes, what do you think that number might be — and how would you find it?' After 4 minutes of silent individual writing, learners share their predictions in a Think-Pair-Share with their desk partner for 2 minutes. The teacher then cold-calls 4–5 pairs to share predictions, recording key words on the board in two columns: CHANGED and STAYED THE SAME. Predictions are NOT evaluated as right or wrong — they are treated as hypotheses to be tested with evidence.	your three answers silently. You have 4 minutes. Go.' Set a visible timer. After 4 minutes: 'Turn to your partner. Compare your predictions. Do you agree on the number? You have 2 minutes.' After pair share, cold-call by name (not volunteers): 'Achieng, what did your pair say changed?' — WAIT 10 seconds. 'Mutua, what stayed the same in your prediction?' — WAIT 10 seconds. 'Njeri, what number did your pair guess connects the two photos, and why?' — WAIT 15 seconds. Record answers verbatim in the two columns. Say: 'These are our hypotheses. We are going to test every single one of them with real measurements today.'	dissonance when they realise they need a precise mathematical tool (ratio) to verify their intuitive guess. Recording predictions publicly on the board creates accountability and makes thinking visible.	books during pair-share to check for engagement. Listen for: (a) learners who correctly identify 'size' as what changed and 'shape/proportions' as what stayed the same — this shows readiness for the ratio activity; (b) learners who guess the number '3' — note these learners to probe their reasoning during evidence gathering; (c) learners who are silent or write 'I don't know' — seat them next to a talkative partner and explicitly prompt: 'Look at just the heights — what do you notice?'
Observe Phase VIDEO: Watch Me Source: TED-Ed Lessons Search ARES for similar videos HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 Search ARES for similar readings	Working in pairs, learners use a 30 cm ruler to measure five pre-labelled corresponding lengths on both Photo A and Photo B. The five lengths are marked on the photo sheet with labelled arrows: (L1) overall width, (L2) overall height, (L3) distance from top of head to chin, (L4) width of the left eye, (L5) width of the nose bridge. Learners record all ten measurements (5 on Photo A, 5 on Photo B) in a structured data table printed on their worksheet. They then compute the ratio Photo B ÷ Photo A for each of the five pairs and record the ratio as a decimal to 2 decimal places. After completing their own table, pairs join another pair to form a group of four and compare ratios, discussing: 'Did you all get the same ratios? What does that tell us?' Each group of four selects	Before measurement begins, demonstrate at the front: 'I am going to measure the overall width of Photo A. I place zero at the left edge — see how I hold the ruler flat — and I read off the measurement at the right edge. I get 35 mm. I record that in column Photo A, row L1.' Model one full entry on the board table. Say: 'Now you measure ALL five lengths on BOTH photos and fill in your table. You have 12 minutes. If you finish early, measure a sixth length of your choice and add it as L6.' Circulate and narrate what you see: 'I can see Kamau's pair has all five measurements for Photo A — good, now move to Photo B.' After 12 minutes, prompt group-of-four discussion: 'Compare your ratio column with your neighbours. Do your ratios agree? If not, re-measure together.' After 5 minutes	Structured Data Table + Group Data Merge (NGSS Science and Engineering Practice 4 — Analysing and Interpreting Data): learners generate their own quantitative evidence, then aggregate it across groups. The data merge on the board makes the constant-ratio pattern visible at scale — no single pair's data is sufficient, but the merged class table makes the pattern undeniable. This mirrors how scientists validate findings through replication.	Observable indicators: (1) Every learner's table has 10 measurements and 5 computed ratios before the group discussion begins — teacher checks during circulation. (2) All five computed ratios are in the range 2.90–3.10 (measurement error is acceptable; ratios outside this range indicate a measurement error and the teacher prompts re-measurement). (3) During group comparison, listen for: 'They're all about 3!' — this is the target insight. If no group voices this, prompt: 'Add up all your ratios and divide by 5. What do you get?' (4) Watch for learners who compute Photo A ÷ Photo B instead — note this as a teaching moment for the Explain phase (direction of the scale factor).

	one spokesperson to report to the class.	of group comparison, cold-call 3 spokespeople: 'Waweru, read me your five ratios.' — WAIT 10 seconds. Record each group's ratios in a class table on the board. Ask: 'Class, what pattern do you see in this table?' — WAIT 15 seconds. Cold-call: 'Otieno, describe the pattern.' — WAIT 10 seconds.		
Explain Phase VIDEO: Watch Me Source: TED-Ed Lessons Search ARES for similar videos HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 Search ARES for similar readings	The teacher formalises the pattern using the class data table on the board. Learners copy a structured note into their exercise books: the definition of Linear Scale Factor (LSF), the formula $LSF = \text{corresponding length in image} \div \text{corresponding length in original}$, and three worked examples using the passport photo data (L1, L2, L3). Learners then work individually on two consolidation tasks: (Task A) Given a photo of a maize seed packet from a Nakumatt supermarket shelf at 40 mm × 60 mm, and an enlarged display poster of the same packet at 200 mm × 300 mm, calculate the LSF. (Task B) A school noticeboard photo of Eliud Kipchoge finishing the Vienna marathon is 120 mm wide. The original print was 40 mm wide. Calculate the LSF. Learners write full working and a sentence conclusion: 'The linear scale factor is ____ because ____.' After 8 minutes of individual work, the teacher cold-calls two learners to present their working on the board.	Begin formalisation: 'Let me give a name to the pattern you just discovered. Every pair of corresponding lengths has the same ratio. We call this ratio the LINEAR SCALE FACTOR.' Write on the board: $LSF = (\text{length in enlarged figure}) \div (\text{length in original figure})$. Then: 'Let's verify it with our Photo data: $105 \div 35 = ?$ ' — cold-call: 'Aisha, calculate that.' — WAIT 10 seconds. ' $135 \div 45 = ?$ ' — cold-call: 'Baraka, your turn.' — WAIT 10 seconds. 'Both give us 3. The LSF = 3. Write that definition in your book now.' Allow 2 minutes for note-writing. Then launch Tasks A and B: 'You have 8 minutes to work individually. Show ALL working. Write a conclusion sentence.' Circulate and identify two learners with correct working to present — tell them quietly: 'I will ask you to come to the board in 8 minutes.' After presentations, address the direction error (if observed earlier): 'If Mutua computed $35 \div 105$, what did he get? What does that number mean?' — WAIT 15 seconds — 'It is the scale factor going FROM the large photo TO the small photo. Both are valid LSFs, but we must be clear about direction. Always state: LSF from A to B.'	Formalisation from Evidence (NGSS Practice 6 — Constructing Explanations): the teacher names and defines the concept AFTER learners have generated the data pattern, not before. This inductive sequence (data → pattern → definition) ensures the abstract formula is anchored in concrete passport photo measurements. The worked examples use Kenyan contexts (Nakumatt maize packet, Kipchoge poster) to reinforce that the LSF is a universal tool, not just a passport-photo trick.	Observable indicators: (1) Both Task A and Task B answers are correct: Task A $LSF = 200 \div 40 = 5$ (or $300 \div 60 = 5$); Task B $LSF = 120 \div 40 = 3$. (2) Learner conclusion sentences explicitly name the direction of scaling (e.g., 'from original to enlarged'). (3) During board presentations, listen for correct vocabulary: 'corresponding lengths,' 'ratio,' 'scale factor.' If learners say 'times bigger' without specifying the factor, prompt: 'Times bigger by how much exactly?' (4) Exit check: ask the class to vote by show of hands: 'Thumbs up if you can explain in one sentence what the linear scale factor tells us.' Count thumbs; re-explain for any learner who signals confusion.
Driving	The class returns to the DQB	Hand out two sticky notes per	Know / Wonder DQB Protocol	Observable indicators: (1) Every

Question Board (DQB) Creation  VIDEO: Watch Me Source: TED-Ed Lessons  Search ARES for similar videos  HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12  Search ARES for similar readings	(initiated in Lesson 1). Each learner writes two sticky-note entries: (1) a BLUE note stating one thing they now know ('I now know that...') and (2) a YELLOW note stating one new question that today's lesson has raised ('I still wonder...'). Pairs share their notes with their neighbour before posting. The teacher reads aloud three blue notes and three yellow notes selected from the board, narrating how today's evidence has partially answered the driving question and identifying which yellow-note questions will be addressed in upcoming lessons (e.g., 'What happens to the AREA when LSF = 3?' will be Lesson 4). The teacher updates a class 'Story So Far' strip on the board: 'We now know: LSF = 3 means every length in Photo B is 3× the corresponding length in Photo A. We still need to know: what happens to area and volume?'	learner. Say: 'In one minute, write your BLUE note — one thing you can now explain about the passport photo puzzle. Then write your YELLOW note — one question today's lesson raised that we haven't answered yet.' Allow 3 minutes of silent writing. 'Show your partner. Does their blue note agree with what you discovered?' Allow 2 minutes of pair comparison. 'Post your notes on the DQB — blues in the KNOW column, yellows in the WONDER column.' Read aloud selected notes, narrating: 'Amina writes: I now know that LSF = 3 means every measurement is exactly 3 times bigger. That is exactly right — and it explains why the photo looks the same but IS bigger.' For yellow notes: 'Odhiambo wonders: does the area also triple? Excellent question — we will investigate that in Lesson 4. I am writing that question here on our DQB under FUTURE EVIDENCE.' Update the Story So Far strip.	(adapted from Project-Based Learning DQB practice): public posting of 'I now know' and 'I still wonder' statements makes individual thinking visible to the whole class, validates curiosity as a mathematical habit of mind, and creates a living record of the unit's evidence-gathering sequence. The teacher's narrative link between yellow notes and future lessons gives learners a sense of purposeful forward momentum.	learner posts at least one blue note and one yellow note — teacher scans the board to ensure full participation. (2) Blue notes should reference the LSF and the passport photo context specifically (not generic statements). (3) Yellow notes that ask about area, volume, or 'what if only one dimension changes' reveal high-level thinking and should be explicitly named and celebrated. (4) If blue notes are vague (e.g., 'I learned about ratios'), the teacher prompts: 'Can you connect that to the passport photo specifically?'
Model Building Phase  VIDEO: Watch Me Source: TED-Ed Lessons  Search ARES for similar videos  HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12  Search ARES for similar readings	Learners open their exercise books to a fresh page labelled 'My Growing Model — Similarity and Enlargement.' They draw a simple annotated diagram showing Photo A (35 mm × 45 mm) and Photo B (105 mm × 135 mm) as two rectangles, with double-headed arrows between corresponding sides. On each arrow they write '× 3 (LSF)'. Below the diagram, they write a one-sentence model statement: 'When a photograph is enlarged with LSF = 3, every length is multiplied by 3, so the shape is preserved but the size increases.' They also draw a third rectangle labelled '?' with the question: 'What would a further	Say: 'In every lesson of this unit, we will update a model in our books. The model starts simple and gets richer as we gather more evidence. Today, draw your two rectangles. Label the dimensions. Draw arrows between corresponding sides and write the LSF on each arrow.' Demonstrate the first rectangle on the board. Allow 5 minutes for individual drawing and annotation. Circulate and give specific praise: 'Wanjiku, I can see you labelled both dimensions on both rectangles — excellent.' Then: 'Draw the third rectangle with LSF = 2 applied to Photo B — what dimensions will it have? Write '?' if you are not sure	Cumulative Annotated Diagram (NGSS Practice 2 — Developing and Using Models): the model is not a one-off task but a living artefact that grows across all 12 lessons. Starting with a simple rectangle diagram ensures accessibility; the '?' third rectangle creates productive anticipation for Lesson 3. Peer checking builds collaborative accountability and catches misconceptions (e.g., missing labels, wrong arrow directions) before they consolidate.	Observable indicators: (1) Diagrams show two rectangles with correct dimensions labelled. (2) Arrows between corresponding sides are present and labelled '× 3 (LSF)'. (3) Model statement is written in the learner's own words and references both 'shape preserved' and 'size increased'. (4) The '?' rectangle is present, indicating the learner has understood that the LSF can be applied repeatedly. (5) Teacher spot-checks 6–8 books during circulation — note any learner who draws the arrows incorrectly (e.g., between non-corresponding sides) as a target for Lesson 3 follow-up on corresponding vertices.

	enlargement with LSF = 2 look like?' (dimensions to be calculated in Lesson 3). Pairs check each other's diagrams for accuracy before the lesson ends.	yet. We will calculate this in Lesson 3.' Invite one learner to share their model under the document camera (or hold it up). Ask the class: 'Does this model answer our driving question yet? Fully or partially?' — WAIT 15 seconds — cold-call: 'Chebet, what do you think?' Close with: 'Our model is a work in progress — just like the photographer's software. Each lesson we add more. Next lesson, we look at what makes two shapes officially SIMILAR, not just resized.'		
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. EVIDENCE QUALITY: Did learners' measured ratios cluster tightly around 3.00, or was there wide variation? If variation was large, what does this tell you about their ruler-reading skills — and should you build a brief measurement-accuracy activity into the start of Lesson 3?
2. PREDICTION PATTERNS: What types of predictions appeared most often on the DQB? Did any learners predict the number 3 before measuring? What reasoning did they give — and how can you use that reasoning as a bridge in Lesson 3 when you formalise the conditions for similarity?
3. PARTICIPATION EQUITY: Were cold-call responses distributed across gender, ethnicity, and ability levels? Were quieter learners able to contribute during pair and group phases? Identify two learners who did not participate verbally today and plan a specific check-in question for them at the start of Lesson 3.
4. MISCONCEPTION WATCH: Did any learners compute the ratio as small $\div$ large (giving 0.33)? If so, this direction-of-scale-factor confusion is a known barrier to the area-ratio lesson (Lesson 4) — address it explicitly at the start of Lesson 3 with a short warm-up: 'Which photo is the original? Which is the enlargement? Which way do we divide?'
5. DQB RICHNESS: Do the yellow 'wonder' notes show curiosity about area, volume, or distortion (what happens when only one dimension changes)? If yes, your class is well-positioned for the upcoming lessons. If most yellow notes are procedural ('How do I calculate this?'), plan an additional Predict prompt in Lesson 3 that pushes conceptual wondering.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students measured five pairs of corresponding lengths on two passport photo prints (35 mm $\times$ 45 mm and 105 mm $\times$ 135 mm) and computed the ratio of each pair.
What did I learn?	Every pair of corresponding lengths shares the same ratio — the Linear Scale Factor (LSF = 3) — meaning the photographer's software multiplied every length by exactly 3, which is why the shape is perfectly preserved even though the size tripled.
How does this explain the phenomenon?	The LSF is calculated as $LSF = (\text{length in enlarged figure}) \div (\text{corresponding length in original figure})$. When the same LSF applies to ALL corresponding lengths, the shape is preserved and the two figures are mathematically similar.

LESSON 3 (80 minutes): Identifying Properties of Similar Figures: Angles, Ratios, and the Rule Behind the Resize

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To deepen learners' understanding of the defining properties of similar figures by systematically measuring angles and side ratios in passport-photo-inspired rectangles and triangles, establishing that similarity requires BOTH equal corresponding angles AND proportional corresponding sides.
Knowledge	Learners will know that: (1) two figures are similar if and only if all corresponding angles are equal AND all ratios of corresponding sides are equal (constant); (2) this constant ratio is the Linear Scale Factor (LSF); (3) changing only one dimension destroys similarity and causes distortion — exactly as seen when a passport photo is stretched on only one axis.
Skills	Learners will be able to: (a) measure corresponding angles in pairs of figures and confirm equality; (b) calculate ratios of corresponding sides and confirm they are equal; (c) state whether two given figures are similar, giving mathematical justification; (d) identify the LSF from a pair of similar figures.
Attitudes	Learners will appreciate precision in mathematical measurement and reasoning, recognising that the photographer's software must obey an exact mathematical rule — not an approximation — to preserve the image perfectly.
Key Inquiry Question	How do we identify and use similar figures in day-to-day situations?
Purpose in Storyline	In Lesson 1 (Predict), learners made initial guesses about what 'same shape, different size' means mathematically. In Lesson 2 (Observe & Explore overview), learners were introduced to the phenomenon and began noticing that the passport print and the enlarged noticeboard print share something deeper than appearance. In THIS lesson (Lesson 3 — deep Observe & Explore), learners gather systematic, measured evidence: they physically measure angles and sides on printed rectangle pairs and triangle pairs, compute ratios, and discover the two-part definition of similarity. This evidence will feed directly into Lesson 4 (Explain), where they formalise the LSF.
Safety Notes	No significant safety hazards. Ensure rulers and protractors are handled carefully and returned after use. If scissors are used to cut printed figures, supervise to avoid injury.

B. LESSON OVERVIEW

Learners work in groups of four with printed 'Passport Photo Investigation Sheets' showing pairs of rectangles and triangles drawn at different scales — some truly similar, some deliberately distorted (one dimension stretched). Using rulers and protractors, they measure every corresponding angle and compute every ratio of corresponding sides. Groups record findings in a structured Evidence Table. A whole-class gallery comparison then surfaces the pattern: truly similar pairs always have equal angles AND a constant side ratio, while distorted pairs break the constant ratio. The lesson closes with learners revising their personal models of 'what similarity means' and posting new questions on the Driving Question Board. Kenya context: the rectangle pairs are labelled with passport photo dimensions in millimetres as used at studios along Nairobi's Kimathi Street; triangle pairs are drawn on a grid referencing a map-enlargement scenario from the Kenya National Examinations Council (KNEC) archive.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Watch Me Source: TED-Ed Lessons  Search ARES for similar videos  HTML: Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12  Search ARES for similar readings	Each learner receives a 'Quick Predict' card showing three pairs of rectangles: Pair A (35×45 mm scaled to 70×90 mm), Pair B (35×45 mm scaled to 105×135 mm), and Pair C (35×45 mm scaled to 70×135 mm — deliberately distorted width-only). Without measuring, learners individually write on a sticky note: (1) Which pairs do you think are 'the same shape'? (2) What single number — if any — connects the sizes? (3) Predict what you will find when you measure the angles and side ratios. Learners stick their predictions on the class Prediction Wall under three columns: 'All three are similar', 'Only A and B', 'I'm not sure yet'.	Distribute Quick Predict cards. Say verbatim: 'Before you touch a ruler today, I want your brain's first answer. There is no wrong prediction — every prediction is data we will test.' Allow 3 minutes of silent individual thinking. Then say: 'Turn to your shoulder partner — you have 90 seconds — convince them of your prediction using only what your eyes tell you.' [Wait 90 seconds.] Cold-call 3 learners from different columns on the Prediction Wall: 'Wanjiku, what did your eyes tell you about Pair C?' [Wait 10 seconds before prompting.] 'Otieno, do you agree or disagree — and why?' [Wait 12 seconds.] Do NOT confirm or deny predictions. Say: 'We are going to gather evidence right now to test every single one of these predictions.'	Prediction-before-evidence (Predict-Observe-Explain cycle launch): By committing predictions to sticky notes before measuring, learners activate prior intuitive knowledge and create cognitive dissonance targets. The Prediction Wall makes thinking visible and gives the class a shared reference to return to after evidence gathering, making the 'surprise' of distortion more salient.	Observable indicator: Are learners' predictions clustered on the Prediction Wall? If >80% predict 'All three are similar', this signals a misconception (that any enlargement preserves similarity) — teacher notes this and plans a targeted cold-call contrast during Phase 2 data-collection debrief. If learners cannot articulate any reasoning beyond 'it looks the same', teacher prompts: 'What exactly would you measure to check?'
Observe Phase  VIDEO: Watch Me Source: TED-Ed Lessons  Search ARES for similar videos  HTML: Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12  Search ARES for similar readings	Groups of four receive: (1) Passport Photo Investigation Sheet with Pairs A, B, C printed at full scale on A3 paper; (2) a Triangle Investigation Sheet with Pairs D (similar triangles) and E (non-similar triangles with one angle changed); (3) an Evidence Table with columns: Figure Pair Corresponding Angles (measure all) Side Lengths (measure all) Ratio Side 1 Ratio Side 2 Ratio Side 3 Are ratios equal? Your verdict: Similar or Not? Reason in one sentence. Roles: Measurer A (angles, rectangle pairs), Measurer B (sides, rectangle pairs), Measurer C (angles & sides, triangle pairs), Recorder (fills Evidence Table, challenges group if data looks inconsistent). Groups measure ALL angles with protractors and ALL sides with rulers to the nearest 0.5 mm,	Before releasing groups, model the first measurement live on the board: 'Watch me measure angle P in Pair A. I place the protractor baseline along this side — the angle reads 90°. I record 90° in the table. Now YOUR job is to measure every angle in every pair.' [Pause 10 seconds, check for questions.] Circulate with a clipboard. Stop at each group for 2 minutes. Ask probing questions — verbatim: 'What do you notice about the angles in Pair C compared to Pair A?' [Wait 12 seconds before accepting answer.] 'Your ratio for Pair C Side 1 is 2.0 but Side 2 is 3.0 — what does that tell you?' [Wait 15 seconds.] 'If the ratios are different, can we call this pair similar? Defend your answer.' Cold-call a different group member each time (rotate: not the recorder). During Gallery Walk:	Structured Data Collection + Gallery Walk with Dot Voting: The Evidence Table scaffolds scientific reasoning — learners cannot jump to a conclusion without filling every cell. The Gallery Walk externalises group thinking so cross-group comparison surfaces natural variation in measurement and prompts self-correction. The dot system transforms passive viewing into active agreement/disagreement — a low-stakes peer-review that mirrors scientific practice. This directly mirrors how a KNEC examiner checks whether a student's diagram satisfies similarity conditions.	Observable indicators: (1) Are all cells in the Evidence Table completed with numeric values (not left blank or filled with 'looks about right')? (2) Do groups correctly compute ratios as fractions/decimals — or are they subtracting side lengths instead (a common additive-thinking error)? Teacher listens for phrases like 'the difference is 35 mm' — if heard, intervene immediately: 'Is similarity about the difference or the multiplier? Let's check.' (3) During Gallery Walk, count orange dots on Pair C — if fewer than half the class flagged Pair C as non-similar, teacher plans a direct challenge question in Phase 3.

	compute ratios as decimals to 2 d.p., and complete the Evidence Table. Teacher circulates and probes. After 20 minutes of group work, each group posts their Evidence Table on the wall for a 5-minute silent Gallery Walk — learners use a 'dot sticker' system: green dot = 'our data agrees with yours', orange dot = 'our data differs — we need to discuss'.	'You have 5 minutes — silent, no talking. Place your dot and move on.' After Gallery Walk, facilitate a 3-minute whole-class discussion: 'Which pairs received the most orange dots? Why do you think groups disagreed?'		
Explain Phase  VIDEO: Watch Me Source: TED-Ed Lessons  Search ARES for similar videos  HTML: Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12  Search ARES for similar readings	Whole-class discussion anchored by the Evidence Tables on the wall. Learners co-construct a 'Similarity Checklist' in their exercise books: Condition 1 — All corresponding angles are equal. Condition 2 — All ratios of corresponding sides are equal (this ratio is the Linear Scale Factor). They then apply the checklist immediately to three new mini-problems projected on the board: (a) Two Kenyan road-sign triangles of different sizes — are they similar? (b) A githeri recipe scaled up: if the pot diameter doubles but the height stays the same, is the new pot geometrically similar to the original? (c) The original passport photo (35×45 mm) and a WhatsApp display picture cropped to 40×40 mm — similar or not? For each, learners write a 2-sentence justification using the Similarity Checklist language. Pairs share justifications; one pair is cold-called per problem.	Open with: 'Look at the Prediction Wall. Raise your hand if your prediction was confirmed by your evidence.' [Wait 8 seconds, scan room.] 'Raise your hand if your evidence surprised you.' [Wait 8 seconds.] 'Nduta, what surprised you about Pair C? Use your Evidence Table numbers.' [Wait 12 seconds.] Guide learners to articulate both conditions by asking: 'Is equal angles enough? Look at your data for Pair E — what happened to the angles there?' [Wait 10 seconds.] 'Is proportional sides enough on its own? Can you imagine a figure where sides are proportional but angles are different?' [Wait 15 seconds — this is a stretch question; accept partial answers.] Write the two conditions on the board as learners dictate them. For the three mini-problems, give 3 minutes of pair-work per problem, then cold-call: 'Kamau, problem (b) — similar or not, and what's your mathematical reason?' [Wait 12 seconds.] Affirm correct use of ratio language; correct additive-difference language explicitly: 'We say the ratio of sides is 2:1, not the difference is 35 mm — similarity is multiplicative.'	Co-constructed Definition + Immediate Application (I Do–We Do–You Do compressed): Rather than presenting the definition top-down, the teacher elicits it from the class's own Evidence Table data — giving learners ownership of the definition. The three varied mini-problems (road signs, cooking pot, WhatsApp photo) immediately transfer the definition to new Kenya-relevant contexts, preventing rote memorisation. The WhatsApp photo problem re-anchors to the passport phenomenon and sets up the distortion concept for Lesson 4.	Observable indicators: (1) Can learners state BOTH conditions without prompting? (2) In written justifications, do learners cite specific ratio values (e.g., ' $35/70 = 0.5$ and $45/90 = 0.5$, so the LSF is 0.5') rather than vague language ('they look the same')? (3) Do learners correctly identify the WhatsApp photo (40×40 vs 35×45) as NOT similar — the key diagnostic for understanding that both dimensions must scale equally? Teacher collects 6 exercise books (one per group) at the end of this phase to check justification quality before Lesson 4.
Driving	Each learner individually writes on	Say: 'You now have evidence.'	Driving Question Board (DQB) —	Observable indicators: (1) 'We now

Question Board (DQB) Creation  VIDEO: Watch Me Source: TED-Ed Lessons  Search ARES for similar videos  HTML: Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12  Search ARES for similar readings	a DQB card: (1) one thing they now KNOW for certain about similarity (evidence from today's measurements); (2) one new question today's lesson raised for them. Cards are posted on the class Driving Question Board under two columns: 'We now know...' and 'We still wonder...'. The class reads the 'We still wonder' cards silently and dot-votes (sticker) on the question they most want answered next lesson. Learners also update their personal 'Passport Photo Puzzle Explanation' — a cumulative written explanation they have been building since Lesson 1 — adding one new paragraph: 'Today I discovered that similarity requires two conditions...'	Let's update our board.' Distribute DQB cards. Give 3 minutes of silent writing. While learners write, teacher reads the Prediction Wall aloud: 'At the start of this lesson, most of us predicted... Our evidence showed... What does that mean for our driving question?' After posting, read 3 'We still wonder' cards aloud. Ask: 'Which of these questions do you think connects most directly to the passport photo puzzle?' [Cold-call 2 learners, wait 10 seconds each.] Close with: 'Next lesson, we will use today's ratio — the Linear Scale Factor — to calculate exactly how area changes when you enlarge a photo. Hold onto that question.'	Public Knowledge Tracking: The DQB is a visible, cumulative record of the class's growing understanding. It prevents the lesson from feeling like a closed unit and maintains the epistemic storyline across 12 lessons. The dot-vote on 'We still wonder' cards gives learners genuine agency in shaping the inquiry direction while allowing the teacher to surface and address persistent misconceptions in the next lesson.	know' cards should reference measured evidence (angles, ratios, specific numbers) — not just restate the textbook definition. (2) 'We still wonder' cards reveal the class's frontier of understanding: anticipated productive questions include 'Does the area also triple?' or 'What if one angle changes but sides stay proportional?' — these signal readiness for LSF-area lessons. Concerning questions include 'What even is a ratio?' — teacher notes these learners for targeted support in Lesson 4.
Model Building Phase  VIDEO: Watch Me Source: TED-Ed Lessons  Search ARES for similar videos  HTML: Trigonometric Functions and Angles of Rotation (Flexbook) Source: CK-12  Search ARES for similar readings	Learners individually revise the 'similarity model' diagram they first drew in Lesson 1 (their Predict-phase sketch of what 'same shape, different size' means). They must now add: (a) annotation arrows showing corresponding angles with the label 'equal'; (b) annotation arrows showing corresponding sides with the ratio written as a fraction and decimal; (c) a red 'X' through any version of the model that showed only size difference without the two conditions. Learners share revised models with their group — the group nominates one model to be photographed (or displayed) as the 'Group Model' on the wall alongside their Evidence Table. A 2-minute whole-class model-comparison closes the lesson: teacher points to two contrasting group models and asks 'What did Group 3 add that Group 1 hasn't shown yet?'	Say: 'Pull out the model you drew in Lesson 1. You have 5 minutes to revise it using today's evidence. Your model must now show both conditions — not just one.' [Circulate, prompt learners who are only re-drawing without annotating: 'Your diagram looks similar to last time — what NEW information from today's Evidence Table can you add to it?'] During group nomination: 'Groups, you have 90 seconds — choose the model that best shows BOTH conditions with actual numbers from your measurements.' During whole-class comparison, cold-call a learner from Group 1 to respond to Group 3's model: 'Achieng, what would you add to your group's model after seeing this?' [Wait 12 seconds.] Close: 'Every lesson, your model will grow. By Lesson 12, it will explain the entire passport photo puzzle — including why the face looks distorted when	Cumulative Model Revision (Progressive Modelling): Returning to and revising the same model across multiple lessons — rather than drawing a new model each time — makes learning progression tangible and builds metacognitive awareness. Learners literally see their understanding change. The red 'X' instruction forces learners to confront and actively reject prior misconceptions rather than simply adding new correct ideas alongside old incorrect ones.	Observable indicators: (1) Revised models must show both angle equality AND ratio equality with specific numbers — a model showing only 'bigger/smaller' arrows is insufficient and teacher provides written feedback. (2) The group nomination process reveals peer consensus on quality — teacher notes if groups consistently nominate the same type of model (e.g., only numerical, no diagram) and plans a model-diversity prompt for Lesson 4. (3) End-of-lesson exit check: teacher scans 6 revised models from across groups and categorises as: Full (both conditions, with numbers) / Partial (one condition only) / Incomplete (pre-lesson level). Target: >70% Full by end of Lesson 3.

		only the width is stretched.'		
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D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) MEASUREMENT ERRORS — Did groups make systematic measurement errors (e.g., misreading the protractor inner vs outer scale, measuring to the nearest 5 mm rather than 0.5 mm)? If yes, build a 5-minute 'precision measurement' micro-lesson into the start of Lesson 4. (2) ADDITIVE THINKING — How many learners used subtraction ('the sides differ by 35 mm') rather than division ('the ratio is 3:1') when comparing sides? This is the most persistent misconception in similarity and must be explicitly addressed in Lesson 4's Explain phase with a direct contrast: 'Similarity is multiplicative — it asks HOW MANY TIMES, not HOW MUCH MORE.' (3) PAIR C DETECTION — Did all groups correctly identify Pair C (the distorted passport photo) as non-similar? If not, use Pair C as the opening warm-up in Lesson 4 before moving to LSF-area calculations. (4) DQB QUALITY — Were the 'We still wonder' questions mathematical (about ratios, areas, scale factors) or superficial ('how does the camera work?')? Mathematical questions signal strong engagement with the phenomenon; superficial questions suggest the phenomenon has not yet been fully activated — consider revisiting the Kimathi Street passport studio context with more physical detail at the start of Lesson 4. (5) MODEL PROGRESSION — What percentage of models reached 'Full' level? If below 70%, pair a 'Full' model learner with a 'Partial' model learner as a think-pair in Lesson 4's opening.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners measured corresponding angles (using protractors) and corresponding side lengths (using rulers) in three pairs of rectangles based on Nairobi passport photo dimensions and two pairs of triangles. They computed ratios of corresponding sides as decimals to 2 d.p. and recorded all findings in a structured Evidence Table. They conducted a Gallery Walk to compare group data using a dot-voting system.
What did I learn?	Two figures are similar if and only if: (1) all corresponding angles are equal, AND (2) all ratios of corresponding sides are equal (this constant ratio is the Linear Scale Factor). Changing only one dimension — as in the distorted Pair C rectangle — destroys similarity because the side ratios become unequal, even though the angles may remain equal in a rectangle. Similarity is a multiplicative relationship, not an additive one.
How does this explain the phenomenon?	Learners explained their Evidence Table findings to peers during the Gallery Walk and to the whole class during the Explain phase, justifying 'similar' or 'not similar' verdicts using specific ratio values. They applied the Similarity Checklist to three new Kenya-relevant contexts (road-sign triangles, a cooking pot, and a WhatsApp profile photo crop), wrote 2-sentence justifications using ratio language, and revised their cumulative similarity model diagrams to show both conditions with annotated measurements.

LESSON 4 (80 minutes): Discovering Similar Figures: Equal Angles and Constant Ratios

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To guide learners to discover the two defining properties of similar figures — equal corresponding angles and constant ratios of corresponding sides — through hands-on measurement of figures cut from Kenyan newspapers and maps.
Knowledge	By the end of the lesson, the learner should be able to: (a) identify corresponding sides and corresponding angles in pairs of similar figures; (b) state that corresponding angles are equal in similar figures; (c) state that ratios of corresponding sides are constant (equal to the linear scale factor) in similar figures; (d) use the formal definition of similarity to determine whether two given figures are similar.
Skills	Measuring lengths with a ruler to the nearest millimetre; measuring angles with a protractor; computing and comparing ratios; tabulating data systematically; constructing reasoned arguments from evidence.
Attitudes	Curiosity in noticing patterns within measured data; precision and care in measurement; collaborative discussion and respect for peers' findings; appreciation that mathematics explains everyday phenomena such as photo resizing.
Key Inquiry Question	How do we identify and use similar figures in day-to-day situations?
Purpose in Storyline	In Lessons 1–3 students established that the passport photo was resized by a factor of 3 in both dimensions and began asking WHY the shape is preserved. In this lesson (Lesson 4 of 12), students gather the critical evidence: by measuring real cut-outs from Kenyan newspapers and maps they discover the two properties that DEFINE similarity — equal angles and constant side ratios. This evidence directly answers the phenomenon question: the photographer's software preserved shape because it kept all angles equal and scaled every length by the SAME ratio.
Safety Notes	Scissors are used to cut newspaper figures — remind learners to cut away from the body and to pass scissors handle-first. Ensure desks are clear of clutter before cutting activity begins.

B. LESSON OVERVIEW

Learners work in groups of four. Each group receives an envelope containing two pre-cut polygon pairs (a rectangle and its enlargement cut from a Kenyan newspaper advertisement, and a triangle pair cut from a printed Kenyan road-map extract). Groups measure all corresponding sides (mm) and all corresponding angles (degrees), record data in a structured table, compute the ratio of each pair of corresponding sides, and look for patterns. Groups then share findings on the board. The teacher orchestrates a whole-class discussion that formalises the two similarity properties. The lesson closes by linking discoveries directly back to the passport photo phenomenon and updating the Driving Question Board.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Parallelogram on the coordinate	Each learner individually examines two printed polygon pairs displayed on the board (a small and a large version of the same	Display the two polygon pairs on the board using a printed A3 sheet. Say: 'Before we touch a ruler, I want your brain's best	Think-Pair-Share: activates prior knowledge and surfaces misconceptions (e.g., 'only lengths change' vs 'everything changes	Observable indicator: every learner has three written sentences in their book before pair-share begins. Teacher scans

plane Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Equations with Variables on Both Sides (Flexbook) Source: CK-12 Search ARES for similar readings	newspaper advertisement logo). Without measuring, learners write a 3-sentence prediction in their exercise books answering: (1) 'What do you think is the same about these two shapes?' (2) 'What do you think is different?' (3) 'How do you think the photographer's software decided how much to enlarge each length?' Learners then share predictions with a shoulder partner (Think-Pair).	guess. Look carefully at these two shapes — one is from a small Kimathi Street newspaper ad, one is the enlarged version. Write three sentences: what is the same, what is different, and how do you think the software decided each new length. You have 3 minutes — silent, individual thinking.' [Start timer. WAIT 3 minutes.] Then: 'Now share with your shoulder partner. You each have 45 seconds.' After pair-share, cold-call 3 pairs: 'Amina, what did you and your partner agree on?' — 'Kamau, did your prediction differ from Amina's? How?' — 'Achieng, what was the most surprising thing your partner said?' Record key prediction words on the board in a 'Predictions' column (e.g., 'angles stay the same', 'lengths double', 'shape unchanged').	proportionally') before evidence is gathered, giving the teacher diagnostic information and giving learners a personal stake in testing their own predictions.	books during the 3-minute window — look specifically for learners who predict 'angles also change' (common misconception) — flag these learners mentally for targeted cold-calling during Phase 3.
Observe Phase  VIDEO: Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Equations with Variables on Both Sides (Flexbook) Source: CK-12 Search ARES for similar readings	Groups of four receive Envelope A (rectangle pair: 35 mm × 50 mm and 70 mm × 100 mm, cut from a Kenyan newspaper advertisement) and Envelope B (triangle pair: scalene triangles with sides 30 mm, 40 mm, 50 mm and 60 mm, 80 mm, 100 mm, printed from a Kenyan road-map extract). Each group completes Measurement Table 1 (rectangles): measure all four sides of each rectangle and all four interior angles; record in the table; compute the ratio (larger ÷ smaller) for each pair of corresponding sides. Then complete Measurement Table 2 (triangles): measure all three sides and all three angles of each triangle; compute the ratio for each pair of corresponding sides. Learners record observations in two columns: 'What I notice about the ANGLES' and 'What I notice	Distribute envelopes and printed measurement tables. Say: 'Your job for the next 20 minutes is to be mathematicians gathering evidence — just like the photographer's software does every time it resizes an image. Measure every side to the nearest millimetre and every angle to the nearest degree. Record everything in your table. Then compute the ratio: larger side divided by smaller side, for EACH pair of corresponding sides. I am not telling you what to find — your data will tell you.' [Circulate. At 7 minutes, pause class for 30 seconds]: 'Quick check — has every group identified which sides CORRESPOND to each other? Corresponding means opposite matching positions. Thumbs up if yes.' [Resume. At 14 minutes, prompt]: 'You should have six	Structured Data Collection with Guided Noticing: the two-column observation journal ('angles' vs 'side ratios') separates the two similarity properties so learners construct each independently from their own data rather than being told. This mirrors how scientists distinguish between different types of measurements before drawing conclusions.	Observable indicators: (1) All ratio values in a group's table are within ±0.05 of 2.0 (rectangle pair) or 2.0 (triangle pair) — teacher checks this during circulation; groups with ratios varying wildly are redirected to re-check whether they identified corresponding sides correctly. (2) Mini-whiteboard sentences: look for language such as 'all angles are the same' and 'all ratios are equal' — these indicate readiness for formalisation. Flag any group whose sentence says 'some ratios are the same' for targeted questioning in Phase 3.

	about the SIDE RATIOS'.	ratios computed by now. Look across your row of ratios — what do you notice? Whisper it to your group.' [WAIT 15 seconds before circulating again.] At 20 minutes, say: 'Calculators down. In the next 2 minutes, each group writes ONE sentence about the angles and ONE sentence about the ratios on your mini-whiteboard — we will share these with the class.'		
Explain Phase  VIDEO: Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Equations with Variables on Both Sides (Flexbook) Source: CK-12  Search ARES for similar readings	Each group's recorder writes their two sentences on the class board in a shared 'Evidence Table'. Teacher facilitates a whole-class gallery-style discussion: learners look for agreement and disagreement across groups' findings. Through structured questioning, the class jointly articulates the two defining properties of similar figures. Learners then copy the formal definition into their books and use it to test two additional figure pairs (one similar, one non-similar) provided on a printed card — they must justify their answer using BOTH properties (angle evidence AND ratio evidence). Learners also revisit their Phase 1 predictions and annotate them: 'Confirmed ✓' or 'I was wrong because...'.	Stand at the board with all groups' sentences displayed. Say: 'I see 6 groups. Let me read these sentences aloud. [Read all.] Raise your hand if your group wrote something about the ANGLES being equal.' [Cold-call one hand]: 'Wanjiku — tell us exactly what your angles showed. What did you measure for the small triangle's angles? The large triangle's?' [WAIT 10 seconds after each answer.] Then: 'Did ANY group find angles that were NOT equal?' [Pause 10 seconds — scan room.] 'Good. Now the ratios — Otieno, read your group's sentence about ratios.' [WAIT 10 seconds.] 'Did every group get the SAME ratio for all three sides of the triangle pair? What was that number?' [Cold-call 3 groups.] Guide class to state both properties clearly. Then say: 'Let me write the formal mathematical definition. Copy this exactly.' [Write and say aloud]: 'Two figures are SIMILAR if and only if: (1) all corresponding angles are EQUAL, and (2) all ratios of corresponding sides are EQUAL — this constant ratio is called the LINEAR SCALE FACTOR.' Then: 'Now apply this definition. Test the two new figure pairs on your card. You must use BOTH conditions as evidence. Five minutes, then I will	Evidence-Based Whole-Class Synthesis (Claim-Evidence-Reasoning): learners publicly compare data across groups to build a shared claim, explicitly connecting each piece of evidence (angle measurements, ratio values) to the reasoning (the formal definition). This strategy makes the mathematical definition feel discovered rather than imposed, and directly connects to the passport phenomenon — the software maintained BOTH conditions.	Observable indicators: (1) During cold-calling on the new figure pairs, learners cite BOTH angle equality AND constant ratio (not just one). (2) Prediction annotation — teacher spot-checks 6 books; learners who wrote 'angles also change' in Phase 1 should now have a written correction. (3) Exit signal: teacher asks 'Show me on your fingers: how many CONDITIONS must be true for two figures to be similar?' — all fingers should show 2.

		cold-call.' [WAIT 5 minutes, circulate.] Cold-call 2 groups for each pair, asking: 'What is your verdict — similar or not similar? Show me your angle evidence first, then your ratio evidence.'		
Driving Question Board (DQB) Creation  VIDEO: Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Equations with Variables on Both Sides (Flexbook) Source: CK-12  Search ARES for similar readings	Each learner receives two sticky notes (one green, one orange). On the GREEN note they write one thing today's evidence ANSWERS about the driving question (e.g., 'The photographer's software keeps all angles equal — that is why the face looks the same'). On the ORANGE note they write one NEW question today's lesson raises (e.g., 'If the scale factor is 3, does the AREA also multiply by 3?'). Learners place notes on the class Driving Question Board under the columns 'Evidence We Now Have' and 'New Questions We Are Asking'. Class reads the orange questions aloud together.	Point to the passport photo prints still displayed from Lesson 1. Say: 'We started this unit asking: how can two photos show the exact same image but have completely different sizes? Today you gathered real evidence. On your green sticky note, write exactly which part of that question you can now answer — use the mathematical language we just formalised. On your orange note, write a NEW question that today's evidence raised for you. You have 2 minutes.' [WAIT 2 minutes — silent individual writing.] Collect notes on DQB. Read 3 orange questions aloud: 'Listen — Zawadi asks: does the area also scale by the same factor of 3? Kipchoge's group asks: what if only the width is scaled — which condition breaks down? Fatuma asks: is there a formula connecting scale factor to area?' Say: 'These are EXACTLY the questions mathematicians ask. They are going on our board and we will answer them in Lessons 5 through 7.'	Driving Question Board (DQB) Protocol: making thinking visible by publicly tracking what is now known versus what is still unknown creates a living map of the unit's intellectual journey, motivates upcoming lessons, and reinforces that science and mathematics are progressive — each lesson answers some questions and opens new ones.	Observable indicators: (1) Green sticky notes contain at least one piece of mathematical language from today ('corresponding angles', 'constant ratio', 'linear scale factor', 'similar figures'). (2) Orange notes generate at least 4 distinct new questions across the class — a low count indicates the lesson did not open sufficient curiosity. Teacher photographs the DQB for tracking across the unit.
Model Building Phase  VIDEO: Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum)  Search ARES	Learners return to the cumulative 'Similarity Model' diagram started in Lesson 1 (a annotated sketch of the two passport photos). Today they add a new annotated layer: they label THREE pairs of corresponding sides on the passport photo diagram with their measurements and computed ratios, and label TWO pairs of	Say: 'Open your cumulative passport photo model from Lesson 1. Today you are adding a new layer — the layer that EXPLAINS why the photos look identical despite different sizes. Add labels for three pairs of corresponding sides — write the actual measurements and compute the ratio next to each pair. Label two	Cumulative Model Revision (Annotated Diagram): adding to the same passport-photo model across every lesson externalises the growth of understanding — learners can literally SEE what they knew in Lesson 1 versus what they know now. The counter-example (distorted photo) deepens understanding by showing what	Observable indicators: (1) The similarity statement in each learner's model references BOTH conditions (not just one). (2) The counter-example sketch is present and the annotation names the specific violated condition ('ratios of corresponding sides are not equal' or equivalent). (3) At least three correctly labelled ratio

for similar videos  HTML: Equations with Variables on Both Sides (Flexbook) Source: CK-12  Search ARES for similar readings	corresponding angles. They write the formal similarity statement: 'Photo A ~ Photo B because (1) all corresponding angles are equal and (2) all ratios of corresponding sides = 3 (the linear scale factor).' Learners also add a 'distorted photo' counter-example sketch (width tripled, height unchanged) and annotate which condition is violated and why the face looks wrong.	corresponding angles. Then write the formal similarity statement at the bottom of your diagram — copy the structure from the board.' [Display structure: 'Figure A ~ Figure B because...']. 'Finally — and this is the most important part — sketch the DISTORTED photo: width tripled, height kept the same. Label it. Then write one sentence: which of our two conditions is violated, and how does that explain why the face looks unrecognisable?' [WAIT 8 minutes for model work. Circulate — check that ratios are computed correctly and that the counter-example is annotated with a specific violated condition.] Close by cold-calling one learner: 'Mwangi — show us your distorted photo sketch and read your sentence about which condition breaks.' [WAIT 10 seconds after answer.] 'Exactly. When the scale factor is different for width versus height, the ratios of corresponding sides are NO LONGER EQUAL — condition 2 fails — and the image distorts. The photographer's software avoids this by applying ONE scale factor to EVERY length. That single number is our linear scale factor.'	happens when similarity conditions are violated, anchoring the formal definition in a memorable visual contrast.	computations appear on the passport photo diagram. Collect 6 models for written feedback before Lesson 5.
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D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) **DATA QUALITY** — Did most groups obtain ratio values within ± 0.05 of the expected scale factor? If not, was the error in identifying corresponding sides or in ruler technique? Plan a 5-minute measurement-skills micro-lesson at the start of Lesson 5 if needed. (2) **MISCONCEPTIONS SURFACED** — Did any groups persist in claiming 'angles also change'? If yes, note names and design a targeted angle-measuring task as a warm-up in Lesson 5. (3) **DQB HARVEST** — Count the number of orange sticky-note questions that mention AREA. If more than half the class raised this question spontaneously, the transition to Lesson 5 (area scale factor) has strong motivation — begin Lesson 5 by reading those notes aloud. (4) **MODEL QUALITY** — Did learners include the distorted-photo counter-example? The counter-example is critical for conceptual depth; if most models omitted it, allocate 5 minutes at the start of Lesson 5 to complete it as a paired task. (5) **PACING** — This lesson targets 80 minutes; if Phase 2 measurement ran long, consider pre-cutting the triangles before Lesson 5's corresponding activity to maintain pace.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners measured corresponding sides and angles of two pairs of polygon cut-outs (a rectangle pair from a Kenyan newspaper advertisement and a triangle pair from a Kenyan road-map extract). They recorded all measurements in a structured table and computed the ratio of each pair of corresponding sides.
What did I learn?	Learners discovered that in both figure pairs: (1) all corresponding angles were equal, and (2) all ratios of corresponding sides were equal (constant), equalling the linear scale factor. They formalised these two findings as the definition of similar figures and applied the definition to classify new figure pairs as similar or non-similar.
How does this explain the phenomenon?	The two defining properties of similarity — equal corresponding angles and constant ratios of corresponding sides — explain why the passport photo looks identical at two different sizes: the photographer's software applied one linear scale factor (3) to every length, preserving all angles and keeping all side ratios equal. When only width is scaled (distorted photo), the ratio condition is violated and the image loses its shape.

LESSON 5 (80 minutes): Linear Scale Factor and Lengths — Explaining the Photographer's Rule

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To develop learners' ability to calculate and apply the Linear Scale Factor (LSF) to find unknown lengths in similar figures, and to connect this mathematical rule to the passport photo phenomenon.
Knowledge	Learners will know that: (a) the Linear Scale Factor (LSF) is the constant ratio of any pair of corresponding lengths in two similar figures; (b) if $LSF = k$, then every length in the enlarged figure is exactly k times the corresponding length in the original; (c) a single LSF simultaneously governs all corresponding lengths in a similar figure — it is not applied selectively.
Skills	Learners will be able to: (a) calculate the LSF given any pair of corresponding lengths; (b) use the LSF to find unknown lengths in similar figures; (c) identify which pairs of sides are corresponding sides before applying the LSF; (d) verify similarity by checking that all length ratios are equal.
Attitudes	Learners will: (a) appreciate precision and consistency as mathematical values — a true enlargement applies one scale factor to ALL lengths; (b) show curiosity about how a single number can govern the shape of an entire image; (c) demonstrate responsibility by checking their scale factor calculations before accepting answers.
Key Inquiry Question	How do we identify and use similar figures in day-to-day situations?
Purpose in Storyline	In Lessons 1–4, learners established WHAT similarity looks like (equal angles, proportional sides) and gathered evidence from the passport photo puzzle. In this Explain lesson (Lesson 5), learners now formalise the mechanism — the Linear Scale Factor — that the photographer's software used to resize the image. They move from describing similarity to calculating with it, directly answering the first part of the driving question: 'What is the mathematics that makes this possible?'
Safety Notes	No physical hazards. Ensure learners do not share rulers carelessly to avoid eye injuries. Remind learners that rulers are measuring tools, not toys.

B. LESSON OVERVIEW

Learners transition from the evidence-gathering phases (Lessons 1–4) into formal explanation. Using the passport photo context as the anchor, learners define the Linear Scale Factor (LSF), practise calculating it from corresponding lengths, use it to find missing lengths in similar figures, and begin to articulate why the photographer's software must apply the SAME scale factor to both dimensions. The lesson uses a Claim–Evidence–Reasoning (CER) sensemaking framework, guided problem-solving with worked examples set in Kenyan contexts, and a gallery walk to consolidate understanding. By the end, learners can explain — in their own words — the mathematical rule that keeps the passport photo's shape perfect regardless of size.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO:	Learners are shown two rectangles drawn on the board: Rectangle A (35 mm × 45 mm, the	Teacher writes the two rectangles' dimensions clearly on the board with a colour-coded label:	Individual Prediction followed by Turn-and-Talk. This activates prior numerical reasoning and surfaces	Observable indicator: Teacher scans exercise books during the 3-minute silent period, noting

Watch Me Source: TED-Ed Lessons Search ARES for similar videos  HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 Search ARES for similar readings	standard passport photo) and Rectangle B (105 mm × 135 mm, the noticeboard print). Working individually in silence for 3 minutes, each learner writes answers to three prompt questions in their exercise books: (1) 'By what number was the width multiplied to go from A to B?' (2) 'By what number was the height multiplied?' (3) 'Do you think these two numbers are always equal in a perfect enlargement — why or why not?' Learners then share predictions with a shoulder partner for 2 minutes. Selected predictions are posted on the Driving Question Board (DQB) sticky-note column labelled 'Our Predictions — Lesson 5'.	'ORIGINAL (passport print)' in blue and 'ENLARGED (noticeboard print)' in red. Teacher says verbatim: 'Before I teach you anything today, I want your brain to wrestle with this. Look at these two rectangles — these are the exact dimensions of the two photos from our story about the student on Kimathi Street. In your book, answer the three questions on the board. You have 3 minutes — silent, individual thinking.' [WAIT — 3 minutes silent work.] After 3 minutes: 'Now turn to your shoulder partner. Compare your two numbers. Did you get the same multiplier for width and height? Discuss.' [WAIT — 2 minutes.] Cold-call 3 pairs: 'Pair at the window — what multiplier did you find for the width?' [WAIT 10 seconds.] 'Pair in the middle row — did you agree or disagree with them?' [WAIT 10 seconds.] 'Last pair near the door — what is your prediction for question 3?' [WAIT 12 seconds.] Teacher does NOT confirm or correct answers yet. Posts a class tally on the board: 'How many found the SAME multiplier for both dimensions?' [Show of hands.] 'How many found DIFFERENT multipliers?' Teacher notes the tally and says: 'Hold those predictions. By the end of this lesson you will know exactly why one of those answers is mathematically impossible in a true similar figure.'	misconceptions (e.g., learners who think the width and height could have different multipliers). Recording predictions publicly on the DQB creates accountability and a reference point for the Explain phase.	whether learners correctly compute $105 \div 35 = 3$ and $135 \div 45 = 3$. A learner who gets different values for the two multipliers reveals a calculation error or conceptual gap about corresponding sides — teacher flags these learners for targeted support in Phase 3. Tally of hands reveals the class split, informing how much scaffolding is needed in the Explain phase.
Observe Phase  VIDEO: Watch Me Source: TED-Ed Lessons Search ARES	Learners receive a printed or drawn Activity Card showing four pairs of similar figures, all framed within Kenyan real-life contexts: (i) two Kenyan passport photo rectangles (already known), (ii) two triangular rooftop trusses from a	Distribute Activity Cards. Teacher says: 'You are now detectives. Each pair of figures in front of you was created by a photographer, an architect, or a designer using the same resizing rule. Your job is to find that rule by measuring and	Data Collection and Pattern Spotting. By computing ratios across multiple real-world similar figure pairs, learners inductively discover that a constant ratio exists between corresponding lengths. Grounding each figure in	Observable indicator: Each group's ratio should equal the LSF stated on the teacher's answer key for that figure pair. Teacher checks that groups are (a) correctly identifying corresponding sides (not mixing non-corresponding

for similar videos  HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12  Search ARES for similar readings	Nyeri hardware catalogue, (iii) two hexagonal floor tiles from a Mombasa hotel renovation plan (given as regular hexagons with one side length each), and (iv) two rectangular maize storage bags labelled with their length and width. For each pair, learners measure or read off ONE pair of corresponding lengths and calculate the ratio (larger $\div$ smaller). They record their ratios in a two-column table: 'Pair of corresponding lengths used' 'Ratio (larger $\div$ smaller)'. Learners work in groups of three for 8 minutes, then each group nominates one spokesperson to share one ratio with the class.	dividing. For each pair, pick ONE pair of corresponding sides, divide the bigger measurement by the smaller, and record your ratio in the table. Work in your groups of three — you have 8 minutes.' [WAIT — 8 minutes group work. Teacher circulates, kneeling to group level, asking probing questions: 'Which side in Figure B corresponds to THAT side in Figure A — how do you know?' and 'Are you dividing bigger by smaller or smaller by bigger — does it matter?'] After 8 minutes: 'Groups, stop. Each group, your spokesperson only — stand when I point to you and give me: the figure pair you used and the ratio you calculated.' Cold-call 4 groups in sequence. [WAIT 10 seconds after each group stands before they speak.] As groups report, teacher writes ratios on the board in a class table. Teacher then asks: 'What do you notice about all these ratios?' [WAIT 12 seconds.] Cold-call 2 learners who have NOT yet spoken. Teacher records the emerging pattern without naming it yet: 'Interesting — you are seeing something. We are going to give this observation a proper mathematical name in the next phase.'	a Kenyan context (Nyeri trusses, Mombasa tiles, maize bags) ensures the pattern feels universal and practical, not artificially constructed.	sides) and (b) dividing consistently (always larger $\div$ smaller or always image $\div$ original). Groups mixing non-corresponding sides will get inconsistent ratios — teacher uses this as a teachable moment during circulation: 'Show me which angle in Figure B matches this angle in Figure A — corresponding sides must be opposite equal angles.'
Explain Phase  VIDEO: Watch Me Source: TED-Ed Lessons  Search ARES for similar videos  HTML: Linear Equations: Deep Dish Pizza (PLIX)	PART A — Formal Definition (8 min): Learners copy and annotate a formal definition of the Linear Scale Factor from the board into their notes, with the passport photo worked example as the anchor. They then individually work through two guided practice problems on similar triangles (a school flag triangular pennant enlarged for a Kisumu inter-school sports day, and a triangular road	PART A: Teacher writes on the board in large text: 'LINEAR SCALE FACTOR (LSF) = (Length in image) $\div$ (Corresponding length in original)'. Say verbatim: 'This is the mathematical name for the rule you all just discovered in your groups. The Linear Scale Factor — we will call it LSF — is the SINGLE number that tells you how many times bigger every length in the image is compared to the	Claim–Evidence–Reasoning (CER) embedded in Part C: Learners make a Claim ('This is NOT a true enlargement'), provide Evidence (the two different ratios they calculated), and give Reasoning (the definition of LSF requires a single constant ratio across ALL corresponding lengths). This moves learners from procedural calculation to conceptual explanation, directly	Observable indicators: (1) In Part A, learners who correctly verify $135 \div 45 = 3$ independently demonstrate procedural fluency with LSF. (2) In Part B, the three board solutions reveal common errors: mixing up which figure is 'image' and which is 'original' (inverted LSF), or failing to multiply/divide correctly. Teacher notes the names of learners with inverted LSF errors for a 2-minute

Source: CK-12  Search ARES for similar readings	sign near the Rift Valley escarpment). PART B — Finding Unknown Lengths (12 min): Learners receive a structured worksheet with three similar-figure problems of increasing difficulty, each requiring them to (1) state the LSF, (2) use LSF to find a missing length, and (3) verify by checking a second pair of corresponding sides. Contexts: (i) rectangular school ID cards (Nairobi school) and a billboard-sized reproduction; (ii) two similar triangular kite frames sold at a Nakuru market stall; (iii) two similar pentagonal plots of land near Lake Victoria shown on a surveyor's map. PART C — The Distortion Challenge (10 min): Learners are shown two rectangles where ONLY the width was scaled by factor 3 but the height was left unchanged (simulating the photographer's mistake from the phenomenon). In groups, learners calculate the two 'ratios' and explain in writing — using the definition of LSF — why this is NOT a true enlargement and why the face would look distorted.	original. Write this definition, then copy the passport photo example I am working on the board.' Teacher works through: $LSF = 105 \div 35 = 3$. Then: 'Now verify: does $135 \div 45$ also equal 3?' [WAIT 8 seconds.] Cold-call one learner: 'Amina — confirm for the class.' After confirmation: 'This is the KEY idea: in a TRUE similar figure, you get the SAME LSF from EVERY pair of corresponding lengths. Not just one pair — EVERY pair.' Teacher writes this as a boxed rule. For the two guided practice problems, teacher models the first (pennant), then says: 'Now YOU do the road sign problem alone — 4 minutes, silent.' [WAIT 4 minutes.] Cold-call 2 learners to share working on the board. PART B: 'Now apply LSF to find missing lengths. Work through problems 1–3 on the worksheet. For each one, write: $LSF = \underline{\hspace{1cm}}$, missing length = $\underline{\hspace{1cm}}$, verification check = $\underline{\hspace{1cm}}$. You have 12 minutes, work individually.' [WAIT 12 minutes, circulate.] After 8 minutes, pause: 'If you are on problem 3, excellent. If you are still on problem 1, skip to problem 2 now — we will revisit problem 1.' After 12 minutes, cold-call 3 different learners to write their solutions to problems 1, 2, and 3 on the board simultaneously. Class checks silently. Teacher asks: 'Raise your hand if your answer for problem 2 matches what is on the board. [Pause.] Hands down. Now — someone whose answer did NOT match — can you tell me where your calculation diverged?' [WAIT 12 seconds.] PART C: Teacher draws the distorted rectangle on the board. Say: 'Here is the photographer's mistake. Width	answering the driving question's 'what is the mathematics' component. Scaffolded worksheet in Part B (three levels of difficulty) ensures all learners access the core skill before advancing.	targeted re-teach at the start of Lesson 6. (3) In Part C, the written CER sentence is the key formative artefact — teacher collects group notebooks and reads 6 sentences during the DQB phase. A complete CER sentence must contain all three components; incomplete sentences indicate learners who can compute but cannot yet reason about why.
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		scaled by 3, height NOT scaled. In your groups, calculate both ratios and write one sentence explaining — using the word LSF — why this is not a true enlargement.' [WAIT 6 minutes group work.] Cold-call 2 groups. Teacher concludes Part C: 'A true enlargement has ONE LSF that applies to ALL lengths equally. The moment you use different multipliers for different dimensions, you break similarity — and you get a distorted face on your passport photo.'		
Driving Question Board (DQB) Creation  VIDEO: Watch Me Source: TED-Ed Lessons  Search ARES for similar videos  HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12  Search ARES for similar readings	Each learner individually writes two sticky notes (or index cards): (1) a BLUE note answering — in one sentence — the question 'What mathematical rule did the photographer's software follow?' using the term LSF; (2) a GREEN note adding one NEW question or remaining puzzle they still have about LSF and similarity. Learners post their blue notes on the DQB column 'We Now Understand' and green notes on the column 'We Still Wonder'. The class does a 3-minute silent gallery walk along the DQB, reading posted notes. Teacher reads aloud 3 blue notes and 2 green notes, highlighting one green note that will be the entry point for Lesson 6 (LSF and areas).	Teacher distributes two sticky notes to each learner and says: 'We started this unit with a puzzle: how can two photos look identical but be completely different sizes? Today you found the mathematical answer. On your BLUE note, write one sentence answering the driving question — use the term Linear Scale Factor. On your GREEN note, write something you are still wondering about.' [WAIT 3 minutes silent writing.] 'Walk to the DQB and post your notes in the correct columns. Then take 3 minutes to read what your classmates wrote — you are not talking, just reading.' [WAIT 3 minutes gallery walk.] Teacher then reads aloud, verbatim, 3 blue notes chosen for clarity and precision. 'Listen to this one — it says: [reads]. Is this a complete answer? What makes it strong?' [WAIT 10 seconds.] Cold-call 1 learner to evaluate the sentence. Then: 'Now listen to this green note: [reads a note about area ratios or volume]. This is a brilliant question — and it is exactly what we are going to investigate in Lesson 6. Hold that question.'	Public Knowledge Construction via the DQB. The two-colour sticky note system makes visible the distinction between consolidated understanding (blue) and productive uncertainty (green). The gallery walk activates peer-to-peer learning without teacher mediation, and the teacher's deliberate selection of a green note as the Lesson 6 entry point models scientific inquiry — new questions arise from new understanding, exactly as in the passport photo phenomenon.	Observable indicator: Blue notes are assessed for three criteria: (1) correct use of the term 'Linear Scale Factor', (2) mention that it is a SINGLE ratio, (3) connection to the passport photo or another real context. Teacher uses a quick mental tally — if fewer than 70% of blue notes meet all three criteria, teacher opens Lesson 6 with a 5-minute whole-class consolidation before advancing to area ratios.

Model Building Phase  VIDEO: Watch Me Source: TED-Ed Lessons  Search ARES for similar videos  HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12  Search ARES for similar readings	Learners return to the cumulative model they began building in Lesson 1 — a labelled diagram of the two passport photos with annotations explaining similarity. In this lesson, they ADD a new layer to their model: (a) they write the $LSF = 3$ prominently on an arrow connecting the two rectangles; (b) they annotate THREE pairs of corresponding lengths (width, height, and one diagonal they calculate using Pythagoras) showing that all three give $LSF = 3$; (c) they write a 'Model Statement' in a box below the diagram: 'A true enlargement is governed by ONE Linear Scale Factor k. Every length in the image $= k \times$ the corresponding length in the original.' Learners who finish early extend by calculating the LSF for a third photo size that the photographer could produce (70 mm $\times$ 90 mm) and annotating it on their model.	Teacher says: 'Open your cumulative model from Lesson 1. Today we are adding the most important mathematical layer. I want you to: first, draw a bold arrow from the small photo to the large photo and label it $LSF = 3$; second, annotate THREE pairs of corresponding lengths showing the same ratio; third, write the model statement I am putting on the board now.' Teacher writes the model statement on the board verbatim. [WAIT 8 minutes for individual model updating.] After 8 minutes: 'If you have added your three annotations and your model statement, raise your hand.' [Scan.] 'Those who have finished — extend your model: could the photographer produce a 70 mm $\times$ 90 mm version? What would be the LSF from the original? Add it to your model.' [WAIT 4 minutes.] Close by cold-calling 2 learners to hold up their models and explain ONE annotation to the class. [WAIT 10 seconds for each learner to gather their thoughts before speaking.] Teacher closes: 'Your model is growing. It now shows WHAT similarity looks like AND the mathematical rule that creates it. Next lesson, we ask: if the LSF for lengths is 3, what is the ratio of the AREAS of these two photos? Your green sticky notes from the board suggest many of you are already wondering this. Come curious.'	Cumulative Model Revision. The model is not a one-off task but a living document that grows across the 12-lesson sequence, mirroring how scientific models are revised as evidence accumulates. Annotating three independent pairs of corresponding lengths (including a calculated diagonal) forces learners to verify LSF is not a one-pair coincidence but a universal property — deepening conceptual understanding beyond a single worked example.	Observable indicator: Teacher examines 6 models during the 8-minute work period, looking for: (1) $LSF = 3$ correctly placed on the arrow; (2) three correctly identified corresponding length pairs (not sides from the same figure); (3) model statement copied accurately. A model that shows LSF applied to only ONE pair of sides, or that shows different LSF values for different pairs (due to mis-identification of corresponding sides), signals a persistent misconception about which sides correspond — teacher writes that learner's name in the lesson record for follow-up in Lesson 6.
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) **CONCEPT CLARITY** — Could learners articulate, in their own words, why applying DIFFERENT multipliers to width and height breaks similarity? If not, the CER writing task in Phase 3 Part C needs more structured sentence frames in Lesson 6's opener. (2) **PROCEDURAL FLUENCY** — What percentage of learners correctly found unknown lengths using LSF in the Part B worksheet without needing peer or teacher support? If fewer than 60%, plan a 10-minute peer-tutoring opener for Lesson 6 before advancing to area ratios. (3) **DQB QUALITY** — Were learners' blue sticky notes precise enough to

distinguish LSF from a general idea of 'ratio'? Strong notes name LSF, give a value, and connect to the phenomenon. Weak notes say only 'it was multiplied by the same number' — these learners need more exposure to mathematical language. (4) MODEL DEPTH — Did learners' models include all three pairs of corresponding lengths, or did they stop at one? Shallow models suggest learners understand LSF as a computational trick rather than a universal property of similarity. (5) TIME MANAGEMENT — Phase 3 is dense (30 minutes, three parts). If Part B ran over time, consider whether the kite or pentagonal plot problem can be assigned as homework and reviewed at the start of Lesson 6, freeing in-class time for Part C's conceptual discussion, which is the higher-priority learning outcome.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed that when corresponding lengths from multiple pairs of similar figures are divided, they consistently produce the same constant ratio — regardless of which pair of corresponding sides is chosen. This pattern held across rectangles (passport photos), triangles (rooftop trusses, pennants, road signs), and other polygons (tiles, kite frames, land plots).
What did I learn?	Learners formalised this observed pattern as the Linear Scale Factor (LSF): the single constant ratio of any corresponding length in the image to the corresponding length in the original. They learned to calculate LSF, use it to find unknown lengths in similar figures, and verify similarity by checking consistency of LSF across multiple pairs of corresponding sides.
How does this explain the phenomenon?	Learners explained — using Claim–Evidence–Reasoning — why the photographer's software must apply the SAME LSF to BOTH the width and height of the passport photo. When only one dimension is scaled (the photographer's mistake), the two 'ratios' differ, the figure is no longer similar to the original, and the face appears distorted. The $LSF = 3$ connecting the 35×45 mm print to the 105×135 mm print is the mathematical rule that preserves shape while changing size.

LESSON 6 (80 minutes): Calculating and Applying the Linear Scale Factor (LSF) to Find Unknown Lengths

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To formalise the Linear Scale Factor (LSF) and use it to calculate unknown lengths in similar figures drawn from real-life Kenyan contexts.
Knowledge	Students will know that: (a) the Linear Scale Factor (LSF) is the single ratio by which every corresponding length in a similar figure is multiplied; (b) $LSF = \text{image length} \div \text{object length}$, and this ratio is the same for every pair of corresponding sides; (c) if $LSF > 1$ the figure is an enlargement, if $LSF < 1$ it is a reduction, and if $LSF = 1$ the figures are congruent; (d) any unknown length in a similar figure can be found by multiplying the original length by the LSF or dividing an image length by the LSF.
Skills	Students will be able to: (a) calculate the LSF from any pair of corresponding lengths; (b) use LSF to find unknown side lengths in similar polygons; (c) verify similarity by confirming all corresponding length ratios are equal; (d) apply LSF to real-life scaling problems including photo resizing, badge enlargement, and recipe card scaling.
Attitudes	Students will appreciate that a single mathematical ratio — the LSF — is sufficient to describe any size change that preserves shape, and will value precision in ratio calculation as the foundation of reliable scaling in design, photography, mapmaking, and engineering.
Key Inquiry Question	How do we identify and use similar figures in day-to-day situations?
Purpose in Storyline	In Lessons 1–5 students observed the passport photo puzzle, measured corresponding sides, and noticed that ratios of corresponding lengths were always equal. In this lesson — the formal pivot of the evidence-gathering sequence — students name that constant ratio the Linear Scale Factor, derive it from their own Lesson 2 measurement data, and practise using it to find unknown lengths in three Kenyan contexts (the passport photo, a school badge, and a recipe card). This directly answers the first part of the driving question: the photographer's software followed the rule 'multiply every length by the same LSF', here $LSF = 3$, and that is what keeps the shape perfectly intact.
Safety Notes	No significant safety concerns. Ensure rulers and compasses are handled carefully during any measurement tasks. Students working on shared devices should follow school ICT guidelines.

B. LESSON OVERVIEW

Lesson 6 is the formal sensemaking pivot of the 12-lesson sequence. Students open by revisiting their Lesson 2 measurement tables and, guided by structured questioning, abstract the repeated ratio into the definition of Linear Scale Factor (LSF). The teacher then leads two worked examples — resizing the Kimathi Street passport photo and scaling a school badge — using a think-aloud protocol. Learners independently tackle a recipe-card scaling problem and a similar-triangles problem set in the Rift Valley. The lesson closes with a DQB update ('LSF tells us how much every length is multiplied') and a cumulative model revision in which students annotate their working similarity diagrams with LSF arrows and ratio labels. Formative assessment is embedded at every phase through mini-whiteboards, cold-calling, and a four-question exit ticket.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Students are shown two unlabelled	1. Display the two rectangles (no	Predict-Before-Instruction with	Observable indicator: Are students

 VIDEO: Watch Me Source: TED-Ed Lessons  Search ARES for similar videos  HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12  Search ARES for similar readings	rectangles on the board: one 4 cm × 6 cm, one 10 cm × 15 cm. Without any calculation, they write on mini-whiteboards: (a) 'Are these two rectangles similar? Yes / No / Not sure — give one reason.' (b) 'If I told you the shorter side of the small rectangle is 4 cm and the shorter side of the large one is 10 cm, predict what single number connects every length in the small rectangle to its partner in the large rectangle.' Students hold up boards. The teacher cold-calls 4 students — one Yes, one No, one Not Sure, and one with a numerical prediction — and records all four predictions on the class Driving Question Board column labelled 'Lesson 6 Predictions'. Teacher then poses the anchoring question: 'Last week in Nairobi, the photographer's software used one number to resize every pixel in the passport photo. Today we are going to name that number and prove it works for every length — not just one pair of sides.'	labels) on the projector. Say: 'Look at these two shapes. Do NOT measure yet — just look.' Wait 30 seconds of silent thinking time. 2. Say: 'On your mini-whiteboard write your answer to Part (a) — similar or not — and your one reason. You have 60 seconds.' 3. After boards are raised, cold-call exactly 4 students. Say to each: 'Read your board aloud and explain your reasoning in one sentence.' Do NOT correct yet — record all responses verbatim on the DQB. 4. Add the measurements verbally: 'Here are the dimensions: small is 4 cm by 6 cm, large is 10 cm by 15 cm.' Pause 15 seconds. 5. Say: 'Now write your numerical prediction for Part (b). What single number do you think connects every length?' Cold-call 3 students and probe: 'How did you arrive at that number?' 6. Anchor to phenomenon: 'Keep your predictions visible. By the end of today you will know whether you were right — and you will be able to use that single number to solve problems the photographer solves every day on Kimathi Street.'	Mini-Whiteboards: by committing a prediction to the board before any teaching, every student activates prior ratio knowledge (from Lessons 2–5) and creates a cognitive hook. The public display of conflicting predictions ('2.5', '1.5', 'not sure') generates productive cognitive dissonance that motivates the formal definition in Phase 2.	able to identify equal ratios ($4/10 = 6/15$) as the basis for similarity, or are they relying only on visual appearance? Teacher scans for students who write 'they look the same' without a ratio reason — these students need targeted questioning in Phase 2. Record the spread of numerical predictions; if most students predict 2.5 (correct), Phase 2 can move briskly to formalisation.
Observe Phase  VIDEO: Watch Me Source: TED-Ed Lessons  Search ARES for similar videos  HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12  Search ARES for similar	Part A (8 min) — Revisiting Lesson 2 Data: Students retrieve their Lesson 2 measurement tables (passport photo dimensions: 35 mm × 45 mm and 105 mm × 135 mm). In pairs, they complete a structured ratio table on a printed/digital worksheet: they calculate width ratio ($105 \div 35$), height ratio ($135 \div 45$), and diagonal ratio (using Pythagoras results from Lesson 4 if available). They record all three ratios and write a one-sentence conclusion: 'Every length ratio equals ____.'	Part A: 1. Say: 'Open your Lesson 2 measurement tables. With your partner, fill in the ratio table on Worksheet 6A — calculate each ratio to 2 decimal places and write your conclusion sentence. You have 6 minutes.' Circulate; look for students who invert the ratio ($35 \div 105$ instead of $105 \div 35$) — prompt: 'Which photo is the image and which is the object? Check the direction of your ratio.' 2. After 6 minutes: cold-call one pair to share all three ratios aloud. Ask the class: 'Does everyone agree?'	Data-to-Definition Bridge: students use their own previously collected measurement data (Lesson 2) rather than textbook numbers to arrive at the definition of LSF. This strategy makes the abstraction feel earned rather than imposed. The think-aloud worked example then models the exact reasoning sequence students will apply independently in Phase 3, reducing cognitive load by separating observation of expert thinking from independent practice.	Observable indicator 1: In the ratio table, all three ratios should equal 3.00. Students who get different values for width and height ratios have made a calculation error OR are using non-corresponding sides — teacher intervenes immediately. Observable indicator 2: During think-aloud, do students annotate their own diagrams with '× 3' arrows without being told? This shows transfer of the strategy. Observable indicator 3: Responses to the distortion question — students should

readings	Part B (12 min) — Teacher Think-Aloud: Worked Example 1 (Passport Photo). Teacher projects the following problem: 'The passport photo studio on Kimathi Street prints a standard photo (35 mm × 45 mm) and a noticeboard enlargement. The width of the enlargement is 105 mm. Find: (i) the LSF, (ii) the height of the enlargement, (iii) whether a 70 mm × 90 mm print from the same studio is also similar to the original.' Teacher solves aloud, narrating every step: 'Step 1 — I identify a pair of corresponding lengths I know fully: width 35 mm → 105 mm. Step 2 — $LSF = \text{image length} \div \text{object length} = 105 \div 35 = 3$. Step 3 — Every other length is multiplied by 3: height = $45 \times 3 = 135$ mm. Step 4 — For the 70 × 90 print: width ratio = $70 \div 35 = 2$, height ratio = $90 \div 45 = 2$. Both ratios equal 2, so yes it is similar, with $LSF = 2$.' Teacher annotates diagram with arrows labelled '× 3' on every side. Teacher then asks: 'What would happen to the face in the photo if the photographer applied $LSF = 3$ to the width but $LSF = 2$ to the height?' Wait 15 seconds. Cold-call 2 students.	Thumbs up / thumbs sideways / thumbs down.' Address any disagreement before moving on. Part B: 3. Project Worked Example 1. Say: 'Watch and listen — I am going to think aloud as I solve this. Your job is to annotate your own diagram on Worksheet 6A as I go.' 4. Narrate each step slowly; pause after Step 2 and ask: 'Why do I divide image by object and not the other way around? Discuss with your partner for 20 seconds.' Cold-call 1 student. 5. After completing Step 4, ask the distortion question. Wait 15 seconds. Cold-call 2 students; affirm: 'Exactly — the face would look squashed or stretched because the two dimensions are no longer in the same ratio. That is the mathematics behind the photographer's distortion problem in our phenomenon.'		connect unequal LSFs to shape change, linking back to the phenomenon.
Explain Phase  VIDEO: Watch Me Source: TED-Ed Lessons  Search ARES for similar videos  HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12  Search ARES	Part A — Worked Example 2, School Badge (Teacher-led, 8 min): Teacher projects a diagram of Nakuru High School's circular badge (diameter 4 cm) alongside an enlarged version for the school gate (diameter 22 cm). The badge contains a triangle inscribed in the circle with sides 2 cm, 3 cm, and 3.5 cm. Problem: 'Find the LSF and use it to find all three sides of the triangle in the enlarged badge.' Students solve in their books	Part A: 1. Project the badge diagram. Say: 'I need a volunteer to read the problem aloud.' Call on a student who has not yet spoken today. 2. Say: 'Set up the LSF calculation in your books — do not solve yet.' Wait 20 seconds. Cold-call 2 students: 'What is your LSF setup? Tell me: image length divided by object length equals what divided by what?' Affirm or correct. Complete the solution narrating aloud. 3. Say: 'Underline	Gradual Release with Targeted Error Analysis: the sequence moves from full teacher modelling (Worked Example 1, Phase 2) → joint solving (Worked Example 2, Part A) → independent practice (Part B). Problem 3 is deliberately constructed around a common student error (inverting the ratio) so that students engage in error analysis — a high-leverage sensemaking move that deepens conceptual understanding of the	Observable indicators: (1) Problem 1 — $LSF = 3.5$, height = 52.5 cm. Errors: students who multiply 35×15 instead of setting up LSF first. (2) Problem 2 — students must match longest side to longest side before calculating $LSF = 350 \div 210 = 5/3$; sides = $120 \times 5/3 = 200$ m and $180 \times 5/3 = 300$ m. Common error: using a non-corresponding pair. (3) Problem 3 — correct $LSF = 6$, height = 360 cm. The teacher's circulation checklist data

for similar readings	simultaneously with the teacher, who stops after setting up the LSF calculation and cold-calls 2 students before completing the solution. Teacher highlights: 'Notice I only needed ONE pair of corresponding lengths to find the LSF — then every other unknown falls out by multiplication.' Part B — Independent Practice (17 min): Students work individually on three problems from Worksheet 6B. Problem 1 (Recipe Card): 'Mama Wanjiku runs a catering business in Kericho. She scales up a recipe card from 10 cm × 15 cm to a laminated poster. If the poster width is 35 cm, find the LSF and the poster height.' Problem 2 (Similar Triangles — Rift Valley): 'Two triangular plots of land near Lake Nakuru are similar. The smaller plot has sides 120 m, 180 m, and 210 m. The longest side of the larger plot is 350 m. Find the LSF and the other two sides of the larger plot.' Problem 3 (Challenge — Reverse LSF): 'A school in Kisumu printed a banner that is 2.4 m wide. This is an enlargement of a design that was 40 cm wide. A student claims the LSF is 0.6 because $40 \div 2.4 \approx 0.6$. Identify and correct the student's error, then find the correct LSF and the height of the banner if the original design was 60 cm tall.' After 17 minutes, teacher leads a whole-class debrief of Problems 1 and 3, cold-calling students who showed errors during circulation.	this key idea in your notes: I only need ONE pair of corresponding lengths to unlock all unknowns.' Part B: 4. Distribute/display Worksheet 6B. Say: 'Work individually and silently for 17 minutes. Show every step — LSF calculation, multiplication or division, and final answer with units. I will be circulating.' 5. Circulate with a checklist: look for (a) correct direction of ratio (image ÷ object), (b) correct identification of corresponding sides in Problem 2 (longest with longest), (c) error identification in Problem 3. Stamp or tick correct setups; write a guiding question in the margin for errors rather than correcting directly: e.g., 'Which is the image — the banner or the design?' 6. At 17 minutes: say 'Pens down.' Cold-call one student for Problem 1 answer (LSF = 3.5, height = 52.5 cm). Cold-call one student who made the Problem 3 error to explain the correction aloud: 'The error is dividing object by image — $LSF = 240 \div 40 = 6$, not 0.6. Banner height = $60 \times 6 = 360$ cm.' 7. Ask the class: 'How does Problem 3's error connect to what would happen in the passport photo studio if the photographer mixed up numerator and denominator?' Wait 15 seconds. Cold-call 1 student.	direction of the ratio. Connecting the error back to the phenomenon ('what would happen to the face?') anchors the mathematics in a memorable real-world consequence.	informs who is cold-called in the debrief and who needs support in the next lesson.
Driving Question Board (DQB) Creation  VIDEO: Watch Me	Students receive a DQB sticky note (or a digital equivalent on the shared class board) and individually write their response to the prompt: 'Complete this sentence using today's	1. Distribute DQB sticky notes. Say: 'You have 3 minutes to write your completion sentence individually. Use the mathematics we established today — include the word LSF and at least one	DQB Sentence Completion with Public Posting: requiring students to complete a sentence that must include 'LSF' and a number forces precise mathematical language rather than vague paraphrase. The	Observable indicator: Can students write a sentence that correctly identifies LSF as the constant multiplier for every length and connects it to shape preservation? Sentences that say

Source: TED-Ed Lessons Search ARES for similar videos HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 Search ARES for similar readings	mathematics: The photographer's software keeps the face looking the same because it _____.' Students post their notes under the DQB column 'What we now know — Lesson 6'. The class reads three posted notes aloud. Teacher then reveals the official DQB update sentence: 'LSF tells us how much every length is multiplied — and because every length is multiplied by the same number, all angles stay the same and the shape is perfectly preserved.' Teacher writes this on the permanent DQB. Students copy it into their science journals under 'Growing Understanding'. Teacher also poses one new 'wondering' question to add to the DQB: 'If the LSF for lengths is 3, what is the ratio for the AREA of the two passport photos? We will investigate this in Lesson 7.'	number.' 2. After 3 minutes, invite students to post notes. Scan quickly and select three: one that is correct and complete, one that is correct but vague, one that has a misconception. Read each aloud without identifying the author. 3. For the vague one, ask the class: 'What could we add to make this more precise mathematically?' For the misconception, ask: 'What evidence from today's lesson would help us revise this?' Wait 10 seconds between questions. 4. Reveal and write the official DQB update sentence. Say: 'Read it aloud with me.' Whole-class choral read. 5. Say: 'Now I am adding one new wondering question to our board — because understanding LSF for lengths immediately raises a new puzzle.' Write the area wondering question. Say: 'Predict silently: do you think the area ratio is also 3? Or something different? Hold that thought until Lesson 7.'	public posting and teacher-facilitated comparison of three notes models how scientists refine explanations through peer review. The 'new wondering' question maintains forward momentum in the storyline and links Lesson 6 directly to Lesson 7 (area ratio = LSF^2).	only 'the ratio is the same' without mentioning multiplication of every length indicate surface-level understanding. Sentences that include 'every length $\times$ LSF = corresponding image length, so angles are unchanged' indicate deep understanding. Teacher photographs the posted notes to inform Lesson 7 planning.
Model Building Phase VIDEO: Watch Me Source: TED-Ed Lessons Search ARES for similar videos HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 Search ARES for similar readings	Students return to their cumulative similarity model — the annotated diagram they have been building since Lesson 1. The model currently shows: two similar figures (passport photos), arrows indicating corresponding vertices, and ratio labels from Lesson 2. Today's task (printed on the model-building instruction card): (a) Add '$\times$ LSF' arrows on every pair of corresponding sides of your diagram, labelled with the numerical LSF value (= 3 for the passport photo). (b) Add a 'Definition Box' to your model: 'Linear Scale Factor (LSF) = image length $\div$ object length = [value]. Every length in the image = corresponding object length $\times$	1. Say: 'Take out your cumulative model. Today we make it speak the language of LSF.' Display the four model-building tasks on the projector. 2. Give 9 minutes for individual model building. Circulate; look specifically for: (a) students who label only one side with LSF instead of all sides, (b) students who write $LSF = 3$ but then label a side incorrectly. Prompt: 'If every length is multiplied by 3, how many arrows should I see on this diagram?' 3. At 9 minutes: say 'Swap with your partner. You have 2 minutes to check — use the three-point checklist on your instruction card.' 4. After swapping: 'If your partner's model needed a red annotation,	Cumulative Model Revision with Peer Verification: annotating the same working model across all 12 lessons makes the growth of understanding visible and concrete. Today's addition — '$\times$ LSF' arrows on every side — transforms the model from a record of observations (Lessons 1–5) into a predictive mathematical tool. The peer verification step using a three-point checklist mirrors scientific peer review and surfaces misconceptions (e.g., labelling only one side) before the exit ticket, giving students a chance to self-correct. The exit ticket provides clean, lesson-specific formative data for the teacher.	Exit ticket expected answers: Q1: $LSF = \text{image length} \div \text{object length}$. Q2: $LSF = 20 \div 8 = 2.5$ (or $30 \div 12 = 2.5$). Q3: height = $12 \times 2.5 = 30$ cm. Q4: The student divided object by image instead of image by object; correct $LSF = 20 \div 8 = 2.5$. Teacher sorts slips into three piles: (Green) all four correct; (Amber) Q1–Q2 correct but Q3 or Q4 wrong; (Red) Q1 or Q2 wrong. Green students are ready for Lesson 7 (area ratio). Amber students need a 5-minute re-teach of the directionality of the ratio at the start of Lesson 7. Red students need a differentiated support activity before Lesson 7.

	LSF.' (c) Add a 'Distortion Warning' annotation: draw a crossed-out version of a distorted photo and label it 'LSF_width $\neq$ LSF_height $\rightarrow$ shape changes $\rightarrow$ NOT similar.' (d) Swap your model with a partner. Partner checks: Are all six sides labelled with the correct scaled value? Is the definition box complete? Is the distortion warning present? Partners sign each other's models if correct, or add a red annotation for anything that needs revision. Exit ticket (last 4 minutes): Teacher distributes a four-question slip. Q1: Write the formula for LSF. Q2: A rectangle 8 cm $\times$ 12 cm is similar to one that is 20 cm $\times$ 30 cm. Find the LSF. Q3: A similar rectangle has a width of 28 cm — find its height. Q4: A student calculates $LSF = 8 \div 20 = 0.4$. What error did they make? Students hand in the slip as they leave.	explain your correction verbally to your partner in one sentence before you hand it back.' 5. Distribute exit ticket slips. Say: 'Last 4 minutes — individual and silent. This is your evidence to me that you can use LSF independently. Answers only need to show your working, not full sentences.' 6. Collect slips at the door. Say: 'Before our next lesson, think about this: if $LSF = 3$ for every length, is the area also 3 times bigger? Or could it be different?' (Plant the Lesson 7 seed.)		
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes in your planner: (1) **RATIO DIRECTION ERROR** — How many exit tickets showed the Q4 inversion error (object $\div$ image)? If more than 25% of students made this error, re-open the question at the start of Lesson 7 using the passport photo context: 'If I divide $35 \div 105$, my LSF is 0.33 — that means I am shrinking the big photo into the small one. Is that the direction I want?' (2) **CONTEXT TRANSFER** — Did students handle the Rift Valley triangular plots (Problem 2) as confidently as the passport photo? If many students struggled to identify corresponding sides in the triangles, plan a short 'matching corresponding vertices' warm-up for Lesson 7 using a Kenyan map-scaling context. (3) **DQB QUALITY** — Were students able to write precise completion sentences using 'LSF' and a number, or were responses vague? Vague responses suggest students have procedural fluency (they can calculate LSF) but have not yet connected the mathematics to the shape-preservation idea. Use the Lesson 7 area investigation to revisit this. (4) **MODEL BUILDING** — Did all students annotate every side (not just one) with the LSF arrow? One-side-only annotation is the most common model error and signals that students see LSF as a property of one pair of sides rather than a property of the whole transformation. (5) **PACING** — The 17-minute independent practice block is ambitious. If the class did not reach Problem 3, use it as the Lesson 7 warm-up so the error-analysis opportunity is not lost.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students revisited their own Lesson 2 measurement data (passport photo: 35 mm $\times$ 45 mm enlarged to 105 mm $\times$ 135 mm) and calculated width ratio, height ratio, and diagonal ratio, finding all three equal to 3. They observed a teacher think-aloud worked example of the Kimathi Street passport photo problem and a school badge problem from Nakuru, watching how a single pair of
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	corresponding lengths unlocks all unknowns.
What did I learn?	Students learned that the constant ratio of any image length to its corresponding object length is called the Linear Scale Factor (LSF), calculated as $LSF = \text{image length} \div \text{object length}$. They learned that $LSF > 1$ indicates enlargement, $LSF < 1$ indicates reduction, and $LSF = 1$ indicates congruence. For the passport photo, $LSF = 105 \div 35 = 3$, meaning every length is multiplied by 3. They also learned that applying different LSFs to different dimensions (e.g., width $\times 3$ but height $\times 2$) destroys similarity and distorts the image.
How does this explain the phenomenon?	Students explained, through DQB sentence completion and cumulative model annotation, that the photographer's software preserves the face perfectly because it multiplies every length — width, height, and all internal distances — by the same LSF (= 3). Because every length scales by the same factor, all ratios between lengths remain unchanged, all angles remain unchanged, and the shape is perfectly preserved. This is the mathematical rule that makes resizing without distortion possible.

LESSON 7 (80 minutes): Area Scale Factor — How Does Enlargement Change the Area of the Passport Photo?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To guide learners to investigate and establish the relationship between the Linear Scale Factor (LSF) and the Area Scale Factor (ASF) using the passport photo enlargement context.
Knowledge	By the end of the lesson, the learner should be able to: (a) calculate the area of similar figures using measured dimensions; (b) derive the Area Scale Factor (ASF) from the Linear Scale Factor (LSF); (c) state and apply the rule $ASF = LSF^2$ to solve problems involving similar figures.
Skills	By the end of the lesson, the learner should be able to: (a) measure and compute areas of similar rectangles and polygons accurately; (b) construct ratio tables comparing linear and area ratios; (c) apply $ASF = LSF^2$ to find unknown areas of similar figures in real-world contexts.
Attitudes	The learner appreciates that mathematical relationships such as $ASF = LSF^2$ are not arbitrary rules but emerge from careful, repeatable investigation — and that noticing patterns is a core scientific and mathematical habit of mind.
Key Inquiry Question	How does the linear scale factor relate to the area and volume of similar figures and solids?
Purpose in Storyline	In Lessons 1–6, learners established what similarity means, identified similar figures, and confirmed that the Linear Scale Factor (LSF) between the standard passport print (35 mm × 45 mm) and the enlarged print (105 mm × 135 mm) is 3. In Lesson 7, learners now OBSERVE and INVESTIGATE what happens to AREA when a figure is enlarged by $LSF = 3$. They measure, calculate, and compare areas of both prints, then test whether the area also scales by 3 — discovering instead that it scales by 9 ($= 3^2$). This critical evidence advances the driving question by revealing a new, non-obvious layer of the mathematics behind the photographer's software.
Safety Notes	No hazardous materials in use. Ensure learners handle rulers carefully. If scissors are used to cut grid paper, supervise and remind learners to pass scissors handle-first.

B. LESSON OVERVIEW

Learners begin by predicting whether the area of the enlarged passport photo is 3 times, 6 times, or 9 times the area of the standard photo, justifying their prediction in writing. They then OBSERVE by measuring both photo rectangles, computing their areas, and building a ratio table. Through guided calculation, learners discover that while lengths scale by $LSF = 3$, areas scale by $ASF = 9 = 3^2$. They extend this finding to a second pair of similar figures (a Kenyan flag motif and its enlargement) to confirm the pattern. In the Explain phase, learners derive $ASF = LSF^2$ algebraically from first principles. The lesson closes with DQB updates and model revision — learners annotate their cumulative similarity model to include the area relationship.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO:	Each learner receives a printed PREDICT card showing a diagram of the two passport photos side-	Distribute PREDICT cards. Say: 'Before we touch any ruler or calculator today, I want you to use	Commit-and-justify prediction: By requiring a written, reasoned commitment before instruction, this	Observable indicator: Each learner has a written prediction with a justification of at least 3 sentences

Watch Me Source: TED-Ed Lessons Search ARES for similar videos  HTML: Proportions to Find Scale Ratios (Flexbook) Source: CK-12 Search ARES for similar readings	by-side with their dimensions labelled: Standard = 35 mm × 45 mm; Enlarged = 105 mm × 135 mm. The card poses three options: 'When the photographer's software enlarged the photo using LSF = 3, the area of the new photo is: (A) 3 times bigger, (B) 6 times bigger, (C) 9 times bigger — or could it be something else entirely?' Learners write a private prediction with a written justification (3–5 sentences minimum) in their exercise books before any discussion. They then share with a shoulder partner and compare reasoning. Pairs are cold-called to share their prediction and rationale with the class.	only what you already know about the passport photo and the scale factor of 3 to make a prediction. Write it down — commit to it. There is no wrong answer right now; what matters is your reasoning.' Allow 5 minutes of silent individual writing. Then say: 'Turn to your shoulder partner. You have 90 seconds — share your prediction and listen carefully to theirs. Do you agree?' After pair discussion, cold-call 4 pairs (aim for at least one prediction from each of the three options). For each pair, ask: 'Walk us through your reasoning — what made you choose that option?' Apply 10–15 seconds WAIT TIME after each cold-call before accepting the answer. Do NOT confirm or deny any prediction. Record all three options on the board under the heading 'OUR PREDICTIONS — we will test these today.' Explicitly say: 'Hold your prediction in your head. We are about to gather evidence that will either support it or force us to revise it.'	strategy activates prior knowledge of LSF and forces learners to articulate their mental models about how scaling affects area. The act of writing the justification surfaces common misconceptions (most learners predict ×3, not ×9) that the Observe phase will then productively challenge.	in their exercise book. Teacher circulates and checks (does not mark) during the 5-minute silent writing. Red flag: Any learner who writes 'I don't know' or leaves the justification blank — the teacher kneels beside that learner and prompts: 'You told me in Lesson 4 that LSF = 3. If every side gets 3 times longer, what do you think happens to the space the photo takes up on the page?'
Observe Phase  VIDEO: Watch Me Source: TED-Ed Lessons Search ARES for similar videos  HTML: Proportions to Find Scale Ratios (Flexbook) Source: CK-12 Search ARES for similar readings	Learners work in groups of 4. Each group receives: (i) a MEASUREMENT WORKSHEET with outlines of Rectangle A (35 mm × 45 mm) and Rectangle B (105 mm × 135 mm) — labelled 'Standard Passport Print' and 'Enlarged Print'; (ii) a RATIO TABLE template with columns: Figure Length (mm) Width (mm) Area (mm²) Ratio of Areas (Enlarged ÷ Standard); (iii) a 1 cm² grid paper sheet. Step 1 (5 min): Each group measures the rectangles on the worksheet with a ruler, records Length and Width in the ratio table, then calculates Area = Length × Width for each	Distribute materials. Say: 'Your job in this phase is NOT to guess — your job is to measure, calculate, and let the numbers tell you the story.' Circulate continuously. At the 5-minute mark, cold-call one group to announce their computed areas for the passport photos: 'Group 3 — what area did you get for the standard print? What about the enlarged print? What is the ratio?' Apply 10 seconds WAIT TIME. If a group gets Area(Standard) = 1,575 mm² and Area(Enlarged) = 14,175 mm² — confirm: 'Good. Now tell me: is that ratio 3? Or something else?' At the 12-minute mark, cold-call a	Data-driven ratio table investigation: By having learners compute the ratio of areas independently across two different pairs of similar figures (passport photos and Kenyan flags), this strategy builds evidence-based induction. The deliberate use of two different LSF values (3 and 2) prevents learners from treating the result as a coincidence and promotes generalisation. The grid-paper visualisation provides a spatial anchor for the numerical result.	Observable indicators: (1) Each group's ratio table has correct numerical values: Passport — Area(Standard) = 1,575 mm², Area(Enlarged) = 14,175 mm², ASF = 9; Flags — Area(Small) = 2,160 mm², Area(Large) = 8,640 mm², ASF = 4. (2) Each group's sentence frame correctly fills in: 'ASF = LSF²'. Red flag: A group that reports ASF = 3 for the passport photos has likely divided one length by the other rather than computing areas. Teacher intervention: 'Show me your area calculation. What is 35 × 45? What is 105 × 135? Now divide those two answers — what do you get?'

	figure. Step 2 (5 min): Groups calculate the Ratio of Areas = Area of Enlarged $\div$ Area of Standard. They record the result. Step 3 (5 min): Groups are then given a second pair of similar figures — a simple rectangular Kenyan flag outline, Small flag: 60 mm $\times$ 36 mm, Large flag: 120 mm $\times$ 72 mm — and repeat the same measurement and ratio calculation. Step 4 (5 min): Groups complete the sentence frames on the worksheet: 'For the passport photos, LSF = ___, ASF = _____. For the flags, LSF = ___, ASF = _____. We notice that ASF = LSF _____. ' Step 5 (5 min): Groups sketch both passport photo rectangles on the 1 cm² grid paper to visualise how many small squares fit inside each rectangle, confirming the area ratio visually.	different group for the flag results. Ask: 'For the flags, LSF = 2. What ASF did you find? Does that match what you found for the passport photos — that ASF = LSF²?' Use the prompt: 'If you finished early, draw a 2$\times$2 square and a 4$\times$4 square on your grid paper. Count the squares inside each. What is the ratio of areas? What was the LSF?' This extension prevents idle time and deepens the pattern recognition. Do NOT yet state the rule ASF = LSF². Say: 'Write down exactly what you found. In Phase 3, your group will explain WHY this is always true.'		
Explain Phase  VIDEO: Watch Me Source: TED-Ed Lessons  Search ARES for similar videos  HTML: Proportions to Find Scale Ratios (Flexbook) Source: CK-12  Search ARES for similar readings	Whole-class sensemaking. A learner volunteer from one group writes their ratio table results on the board. The teacher facilitates a structured algebraic derivation with learner input. Learners follow along and construct the same derivation in their exercise books. The derivation proceeds: 'Let a rectangle have sides a and b. Area = ab. When enlarged by LSF = k, new sides are ka and kb. New area = (ka)(kb) = k²ab = k² $\times$ (original area). Therefore ASF = k² = LSF².' Learners then apply the rule to two problems: Problem 1 — 'A maize farm plot is similar in shape to a smaller test plot. The test plot has area 120 m². The LSF from test plot to farm is 5. What is the area of the farm?' Problem 2 — 'Two similar triangular road signs near Nakuru highway have LSF = 4. The smaller sign has	Invite a volunteer to write ratio table results on the board. Say: 'Let's now build the mathematical argument for WHY ASF always equals LSF squared — not just for these photos, but for ANY similar figure, anywhere in the world.' Write the general rectangle derivation step by step on the board, pausing after each line. After writing 'New area = (ka)(kb)', ask the class: 'Before I expand this — what do you think happens when I multiply (ka) by (kb)? Write it down first.' Apply 12 seconds WAIT TIME. Cold-call 3 learners. After the derivation is complete, say: 'This is the algebraic proof. It tells us ASF = LSF² is not a coincidence — it is a mathematical certainty.' For Problem 1, say: 'A farmer in Kericho has a test plot and a main farm that are similar in shape. Use our new rule to find the	Algebraic generalisation from specific cases (Concrete–Representational–Abstract progression): Learners move from measured numbers (concrete) $\rightarrow$ ratio table (representational) $\rightarrow$ algebraic proof (abstract). This CRA sequence ensures that the abstract formula ASF = LSF² is not a memorised rule but a derived truth grounded in the data learners collected themselves. The two applied problems (maize farm, road sign) re-anchor the abstract formula in Kenyan real-world contexts.	Observable indicators: (1) Each learner's exercise book contains the completed algebraic derivation with correct notation. (2) Problem 1 answer: ASF = 5² = 25; Farm area = 25 $\times$ 120 = 3,000 m². (3) Problem 2 answer: ASF = 4² = 16; Larger sign area = 16 $\times$ 900 = 14,400 cm². Red flag: Learners who compute 5 $\times$ 120 = 600 m² for Problem 1 are still using LSF instead of ASF — teacher circulates and asks: 'Which scale factor applies to areas? Is it the same as the one for lengths?'

	area 900 cm ² . What is the area of the larger sign?' Learners work individually for 5 minutes, then compare answers with their partner.	farm's area.' Cold-call a learner for the answer after 5 minutes of individual working time: 'Tell me your answer AND show me the rule you used.' After Problem 2, ask: 'If the sign painter needs to buy paint proportional to area, how many times more paint does he need for the larger sign? Does this surprise you?' Apply 10 seconds WAIT TIME before accepting responses.		
Driving Question Board (DQB) Creation  VIDEO: Watch Me Source: TED-Ed Lessons  Search ARES for similar videos  HTML: Proportions to Find Scale Ratios (Flexbook) Source: CK-12  Search ARES for similar readings	Learners return to the class Driving Question Board. Each group reviews the sticky notes posted in earlier lessons. Guided by the lesson's new evidence, groups write 1–2 new sticky notes responding to: (a) WHAT WE NOW KNOW — 'Today's evidence tells us that when $LSF = 3$, $ASF = 9$, so the enlarged passport photo takes up 9 times as much paper area as the standard photo.' (b) NEW QUESTIONS — e.g., 'If area scales by LSF^2 , what happens to VOLUME when a 3D object is enlarged? Does it scale by LSF^3 ?' Groups post their notes and the teacher orchestrates a 3-minute whole-class gallery read of all new contributions.	Say: 'Our Driving Question asks: What mathematics did the photographer's software follow to resize the image perfectly? Today we added a new layer of that answer. Go to the DQB and post what you now know and what new question today's evidence opened up for you.' Circulate during the 5-minute group writing. Then orchestrate the gallery read: 'I will point to each new note. Group that wrote it — read it aloud in one sentence.' After gallery read, ask the class: 'Several groups asked about VOLUME. That is exactly where this storyline goes next. Keep that question — it is on our board and we will answer it.' Point to the DQB column labelled 'Volume Scale Factor' if pre-labelled, or write 'VSF — next investigation' on the board.	Driving Question Board as cumulative knowledge tracker: The DQB makes the progression of evidence visible to learners. By physically posting 'WHAT WE NOW KNOW' notes alongside 'NEW QUESTIONS', learners experience science and mathematics as a growing, iterative process — not a sequence of isolated facts. The new questions about volume create productive intellectual tension that motivates Lesson 8.	Observable indicator: Each group posts at least one 'WHAT WE NOW KNOW' note that correctly states the $ASF = LSF^2$ relationship in their own words with a specific numerical example. Red flag: A group whose note says only 'area gets bigger' without a numerical claim — teacher prompts: 'Can you add a number to that? By how much does the area get bigger when $LSF = 3$?'
Model Building Phase  VIDEO: Watch Me Source: TED-Ed Lessons  Search ARES for similar videos  HTML:	Each learner opens their cumulative SIMILARITY MODEL — a diagram they have been building across all lessons, showing two similar rectangles (the passport photos) with annotations. In Lesson 7, learners add a new annotation layer: they label both rectangles with their computed areas (1,575 mm ² and 14,175 mm ²) and draw a double-	Say: 'Take out your Similarity Model. Today you discovered something the photographer's software knows but never tells you: the enlarged photo uses 9 times more paper than the small one, not 3 times. Add that to your model now — label the areas, label the ASF, and write your model statement.' Circulate and check that: (a) area values are	Cumulative annotated model revision: Requiring learners to physically annotate their own diagrams with numerical evidence (areas, ASF) and a written model statement embeds the new knowledge into a growing conceptual structure. The model is not a worksheet — it is a learner-owned artefact that develops across 12 lessons, making the	Observable indicator: Each learner's model contains (a) correctly labelled areas for both passport photo rectangles, (b) the annotation ' $ASF = 9 = 3^2 = LSF^2$ ', and (c) a model statement of at least 2 sentences that references the passport photo phenomenon. Exit check: Teacher collects 6 models (one per group, randomly selected) to review before Lesson

Proportions to Find Scale Ratios (Flexbook) Source: CK-12 Search ARES for similar readings	headed arrow labelled 'ASF = $9 = 3^2 = \text{LSF}^2$ '. Below the diagram, learners write a model statement: 'When two figures are similar with Linear Scale Factor k , their areas are related by Area Scale Factor k^2 . This means the enlarged passport photo uses $9\times$ more paper, even though it looks identical in shape to the standard one.' Learners who finish early extend the model by sketching a 3D box and annotating it with a question mark: 'What would VSF = ?'	correctly labelled, (b) the ASF annotation reads 'ASF = $9 = 3^2 = \text{LSF}^2$ ', (c) the model statement connects back to the passport photo phenomenon. After 7 minutes, cold-call 2 learners to read their model statements aloud. Ask the class: 'Does anyone's model statement say something different — or something MORE — than what was just read? Tell us.' Apply 12 seconds WAIT TIME. Close by saying: 'Your model is growing. Every lesson, we add one more layer. Next lesson, we will ask: if we built a 3D model of the photo studio — a miniature box — and scaled it up by $\text{LSF} = 3$, what happens to its volume? Your model will need a new annotation.'	progression of understanding tangible and reviewable at any point.	8 and identify any persistent misconceptions about the difference between LSF and ASF.
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did the majority of learners predict $\times 3$ in Phase 1? If so, was the productive surprise of finding $\times 9$ in Phase 2 clearly articulated — did learners verbalise WHY their prediction was wrong and WHY the correct answer is $\times 9$? (2) Were learners able to complete the algebraic derivation $(ka)(kb) = k^2ab$ independently, or did most require heavy scaffolding? If scaffolding was heavy, consider providing a partially completed derivation frame in Lesson 8's volume equivalent. (3) Review the 6 collected cumulative models: do learners correctly distinguish between LSF (applies to lengths) and ASF (applies to areas)? Any learner who writes $\text{ASF} = \text{LSF} \times 2$ rather than $\text{ASF} = \text{LSF}^2$ needs a targeted 5-minute intervention at the start of Lesson 8. (4) Did the Kenyan context examples (Kericho farm plot, Nakuru highway sign, Kenyan flag) feel authentic to learners and prompt engagement — or did they feel forced? Adjust the Problem 3 context in Lesson 8 accordingly. (5) Were the DQB questions about volume specific and mathematically directed, or vague? If vague, model a stronger question on the board at the start of Lesson 8: 'If $\text{LSF} = 3$ and $\text{ASF} = 3^2 = 9$, then $\text{VSF} = 3^3 = ?$ '

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners measured the dimensions of the standard passport photo (35 mm $\times$ 45 mm) and the enlarged passport photo (105 mm $\times$ 135 mm), calculated the area of each, and computed the ratio of areas. They repeated this for a pair of similar Kenyan flag rectangles ($\text{LSF} = 2$). In both cases they recorded the ratio of areas in a structured ratio table.
What did I learn?	Learners derived and confirmed that the Area Scale Factor (ASF) equals the square of the Linear Scale Factor (LSF): $\text{ASF} = \text{LSF}^2$. For the passport photos ($\text{LSF} = 3$), $\text{ASF} = 9$ — the enlarged photo uses 9 times more paper area, not 3 times. This relationship was proved algebraically: if sides scale by k , area scales by k^2 because $\text{Area} = (ka)(kb) = k^2(ab)$.
How does this explain the phenomenon?	Learners applied $\text{ASF} = \text{LSF}^2$ to two real-world problems: a maize farm plot in Kericho ($\text{LSF} = 5$, area = 3,000 m^2) and a road sign near Nakuru ($\text{LSF} = 4$, area = 14,400 cm^2). They updated their cumulative Similarity Models with area labels and the annotation $\text{ASF} = \text{LSF}^2$, and posted new DQB notes asking what happens to volume when a 3D object is enlarged — setting up the Volume

	Scale Factor investigation in Lesson 8.
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LESSON 8 (80 minutes): Area Scale Factor: Why Does the Larger Passport Photo Use Nine Times as Much Paper?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To enable learners to discover, derive, and apply the relationship between the Linear Scale Factor (LSF) and the Area Scale Factor (ASF) of similar figures.
Knowledge	Learners will know that for any two similar figures with linear scale factor k , the ratio of their areas is k^2 ($ASF = LSF^2$). They will know that this relationship holds for all similar figures regardless of shape.
Skills	Learners will be able to: (a) draw pairs of similar rectangles on graph paper using given LSFs; (b) calculate areas by counting squares; (c) tabulate LSF vs. area ratio data and identify the pattern $ASF = LSF^2$; (d) apply ASF to solve problems involving similar figures, including the passport photo context.
Attitudes	Learners appreciate that mathematics reveals precise, predictable patterns hidden in everyday observations — such as why a photographer uses more paper for an enlarged print — and value systematic data collection as a path to mathematical discovery.
Key Inquiry Question	How does the linear scale factor relate to the area of similar figures?
Purpose in Storyline	In Lessons 1–7 students established what similarity means, how to identify similar figures, and how to calculate and apply the Linear Scale Factor (LSF). This lesson advances the storyline by asking: if LSF tells us how lengths scale, what happens to AREA? Students gather their own evidence by drawing Kenyan farm-plot rectangles on graph paper and counting squares, then discover $ASF = LSF^2$ inductively before the teacher formalises it. The lesson closes by returning to the passport photo: the larger print is $3\times$ wider and $3\times$ taller, so it uses $3^2 = 9$ times as much paper — a satisfying, concrete answer to a real cost-and-material question.
Safety Notes	No safety concerns for this lesson. Ensure rulers and pencils are used carefully on graph paper. Remind learners to keep graph paper flat on desks to avoid inaccurate drawings.

B. LESSON OVERVIEW

Learners begin by predicting whether the area of a scaled figure grows by the same factor as the lengths. They then conduct a structured graph-paper investigation: drawing similar rectangles (modelled as Kenyan farm plots) with $LSF = 2, 3$, and 4 , counting grid squares to find areas, and recording results in a class data table. Through pattern analysis they discover that $ASF = LSF^2$. The teacher formalises this with an algebraic proof using a general rectangle. Learners apply the relationship to the anchoring phenomenon (passport photo) and to real Kenyan farm and map problems. The lesson closes with DQB updates and a cumulative model revision.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Picture graphs Source: Khan Academy (English)	Each learner receives a half-sheet with two rectangles: a small 'farm plot A' ($2\text{ cm} \times 3\text{ cm}$, drawn on graph paper) and a prompt stating that farm plot B is similar to A with	Distribute the half-sheets face-down. Say: 'Before we draw a single line, I want your honest mathematical instinct — no calculators, no rulers yet. Flip your	Prediction Commitment with Class Tally. By committing predictions in writing before evidence, learners activate prior knowledge about scaling and surface a common	Observable indicator: At least 70% of learners write a numerical prediction for all three questions within 3 minutes. Teacher scans books during circulation — flag

- US curriculum) Search ARES for similar videos HTML: Proportions to Find Scale Ratios (Flexbook) Source: CK-12 Search ARES for similar readings	LSF = 3. Before drawing or calculating anything, learners write individual predictions in their exercise books answering THREE questions: (1) What will the dimensions of Plot B be? (2) Will the area of Plot B be 3 times the area of Plot A? More? Less? (3) The passport photo studio enlarges a 35 mm × 45 mm photo with LSF = 3 — how much more paper do you think the larger print needs? Learners are told their predictions will be tested with evidence in the next phase and they must NOT change their written predictions once the activity begins.	sheet over and write your three predictions now. You have 3 minutes.' [Set a visible 3-minute timer.] Circulate silently — DO NOT correct or affirm any prediction. After 3 minutes say: 'Hands up — who predicted the area grows by exactly 3 times, the same as the LSF?' [Count hands aloud: 'I see 8 hands.'] 'Who predicted it grows by MORE than 3 times?' [Count.] 'Who predicted LESS than 3 times?' [Count.] Record the tally on the board under the heading: CLASS PREDICTION TALLY. Say: 'We have three different camps. Today's evidence will tell us who is right — and why.' Cold-call 2 learners from different camps: 'Amina, tell me the reasoning behind your prediction' [WAIT 12 seconds]. 'Kipkemoi, do you agree or disagree with Amina, and why?' [WAIT 12 seconds]. Do not resolve the disagreement — preserve productive tension.	misconception (ASF = LSF rather than LSF²). The public tally creates a genuine question to resolve, raising cognitive engagement and giving the teacher a live misconception map.	any learner who leaves questions blank for targeted support. The tally reveals the distribution of the key misconception (ASF = LSF) which guides emphasis during Phase 3.
Observe Phase VIDEO: Picture graphs Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Proportions to Find Scale Ratios (Flexbook) Source: CK-12 Search ARES for similar readings	Groups of 3 receive: 1 cm graph paper (A3 sheet), rulers, pencils, and an Investigation Card. The card instructs them to draw a 'Base Farm Plot' (2 cm × 3 cm) labelled Plot A in the top-left corner. They then draw three enlarged similar plots — Plot B (LSF = 2), Plot C (LSF = 3), Plot D (LSF = 4) — anywhere on the same sheet, label each, and complete a Results Table: Plot Dimensions Area (sq cm, counted) LSF (compared to A) Area ratio (Plot ÷ Plot A) . After completing the table, groups calculate the Area Ratio for each plot and look for a pattern relating LSF to Area Ratio. Each group also answers: 'How did you count	Before groups start, say: 'The rectangles represent farm plots near Nakuru that a county land office wants to model at different scales. Your job is to act like land surveyors — measure and count carefully, because your data will feed a class discovery.' [Distribute materials.] 'You have 20 minutes. The only rule: area must be found by COUNTING squares, not just multiplying — I want to see your tally marks inside each rectangle.' [Circulate every 3–4 minutes.] At the 8-minute mark, visit any group stuck on Plot D and prompt: 'Plot A is 2 by 3. If LSF is 4, what does EVERY length get multiplied by? So what are Plot D's dimensions?' [WAIT 10 seconds — do not	Data-Driven Inductive Discovery with Class Consolidation Table. Requiring square-counting (not formula application) ensures learners engage with what area physically means on the grid. The class table aggregates evidence from all groups, making the pattern robust — no single group's potential measurement error dominates. This mirrors how scientists share data before drawing conclusions.	Observable indicators: (1) Each rectangle has visible tally marks or shading inside it (evidence of counting, not guessing). (2) The Results Table is fully completed with correct dimensions — Plot B: 4×6, Plot C: 6×9, Plot D: 8×12. (3) The area ratios are 4, 9, 16 respectively. Teacher checks at least 4 group tables during circulation. If a group records area ratio = 2, 3, 4 (the misconception), ask: 'Read out your actual counted area for Plot B, then for Plot A. Now divide. What do you get?' — redirect to the data rather than correcting directly.

	the area — what method did you use, and why is counting squares more convincing than just using a formula here?' A class data consolidation table is drawn on the board; one reporter per group fills in their row.	answer for them.] At the 15-minute mark, say: 'You should have your table complete. Now look at the Area Ratio column — what do you notice about those numbers? Write one sentence as a group.' [WAIT while groups write.] After 20 minutes, say: 'Reporters — come fill in your group's row.' Cold-call the reporter from Group 3: 'Before you write, tell me how your group counted the squares for Plot C.' [WAIT 12 seconds.] After the class table is filled, say: 'Look at the LSF column and the Area Ratio column together. Turn to the person next to you and describe the pattern in one sentence. Go.' [WAIT 15 seconds for pair talk.] Cold-call 3 pairs to share.		
Explain Phase  VIDEO: Picture graphs Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Proportions to Find Scale Ratios (Flexbook) Source: CK-12  Search ARES for similar readings	The teacher leads a whole-class discussion anchored on the completed class table, guiding learners to articulate the pattern verbally before writing it algebraically. Learners copy a structured two-column proof into their notes showing algebraically why $ASF = LSF^2$ for any rectangle with dimensions l and w, scaled by factor k. They then complete three tiered practice problems: (TIER 1) A shamba in Kericho is $5\text{ m} \times 8\text{ m}$. A similar shamba has $LSF = 4$. Find the area of the larger shamba. (TIER 2) Two similar triangular fields near Lake Victoria have areas 50 m^2 and 200 m^2. Find the LSF. (TIER 3 — Return to Phenomenon) The passport photo studio on Kimathi Street produces a standard print ($35\text{ mm} \times 45\text{ mm}$) and an enlarged print with $LSF = 3$. (a) Find the area of each print. (b) Photo paper costs Ksh 0.08 per mm^2. How much more does the enlarged print cost to produce?	Point to the class table. Say: 'Let's read the Area Ratio column aloud together: 4, 9, 16. Now read the LSF column: 2, 3, 4. Korir, what is 2 squared?' [WAIT 10 seconds.] '3 squared?' [WAIT 10 seconds.] '4 squared?' [WAIT 10 seconds.] 'So what is the relationship between LSF and Area Ratio?' [WAIT 12 seconds — cold-call Wanjiku.] Write on the board: $ASF = LSF^2$. Say: 'This is the Area Scale Factor rule. But why does squaring appear? Let me show you algebraically — and I want you to tell me each step.' Write the proof step by step, pausing at each line: 'Original rectangle: length = l, width = w. Area of original = ?' [WAIT — cold-call.] 'Scaled rectangle: length = kl, width = kw. Area of scaled = ?' [WAIT — cold-call.] 'Area ratio = $k^2lw \div lw = ?$' [WAIT 10 seconds.] 'So $ASF = k^2$. The k^2 comes from the fact that BOTH dimensions are multiplied by k — area is two-dimensional, so k	Socratic Step-by-Step Algebraic Proof followed by Tiered Application. The Socratic proof technique requires learners to supply each algebraic step, keeping them as active participants rather than passive note-takers. Tiered problems ensure all learners access the concept (Tier 1) while high-readiness learners extend to inverse problems (Tier 2) and the anchoring phenomenon (Tier 3), differentiating without tracking.	Observable indicators: (1) During the proof, at least 6 different learners successfully supply a step when cold-called (teacher tracks names). (2) Tier 1: learners correctly state the larger shamba area = $5 \times 8 \times 16 = 640\text{ m}^2$. (3) Tier 2: learners correctly find $LSF = \sqrt{(200/50)} = 2$. (4) Tier 3: learners correctly compute standard area = 1575 mm^2, enlarged area = 14175 mm^2, cost difference = Ksh 1,008. If learners in Tier 3 forget to apply ASF and instead multiply by 3 (not 9), ask: 'You said the area is 3 times bigger. But 3 is the LSF — what is the AREA scale factor?' — redirect to the rule.

	Learners compare their Phase 1 predictions with the discovered rule and write a one-sentence 'I was right/wrong because...' reflection.	appears TWICE.' Distribute the tiered practice. Say: 'Everyone starts at Tier 1. If you finish Tier 1 in 5 minutes, move to Tier 2, then Tier 3. Tier 3 takes us back to our passport photo — I expect everyone to reach it.' [Allow 12 minutes.] Cold-call Tier 3 answers: 'Achieng, what is the area of the standard passport print?' [WAIT 12 seconds.] 'Mwangi, how did you find the cost difference?' [WAIT 12 seconds.] Close by returning to the Class Prediction Tally: 'Look at our tally from the start of class. Who had the right instinct? Why is area NOT the same as length when we scale?'		
Driving Question Board (DQB) Creation  VIDEO: Picture graphs Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Proportions to Find Scale Ratios (Flexbook) Source: CK-12  Search ARES for similar readings	Each learner receives a sticky note. They write: (1) ONE thing they can now answer on the Driving Question Board (e.g., 'I can now explain why the enlarged passport photo costs 9 times more to print, not 3 times more'); (2) ONE new question that today's lesson raised (e.g., 'If LSF^2 gives area ratio, does LSF^3 give volume ratio for 3D objects?'). Learners post their sticky notes on the class DQB under the columns: 'What we now understand' and 'New questions we have'. The teacher reads 3 selected notes aloud and connects them explicitly to the storyline: the photographer, the paper cost, and what comes next in the unit.	Say: 'We've just discovered that scaling a shape by k multiplies its area by k^2 . Let's update our Driving Question Board.' Distribute sticky notes. 'You have 4 minutes: write what you can NOW answer about our passport photo puzzle on the green side, and a new question the lesson raised on the yellow side.' [Set 4-minute timer. Circulate and read notes as learners write — do not correct, but nudge vague notes: 'Can you be more specific about which number — 3 or 9?'] After posting, read aloud: the most precise 'now understand' note, the most surprising 'new question' note, and one note that directly references the phenomenon. Say: 'This question about volume — [read the note] — is exactly where our storyline is heading. The passport photo is flat, a 2D object. But what about a 3D package or a water tank? That is Lesson 9.' Point to the original driving question displayed in the room: 'We said the mathematics behind the photographer's software is the	Driving Question Board as a Living Storyline Tracker. The DQB makes metacognitive progress visible and communal — learners see that understanding grows lesson by lesson. Connecting 'new questions' explicitly to the next lesson's content models scientific inquiry as an ongoing, self-correcting process rather than a collection of isolated facts.	Observable indicators: (1) At least 85% of learners post both a 'now understand' and a 'new question' note within 4 minutes. (2) The 'now understand' notes reference a specific number (9 , k^2 , LSF^2) rather than vague language like 'I understand scaling better'. (3) At least 3 learners independently generate the conjecture about volume (LSF^3), indicating readiness for Lesson 9 extension.

		Linear Scale Factor. Today we added a second layer: LSF controls lengths, but LSF^2 controls areas. Our model is getting more powerful.'		
Model Building Phase VIDEO: Picture graphs Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Proportions to Find Scale Ratios (Flexbook) Source: CK-12 Search ARES for similar readings	Learners retrieve their cumulative similarity model — a diagram they have been adding to since Lesson 1, which currently shows: the passport photo pair labelled with $LSF = 3$ and corresponding side lengths. Today they add a new layer to the model: they annotate both photos with their areas (1575 mm^2 and 14175 mm^2), draw a double-headed arrow labelled ' $ASF = LSF^2 = 3^2 = 9$ ', and write the generalisation box: 'For any two similar figures with $LSF = k$: all lengths scale by k ; all areas scale by k^2 .' Learners also add a 'cost implication' annotation: 'Paper cost scales by k^2 — the studio uses $9\times$ as much paper, not $3\times$.' They then write a 3-sentence exit reflection: 'Before today I thought... Now I know... This changes my understanding of the passport photo because...'	Say: 'Take out your cumulative model. You've been building this since Lesson 1. Look at what it shows so far — lengths, ratios, LSF. Today we add the AREA layer.' [Allow 1 minute for learners to locate their models.] 'Add the areas to both photos, draw your ASF arrow, and write the generalisation box. You have 6 minutes.' [Circulate — specifically check that: (a) areas are correctly calculated, not just labelled k^2 ; (b) the generalisation box uses k not a specific number, showing abstraction.] After 6 minutes: 'Now write your 3-sentence exit reflection. Be honest — if you thought $ASF = LSF$ before today, say so.' [Allow 3 minutes.] Collect exit reflections as learners leave — these are the primary summative evidence for this lesson. Before dismissal, say: 'The photographer on Kimathi Street didn't just make the photo 3 times bigger. She made it 9 times the area. That is why a large print costs so much more. And that is the power of understanding $ASF = LSF^2$. Next lesson, we will ask: what happens to VOLUME when we scale a 3D object?'	Cumulative Model Revision with Layered Annotation. The cumulative model externalises the growing conceptual framework lesson by lesson, making the architecture of understanding visible. Adding a new layer (area) to an existing layer (lengths) helps learners see mathematics as a coherent, building structure rather than isolated rules. The distinction between annotating with specific numbers AND with the generalisation (k^2) pushes learners from concrete to abstract.	Observable indicators: (1) The model correctly labels both photo areas (1575 mm^2 and 14175 mm^2). (2) The ASF arrow is labelled with both the specific value (9) and the general form ($LSF^2 = k^2$). (3) The exit reflection 'Now I know...' sentence contains a specific mathematical statement involving k^2 or ASF — not just 'area gets bigger'. (4) Exit reflections collected: teacher sorts into 3 piles (secure, developing, needs support) before Lesson 9 to inform grouping.

D. TEACHER REFLECTION

After collecting exit reflections, sort them into three piles: SECURE (correctly states $ASF = LSF^2$ with at least one correct calculation), DEVELOPING (states the rule but makes arithmetic errors or applies it only to the specific passport photo without generalising), and NEEDS SUPPORT (still equates ASF with LSF, or leaves the generalisation box blank). Count each pile. If more than 30% of learners fall in NEEDS SUPPORT, open Lesson 9 with a 5-minute retrieval activity: display the class data table from today and ask: 'What pattern did our farm-plot data show us?' before introducing volume. Reflect: Did the graph-paper counting phase produce genuine

discovery, or did learners feel they were confirming what you had already told them? If discovery felt scripted, consider withholding the column headers on the Results Table next time and letting groups design their own data structure. Note which learners generated the LSF^3 conjecture on the DQB — these learners can be given an extension task at the start of Lesson 9 investigating a 3D cuboid before the main lesson begins.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners drew similar rectangles (Kenyan farm plots) with $LSF = 2, 3$, and 4 on graph paper, counted grid squares to find areas, and recorded results in a class data table showing area ratios of $4, 9$, and 16 respectively.
What did I learn?	For any two similar figures with linear scale factor k , the Area Scale Factor is k^2 ($ASF = LSF^2$). This means the enlarged passport photo ($LSF = 3$) has $3^2 = 9$ times the area of the standard print, which is why it uses significantly more paper and costs more to produce.
How does this explain the phenomenon?	The squaring relationship arises because area is two-dimensional: when both length and width are each multiplied by k , the area is multiplied by $k \times k = k^2$. An algebraic proof for a general rectangle ($l \times w$ scaled to $kl \times kw$) confirms the pattern discovered from the graph-paper data.

LESSON 9 (40 minutes): Locating the Centre of Enlargement: Where Was the Camera?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To enable learners to locate the centre of enlargement given an object and its image, using the collinearity property of corresponding points, and to connect this to the real-world position of a camera or projector relative to its subject.
Knowledge	Learners will know that the centre of enlargement is the unique point through which lines joining each object point to its corresponding image point all pass; that this property is called collinearity; and that the centre can be located by construction on a grid or by algebraic coordinate methods.
Skills	Learners will be able to: (1) draw lines through pairs of corresponding points on an object and its image to locate the centre of enlargement by construction; (2) use coordinate geometry to calculate the centre of enlargement algebraically; (3) verify the centre by confirming that the ratio of distances from the centre matches the linear scale factor.
Attitudes	Learners will appreciate that mathematics allows us to reverse-engineer real-world processes — such as determining where a camera or projector was positioned — and will show persistence when working through multi-step geometric reasoning.
Key Inquiry Question	How do we identify and use similar figures in day-to-day situations? (This lesson extends that inquiry by asking: given two similar figures, can we find the exact point from which one was enlarged to produce the other?)
Purpose in Storyline	Lessons 7 and 8 taught learners to construct enlargements given the centre and scale factor. Lesson 9 reverses the process: given the passport photo (object) and the noticeboard enlargement (image), where exactly was the optical centre — the camera lens? This reverse-engineering step deepens understanding of the enlargement relationship and prepares learners for Lesson 10, where they must choose an appropriate centre when producing their own scale drawings.
Safety Notes	No physical hazards. Ensure rulers and pencils are used carefully. Graph paper edges can be sharp; remind learners to handle paper stacks with care.

B. LESSON OVERVIEW

Learners begin by predicting whether the centre of enlargement can be found from the object and image alone, without being told where it is. They then observe a worked construction on a Kenyan map-grid context, drawing lines through corresponding vertices to find the intersection point. In the Explain phase, the teacher formalises the collinearity property and introduces the coordinate algebraic method. Learners apply both methods to problems set in Kenyan contexts — locating the camera position for a school noticeboard photo and finding the projection centre for a Huduma Centre digital display. The lesson closes with a DQB update and a model-revision activity in which learners annotate the passport photo diagram with the newly discovered camera-centre concept.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO:	Learners are shown two congruent printed grids side by side on the board. On the left grid is a simple	Distribute the printed grid sheet (or direct learners to the matching page in their exercise books where	Prediction journaling with public recording of key vocabulary. The teacher maps learner language	Listen for whether learners spontaneously mention 'lines', 'ratios', or 'intersection' — this

Equivalent ratios Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Points in the Coordinate Plane (Flexbook) Source: CK-12 Search ARES for similar readings	quadrilateral labelled ABCD (representing a passport photo outline, 35 mm × 45 mm to scale). On the right grid is a larger quadrilateral A'B'C'D' (the noticeboard print, LSF = 3). The teacher asks: 'Both photos came from one camera click. The camera had a lens — a single point through which all the light passed. That point is the centre of enlargement. Without me telling you where it is, do you think it is possible to find it using only the two photos and a ruler? Write your prediction: YES or NO, and one sentence explaining your reasoning.' Learners write individually for 90 seconds, then share with a shoulder partner for 30 seconds.	the two quadrilaterals are pre-drawn to save time). Say: 'Before we do any mathematics, I want your instinct. A studio photographer along Kimathi Street takes your photo. You can see the small print and the large print. Can you work backwards to find where the camera lens was? Write YES or NO and your reason.' Allow 90 seconds of silent individual thinking. Then: 'Share with the person next to you — 30 seconds.' Cold-call three learners: one YES, one NO, one unsure. Record key prediction words on the board (for example: 'straight lines', 'ratio', 'impossible without more information'). Do NOT confirm or deny yet. Say: 'Hold that prediction. We are going to test it today.'	onto mathematical terms on the board without evaluation, creating a word-bridge between intuition and formal geometry.	reveals prior knowledge from Lessons 7 and 8. Learners who say 'impossible' likely have not yet connected the collinearity property from the construction lessons; note these learners for targeted questioning during the Observe phase.
Observe Phase  VIDEO: Equivalent ratios Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Points in the Coordinate Plane (Flexbook) Source: CK-12 Search ARES for similar readings	The teacher reveals a large pre-drawn diagram on the board (or projects it): a simple triangle PQR and its enlarged image P'Q'R' on a coordinate grid, both drawn clearly with corresponding vertices labelled. (Context: 'Imagine P, Q, R are three corner posts of a school compound boundary in Kisumu, and P'Q'R' is the same compound as it appears on a Kenya Survey map extract.') Learners are instructed: 'Using your ruler, draw a straight line from P through P' and extend it in both directions. Then draw a line from Q through Q'. Then from R through R'. Mark the point where all three lines meet. Label it O.' Learners work on their own copies of the diagram (pre-printed or copied from the board) for 5 minutes. The teacher circulates, asking: 'Do all three lines meet at one point? What does that tell you?' After	Before learners begin drawing, say: 'I want complete silence for the first two minutes — just ruler and pencil. Let the diagram speak to you.' [WAIT 2 minutes.] Circulate and observe without commenting. After 2 minutes, say quietly to individuals: 'What do you notice about where your lines are heading?' After 5 minutes, call the class back. Ask: 'Raise your hand if all three lines met at exactly one point.' Pause and scan. [WAIT 10 seconds.] 'For those whose lines did not meet at one point — what do you think went wrong? Check whether you drew the line through the CORRESPONDING vertices.' Address any mismatches. Then ask: 'What is the ratio OP' to OP? Compare it to the LSF you calculated from the side lengths. What do you find?' [WAIT 15 seconds before taking responses.] Expected answer: they are equal.	Guided construction with structured measurement and ratio tabulation. The 'silent start' respects individual processing time before collaborative sense-making. The ratio table creates a data-driven discovery moment analogous to the area investigation in Lesson 4.	Circulate and check: (1) Are learners drawing lines through corresponding vertices (P to P', not P to Q')? A common error is connecting non-corresponding points. (2) Are the three lines concurrent? Non-concurrence indicates a construction error or a pre-drawn diagram error — check the master copy. (3) Is the ratio OP'/OP consistent with LSF? If not, learners may be measuring from the wrong endpoint. Use these checks to identify misconceptions before the Explain phase.

	construction, learners measure the distance OP and OP', then OQ and OQ'. They record the ratios OP'/OP and OQ'/OQ in a mini-table and compare them to the LSF they can calculate from the side lengths.	Record on the board: 'OP prime divided by OP equals LSF.' Say: 'This is not a coincidence. This is a mathematical law. Let us explain why.'		
Explain Phase  VIDEO: Equivalent ratios Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Points in the Coordinate Plane (Flexbook) Source: CK-12  Search ARES for similar readings	The teacher formalises the collinearity property: for any enlargement with centre O and linear scale factor k, the points O, A, and A' are collinear for every pair of corresponding points A and A', and $OA'/OA = k$. Learners copy a clean diagram and the formal statement into their notes. The teacher then introduces the coordinate method: given $A = (x_1, y_1)$ and $A' = (x_1', y_1')$, and $B = (x_2, y_2)$ and $B' = (x_2', y_2')$, the centre $O = (a, b)$ lies on the line through A and A' AND on the line through B and B'. Learners find O by writing two line equations and solving simultaneously. Worked Example 1 (construction method): 'A Huduma Centre in Nakuru has a display screen. The original design logo has vertices at $(2, 1)$, $(4, 1)$, and $(4, 4)$. The enlarged projected image has vertices at $(8, 7)$, $(14, 7)$, and $(14, 16)$. Find the centre of projection.' Learners draw both triangles on their grid, draw lines through corresponding vertices, and identify the intersection as the centre. Then verify algebraically. Worked Example 2 (coordinate method only, no grid): 'A photographer in a studio on Kimathi Street photographs a rectangular ID document. Corresponding corners of the object and image are $A(1, 2)$ and $A'(4, 8)$, and $B(3, 2)$ and $B'(10, 8)$. Find the centre of enlargement algebraically.' Learners write	Begin with: 'In Lessons 7 and 8, we were given the centre and asked to find the image. Today we reverse the process — we are given the object and image and asked to find the centre. This is called reverse-engineering, and it is exactly what an investigator does.' Write on the board: 'KEY PROPERTY — O, A, A' are collinear and OA prime divided by OA equals k.' Say: 'Write this in your own words in your notes before I write the formal version.' [WAIT 30 seconds.] For Worked Example 1, say: 'I will model the first two lines. Then I want you to draw the third line yourself and find where they all meet.' After learners find the intersection, confirm: 'So the projector in the Nakuru Huduma Centre was positioned at this coordinate. Does that make sense geometrically? Is it inside or outside the triangles?' [WAIT 15 seconds.] For Worked Example 2, say: 'Now no grid. We use algebra. With a partner, write the equation of the line through $A(1,2)$ and $A'(4,8)$.' [WAIT 90 seconds.] Take one pair's response, correct if needed, then say: 'Now write the line through B and B'. Solve simultaneously. You have 2 minutes.' Circulate, checking gradient calculations and simultaneous equation setup. Bring class back: 'What coordinates did you get? Does the LSF from OA prime over OA match	Gradual release of responsibility: teacher models the construction method fully in Example 1, then learners complete part of it independently; in Example 2, learners work in pairs on the algebraic method with teacher circulating. The Nakuru Huduma Centre and Kimathi Street photographer contexts anchor the algebra in recognisable Kenyan spaces, maintaining the phenomenon connection throughout.	Use mini-whiteboards or exercise books held up on request: after Example 2, ask all pairs to write their centre coordinates and hold up. Scan for: (a) correct gradient calculation (rise over run from A to A'); (b) correct simultaneous solution; (c) correct LSF verification. Common errors to watch: using run over rise for gradient; finding the midpoint instead of solving for the intersection; forgetting to verify using the second pair of corresponding points. Address errors immediately with: 'Check your gradient formula — which coordinate goes in the numerator?'

	equations of lines AA' and BB', solve simultaneously, and interpret the result as the camera lens position. They also calculate the LSF from OA'/OA and confirm it matches the ratio of corresponding side lengths.	what you expect?' Consolidate: 'This algebraic method is powerful because it works even when the lines would intersect off the page.'		
Driving Question Board (DQB) Creation  VIDEO: DNA as the "transforming principle" Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Points in the Coordinate Plane (Flexbook) Source: CK-12  Search ARES for similar readings	Learners return to the Driving Question Board displayed at the front of the room. The teacher reads aloud the current state of the board (accumulated from Lessons 1 to 8): the LSF connection, $ASF = LSF^2$, $VSF = LSF^3$, positive and negative scale factors. The teacher then asks: 'Our driving question asks what mathematical rule the photographer's software followed. We now know about LSF, ASF, and VSF. But today we discovered something new. What new evidence does Lesson 9 add to our answer?' Learners write one sentence on a sticky note (or in their books if sticky notes are unavailable) and selected learners read theirs aloud. The teacher adds a new card to the DQB reading: 'The centre of enlargement is the unique point from which every length is scaled — it corresponds to the camera lens position. Given any object and its image, we can locate this centre by the collinearity property.' Learners copy the updated DQB summary into their notes.	Stand beside the DQB and say: 'Let us look at what our board says so far.' Read the existing cards aloud slowly. Then say: 'The photographer pressed one button and the software resized the image. We now know the software had to: (1) apply an LSF to every length, (2) scale areas by LSF squared, (3) keep all angles equal, and (4) do all of this from a single fixed point — the centre of enlargement. Write me one sentence: what new mathematical idea from today belongs on this board?' [WAIT 60 seconds of silent writing.] Take 3 to 4 responses. Affirm and synthesise. Add the new DQB card. Say: 'Next lesson, you will create your own scale drawings of this classroom. You will need to choose your own centre. Today is the foundation for that.'	Public knowledge consolidation through the DQB ritual. The DQB functions as a class-constructed explanation that grows across the 12-lesson sequence, giving learners a sense of cumulative progress. Reading prior cards aloud activates retrieval practice across previous lessons.	Read the sticky-note sentences to gauge whether learners have grasped the key idea (collinearity and its connection to the camera lens). Misconceptions to listen for: (1) 'The centre is always at the origin' — address by pointing to Example 1 where the centre was at a different point; (2) 'The centre is the midpoint of A and A'' — address by distinguishing ratio from midpoint. Use these misconceptions to decide whether a brief re-teach is needed at the start of Lesson 10.
Model Building Phase  VIDEO: Equivalent ratios Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners open to the diagram they have been building across the unit — the annotated passport photo enlargement model. In Lesson 7 they added the centre of enlargement to the diagram; in Lesson 8 they showed what happens with a negative LSF. Today they annotate the model	Say: 'Take out your unit model — the annotated passport photo diagram. We are going to add today's discovery.' Give learners 3 minutes to draw the collinearity lines and write the caption. Circulate and prompt: 'Is your centre inside both photos or outside? What does that tell you	Cumulative model annotation reinforces the storyline structure and gives learners a visual artefact that spans all nine lessons. The independent practice problem followed by partner comparison uses the peer-explanation strategy (learners who explain consolidate understanding; learners who listen	Collect the independent practice problem solutions (or circulate and photograph selected books) to assess: (1) correct identification of the line equations through A, A' and B, B'; (2) correct simultaneous solution for the centre; (3) correct LSF calculation and verification. Exit-ticket prompt (spoken, not

 HTML: Points in the Coordinate Plane (Flexbook) Source: CK-12  Search ARES for similar readings	further: (1) they draw lines from the centre O through each corner of the small photo to the corresponding corner of the large photo, labelling the distances OA and OA' and writing $OA'/OA = 3$ (the LSF); (2) they write a caption: 'The centre of enlargement corresponds to the camera lens. Given both prints, we can locate the lens position by the collinearity property: O, A, and A prime are always collinear.' Learners also attempt a short independent practice problem: 'A photo studio in Eldoret produces a small print with one corner at A(2, 3) and the corresponding corner of the large print at A'(8, 15). A second pair of corresponding corners is B(5, 3) and B'(17, 15). Locate the centre of enlargement and state the LSF.' Learners work individually for 5 minutes, then compare with a partner.	about the scale factor?' Then say: 'Now try the Eldoret studio problem independently. 5 minutes, no talking.' [WAIT 5 minutes — maintain silence and circulate.] After 5 minutes: 'Compare your centre coordinates and LSF with your partner. If you disagree, work through each other's method and decide who is correct.' Allow 2 minutes of partner discussion. Bring class together: 'What centre did you find? What is the LSF?' Take one pair's answer, invite a second pair to confirm or challenge, then display the correct solution. Ask: 'Does the ratio OA prime over OA equal your LSF? Check it.' Close with: 'You can now not only construct enlargements but also reverse-engineer them — you can find the centre from the object and image alone. A forensic cartographer at the Kenya National Mapping Authority uses exactly this skill when verifying whether two maps are genuine enlargements of each other.'	receive an alternative explanation). The forensic cartography connection to the Kenya National Mapping Authority gives the skill professional relevance.	written, to save time): 'In one sentence, tell your partner: what is the one property that always lets us find the centre of enlargement?' Listen for: 'corresponding points and the centre are always on the same straight line' or equivalent phrasing. If fewer than 70% of pairs articulate this correctly, open Lesson 10 with a 3-minute re-teach of the collinearity property before moving to the scale drawing activity.
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D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) Were learners able to draw accurate lines through corresponding vertices, or did construction errors (such as connecting non-corresponding points) undermine the discovery? If so, consider providing a pre-drawn diagram with vertex labels more clearly matched in Lesson 10. (2) Did learners successfully set up line equations for the coordinate method, or did gradient calculation errors block progress? If gradient errors were common, a brief retrieval activity on gradient formula at the start of Lesson 10 would be beneficial. (3) Did the Kenyan contexts (Kimathi Street studio, Nakuru Huduma Centre, Eldoret studio, Kenya National Mapping Authority) feel authentic and engaging, or did learners treat them as decoration? If the latter, invite learners to suggest their own local contexts for the next problem set. (4) Did the collinearity property emerge naturally from the construction, or did it feel handed down? If it felt imposed, increase the observation phase next time by having learners construct four or five lines before the formal explanation, to let the pattern become undeniable. (5) Is the cumulative model diagram becoming too cluttered? Consider asking learners to produce a clean final version of the model diagram as part of the Lesson 10 preparation homework.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	When we drew straight lines through each pair of corresponding vertices of the small passport photo and the large noticeboard print, all three lines met at exactly one point, which we called the centre of enlargement. The ratio of the distance from the centre to
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	the image point divided by the distance from the centre to the object point equalled the linear scale factor (3 for the passport photo).
What did I learn?	The centre of enlargement is the unique point through which all lines joining corresponding object and image points pass (collinearity property). Given any object and its enlarged or reduced image, we can locate the centre by construction (drawing lines through corresponding points and finding their intersection) or by algebra (finding the intersection of two such lines using coordinate geometry). This centre corresponds to the position of the camera lens or projector in real-world enlargement situations.
How does this explain the phenomenon?	The passport photographer's software resized the image by applying a single linear scale factor ($LSF = 3$) from a fixed centre point — equivalent to the camera lens position. Every point in the original photo was moved along the straight line from the centre through that point, to a new position exactly 3 times as far from the centre. This is why the shape is perfectly preserved: all angles remain equal and all length ratios remain constant. By knowing the small print and the large print, we can reverse-engineer exactly where the camera was positioned, using the collinearity property of the centre of enlargement.

LESSON 10 (40 minutes): MODEL BUILDING: Constructing a Scale Drawing of Our School Space

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To apply knowledge of linear scale factor and similarity to create an accurate scale drawing of a real classroom or school compound space, bridging the abstract mathematics of enlargement to the everyday world of architecture, photography, and design.
Knowledge	Learners will know how to select an appropriate linear scale factor for a real-world drawing, convert actual measurements to scaled measurements using the chosen LSF, and label a scale drawing with correct dimensions and scale notation.
Skills	Learners will be able to measure real dimensions using a tape measure or metre rule, apply the linear scale factor to calculate drawing dimensions, construct an accurate and labelled floor plan on graph paper, and justify their choice of scale to an audience.
Attitudes	Learners will appreciate the practical relevance of mathematical similarity in professional fields such as architecture, urban planning, and photography, and will demonstrate collaborative responsibility and precision in group measurement and drawing tasks.
Key Inquiry Question	How do we identify and use similar figures in day-to-day situations?
Purpose in Storyline	This lesson is the hands-on model-building peak of the evidence-gathering sequence. After nine lessons developing the theory of LSF, ASF, VSF, enlargement constructions, and locating centres of enlargement, learners now apply every concept at architectural scale. The passport photo studio software that resized the students photo used exactly the same mathematical principle that learners will use today to create their floor plan — choosing one number (LSF) that scales every length uniformly. This lesson makes the phenomenon concrete and personal: learners become the photographer or architect who must choose the right scale.
Safety Notes	Ensure learners handle metre rules and tape measures carefully indoors. No learner should run in the corridor during measurement. Groups working outside should remain within the school compound boundary and one group member should act as safety observer. Sharp pencils and compasses should be used responsibly.

B. LESSON OVERVIEW

Groups of 4 to 5 learners measure the actual dimensions of their classroom or an assigned section of the school compound, select an appropriate linear scale factor, convert measurements, and construct a labelled floor plan on A3 graph paper. Each group presents their scale, justifies it, and explains how their drawing is mathematically similar to the real space. The lesson advances the driving question by demonstrating that the same LSF principle used in the passport photo studio operates at every scale — from a 35 mm print to a 12-metre classroom wall.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Multi-step word problem with Pythagorean	The teacher displays two images side by side on the board: the original passport photo pair (35 mm x 45 mm and 105 mm x 135 mm) from Lesson 1, and a simple	Display the two image pairs and say: 'We have seen how a photographer in a Kimathi Street studio used one number — the linear scale factor — to make a	Prediction anchoring: learners activate prior knowledge of LSF and connect it explicitly to the new architectural context before any measurement begins. The side-by-	Teacher scans written predictions during pair-share to check whether learners can articulate the concept of LSF as a ratio and whether they understand that the choice of scale

theorem Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 Search ARES for similar readings	architect floor plan of a single room next to a sketch of the actual room. The teacher asks: 'Today you will create a floor plan of our classroom. Before you touch a ruler, predict in your exercise book: what single number will connect the real classroom to your drawing? How will you choose it? Write your prediction in the form — my scale will be 1 : ____ because ____.' Learners write individually for 90 seconds, then share with a partner using Think-Pair-Share.	perfectly similar photo three times as large. Architects do exactly the same thing when they draw floor plans of buildings. Look at this floor plan of a room. The room is real and large; the drawing is small. What is the connection?' WAIT 20 seconds. Then: 'In your book, write your prediction: what number will connect your classroom to your drawing today, and how will you decide what that number should be?' After Think-Pair-Share, cold-call two or three pairs. Record their predicted scales on the board without evaluation. Say: 'Keep your predictions — we will return to them at the end.'	side display of the passport photo and the floor plan creates a visual bridge between the familiar phenomenon and the new application.	must fit the paper size. Common misconceptions to watch for: learners who write a scale of 1:1 (not understanding reduction) or who choose a scale without considering paper dimensions.
Observe Phase  VIDEO: Multi-step word problem with Pythagorean theorem Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 Search ARES for similar readings	Groups receive a measurement recording sheet (pre-prepared by teacher) listing the features to measure: length of classroom, width of classroom, height of one wall, width and height of one door, width and height of one window, and the dimensions of one desk. Groups use tape measures or metre rules to measure and record all values in millimetres in their table. They also sketch a rough free-hand outline of the classroom noting where doors and windows are positioned. The teacher circulates and prompts precision: 'Record to the nearest 10 mm. Does your partner agree with your reading?' After measurements are complete, groups return to their desks.	Distribute tape measures and recording sheets. Say: 'You are now working exactly as a surveyor or architect works before drawing a floor plan. Record every measurement in millimetres — the same unit used in the passport photo problem.' Circulate continuously. At each group, crouch to eye level and ask: 'What is the longest measurement you have found so far? Now look at your graph paper. If you drew that length at full size, would it fit?' WAIT 15 seconds for the group to discuss. Then: 'That realisation will help you choose your scale. Keep measuring.' After groups return, say: 'Before we do any calculations, tell me — what was the largest single dimension you measured?' Take two or three responses and write the largest value on the board.	Concrete data collection grounds the abstract concept of scale in real, tactile experience. By measuring real objects — a door, a window, a desk — learners encounter the same proportional relationships they have studied in photographs and maps, now at the scale of their own lived environment. Recording in millimetres maintains consistency with the passport photo units.	Teacher checks that groups are recording in consistent units and that their rough sketch shows spatial relationships of features. Ask targeted questions: 'Is your classroom longer or wider? By how much? What ratio is that?' Groups that finish early are asked: 'If I told you the scale is 1:50, what would the length of the classroom be on paper? Is that reasonable for your graph paper?'
Explain Phase  VIDEO: Multi-step word	Back at their desks, groups complete three structured calculation steps. Step 1 —	Before groups begin Step 1, say to the whole class: 'You need one number that will shrink your	The three-step structured task (choose scale, convert, construct) mirrors the explicit modelling cycle.	Teacher checks two things at each group during construction: (1) that all scaled dimensions are

problem with Pythagorean theorem Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 Search ARES for similar readings	Choose scale: they examine their graph paper dimensions (provided as A3, approximately 420 mm x 594 mm) and their largest classroom measurement, then calculate the minimum scale factor needed so the longest dimension fits on the paper with a margin. Step 2 — Convert all measurements: using their chosen scale (e.g. 1:50 meaning LSF = 1/50), they divide every real measurement by 50 (or their chosen denominator) to get the drawing measurement in millimetres. They complete a two-column table: Actual dimension (mm) and Drawing dimension (mm). Step 3 — Construct the floor plan: using pencil, ruler, and set square, they draw the outline of the classroom to scale on graph paper, then add doors, windows, and one desk, all to the same scale. Every feature is labelled with its actual dimension and the scale is written clearly in the title block: e.g. Scale 1:50.	classroom onto this sheet. That number is the linear scale factor. The rule is simple: drawing dimension equals actual dimension divided by the denominator of your scale. If I choose 1:50, every millimetre of actual space becomes 1/50 of a millimetre on paper. Let us check together: the longest classroom dimension on the board is ____ mm. If I choose 1:50, what is the drawing length?' Call a learner to compute. WAIT 15 seconds. 'Is that smaller than 420 mm? Good. If it is larger, what scale would you try next?' During construction, circulate and use targeted prompts: 'Show me your scale. Now check: is the door width on your drawing exactly the actual door width divided by your denominator?' If a group has inconsistent measurements, ask: 'If the classroom is 8 000 mm long and 6 000 mm wide in real life, and your drawing shows the length as 160 mm, what scale have you used? Now use that same scale for the width.' Do not correct errors directly — prompt the group to self-correct using the LSF relationship.	Learners experience the same cognitive process as the passport photo software: select one LSF, apply it uniformly to every dimension, and produce a similar figure. The labelled title block with scale notation connects to professional cartographic and architectural practice visible on Kenyan survey maps.	computed from the same LSF — inconsistency here reveals conceptual confusion between LSF and arbitrary resizing; (2) that angles are preserved — the floor plan should have right angles matching the real room. Ask: 'What angle is the corner of your classroom in real life? What angle should it be on your drawing? Why?' This probes understanding that similarity preserves angles.
Driving Question Board (DQB) Creation  VIDEO: Multi-step word problem with Pythagorean theorem Source: Khan Academy (English - US curriculum) Search ARES for similar videos	Each group writes one new sticky note or card for the Driving Question Board. The prompt is: 'Complete this sentence — Today I proved that the passport photo principle works at ____ scale because ____.' Groups post their cards. The teacher reads two or three aloud and the class discusses whether they answer or extend the driving question: Why does resizing a passport photo change its size but never its shape? A class scribe adds one	Say: 'For nine lessons we have been gathering evidence to answer our driving question. Today you have built physical evidence with your own hands. Write your sticky note — what did your floor plan prove about the passport photo principle?' WAIT 90 seconds for groups to write. Collect and post the cards. Read two contrasting cards aloud — one that mentions a specific measurement and one that speaks generally. Ask the class: 'Which of	The DQB update at this stage serves as a metacognitive consolidation moment. Learners explicitly connect the hands-on scale drawing activity back to the original driving question and to nine prior lessons of accumulated evidence. The collaborative summary sentence builds shared disciplinary language and advances the storyline toward the final two lessons.	The sticky notes function as an exit-ticket proxy for this phase. Teacher reads all cards after the lesson and sorts them into three categories: (A) cards that correctly name LSF and explain uniform scaling; (B) cards that describe the activity without mathematical connection; (C) cards that reveal residual misconceptions (e.g. thinking scale drawings are approximate rather than mathematically exact). Category B and C cards inform the opening of

 HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12  Search ARES for similar readings	agreed summary sentence below the existing DQB cards: 'LSF controls every length uniformly — whether the object is a 35 mm photo or a 12-metre classroom wall — so the shape is always preserved.'	these two cards gives stronger evidence? Why?' WAIT 20 seconds. Then guide the class to construct the summary sentence collaboratively. Add it to the DQB with a marker and say: 'Our board now shows evidence from photos, maps, farm plots, water tanks, and a classroom. The LSF is the same idea in every case.'		Lesson 11.
Model Building Phase  VIDEO: Multi-step word problem with Pythagorean theorem Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12  Search ARES for similar readings	Each group selects one spokesperson to present their completed floor plan (or their best-progress version) to the class in a 90-second gallery-walk style presentation. The spokesperson states: (a) the actual dimensions of the room, (b) the scale chosen and why, (c) one example of a converted measurement, and (d) how they would use the area scale factor to calculate the actual floor area from the drawing area. Listening groups use a peer-feedback card to record one strength and one question for the presenting group. After all presentations, the teacher leads a 3-minute whole-class synthesis: 'Every group used a different scale but every group produced a drawing that is mathematically similar to the real classroom. What does that tell us about the LSF?'	Signal the start of presentations: 'You have 90 seconds. Begin with your scale ratio, justify it, and end with the area scale factor calculation.' Time each group strictly. After each presentation, say to the listening groups: 'Write your one question on the feedback card now — you have 20 seconds.' WAIT 20 seconds. Collect one question from a listening group and direct it back to the presenting group: 'Group B asks — if you had chosen a scale of 1:100 instead, would your floor plan still be similar to the real classroom? Answer in one sentence.' For the synthesis, ask: 'Three groups chose different scales. All three floor plans are similar to the real classroom. Is it possible for two drawings made with different scales to both be similar to the same real object? Discuss with your partner.' WAIT 30 seconds. Take responses and affirm: 'Yes — any uniform scale produces a similar figure, because similarity depends on equal angles and proportional sides, not on which specific ratio you choose.'	The gallery-walk presentation structure serves dual purposes: it consolidates learning through articulation (learners must explain their mathematical choices publicly) and it surfaces the generalisation that similarity is a class of relationships, not a single fixed scale. The area scale factor question in point (d) of each presentation keeps the connection alive between LSF and ASF established in Lesson 4.	Teacher uses the four-point presentation rubric mentally during each 90-second slot: (1) correct scale stated with units; (2) conversion calculation correct; (3) drawing dimensions proportional and right angles preserved; (4) area scale factor correctly computed as LSF squared. Groups that cannot compute the area scale factor or who confuse ASF with LSF are noted for targeted support in Lesson 11. The peer feedback cards are collected by the teacher for review.

D. TEACHER REFLECTION

After the lesson, consider the following questions: Did every group successfully choose a scale that fit their graph paper, and if not, what was the blocking misconception — unit confusion, arithmetic error, or conceptual misunderstanding of ratio? During the Explain Phase, which groups needed the most support in applying a consistent

LSF to all features — and does that suggest the need for more scaffolding in Lesson 11 problems? Did the DQB summary sentence generated by the class accurately capture the key insight — that LSF operates uniformly regardless of the physical scale of the object — or was it too vague? In the Model Building presentations, could learners correctly calculate the area scale factor from their chosen LSF? If not, plan a brief retrieval activity at the start of Lesson 11. Finally, consider whether the measurement activity gave all learners equitable participation — were quieter group members holding the tape measure and recording, or did one learner dominate? Adjust group roles in Lesson 11 accordingly.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that when we applied one linear scale factor to every dimension of our classroom, the resulting floor plan had exactly the same shape as the real room — all angles were preserved and all length ratios were constant — just as the passport photo studio software preserved the shape of the students face by scaling every pixel distance by the same factor of 3.
What did I learn?	We learned that a scale drawing is a real-life example of a similar figure: the linear scale factor (written as 1 : n) controls every length uniformly, so any real space can be reduced to fit on paper without distortion, and the area of the drawing relates to the actual area by the square of the LSF.
How does this explain the phenomenon?	We explained that the passport photo principle — one number scales everything equally — works at every physical scale, from a 35 mm photograph to a 12-metre classroom wall. This is why architects, cartographers, and photographers all use the same mathematical idea: choose one LSF, apply it to every length, and the shape of the object is perfectly preserved in the new size.

LESSON 11 (80 minutes): EXPLAIN & APPLY: Mixed Problems — LSF, ASF, and VSF

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate and integrate understanding of Linear Scale Factor (LSF), Area Scale Factor (ASF), and Volume Scale Factor (VSF) by solving mixed, multi-step problems drawn from real-life Kenyan contexts.
Knowledge	Learners know that: (a) $ASF = LSF^2$, (b) $VSF = LSF^3$, (c) the LSF is the single multiplier that scales every length in a similar figure or solid, and (d) angles are preserved under enlargement while all lengths scale by the same ratio.
Skills	Learners are able to: (a) identify the correct scale factor type (LSF, ASF, or VSF) needed to solve a given problem; (b) move fluently between LSF, ASF, and VSF; (c) set up and solve multi-step problems involving similar figures and solids; (d) justify solution pathways using mathematical reasoning.
Attitudes	Learners appreciate that a single mathematical relationship — the Linear Scale Factor — controls size, area, and volume simultaneously, making it a powerful tool for real-world design, photography, architecture, and engineering.
Key Inquiry Question	How does the linear scale factor relate to the area and volume of similar figures and solids?
Purpose in Storyline	Lessons 1–10 built understanding of similarity, LSF, ASF, and VSF in separate contexts. Lesson 11 is the integration lesson: learners now face mixed problems where they must first diagnose which scale factor is relevant before computing — mirroring the complexity of the passport-photo phenomenon and real design challenges. This lesson advances the driving question from 'what is the rule?' to 'how do I choose and apply the right rule in any situation?'
Safety Notes	No physical hazards. Ensure learners do not share calculators in ways that delay individual thinking — each learner should attempt every step independently before peer discussion.

B. LESSON OVERVIEW

Learners open by revisiting the passport-photo phenomenon and articulating — in their own words — the three scale factor relationships established across Lessons 1–10. The teacher then launches a structured 'Diagnosis Before Calculation' routine: for each mixed problem, learners must first publicly commit to which scale factor they need and why, before any arithmetic begins. Problems are drawn from Kenyan contexts: enlarging a school noticeboard photo, scaling a githeri cooking pot, comparing maize storage silos in the Rift Valley, and resizing a school athletics track plan. The lesson closes with learners updating their cumulative models and Driving Question Board, recording precisely how LSF, ASF, and VSF are connected as a unified system.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: WATCH: Railroads and the Industrial	Learners receive a 'Scale Factor Sorting Card' containing six short problem stems — two involving lengths only, two involving areas, two involving volumes — all set in	Distribute cards and sticky notes. Say verbatim: 'Before we calculate anything today, we are going to think like diagnosticians — like a doctor deciding which test to run	Prediction Commitment (Anticipation Guide variant): By forcing a public, written commitment before instruction, this strategy activates prior knowledge,	Observable indicator: Are learners correctly distinguishing area and volume problem stems from length problems? A learner is on track if they identify 'area', 'surface',

Revolution Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 Search ARES for similar readings	Kenyan contexts (passport photos, maize silos, cooking pots, school plans). Without doing any calculations, learners read each stem and write on a sticky note: 'LSF', 'ASF', or 'VSF' — whichever they predict is needed — along with a one-sentence justification. Sticky notes are posted on a class sorting wall under three columns.	before treating a patient. Read each problem stem carefully. Write only the type of scale factor you think you need, and one reason why. You have 6 minutes of silent thinking time — no calculations yet.' Circulate and observe written justifications. After posting, cold-call 4 learners (one per quadrant of the room) to read their prediction for Problem 3 (a volume problem). Ask: 'What word in the problem stem told you it was a volume situation?' Wait 12 seconds after each response. Do NOT correct errors yet — preserve predictions for comparison at the end of the lesson.	surfaces misconceptions (e.g., learners who use LSF² for volume problems), and creates cognitive dissonance that drives engagement during the Observe phase. The sorting wall makes class-level misconceptions visible.	'covers', 'painted' as ASF triggers and 'volume', 'holds', 'capacity', 'fills' as VSF triggers. Flag any learner who writes 'LSF' for all six — this signals a foundational gap requiring targeted support during Phase 3.
Observe Phase  VIDEO: WATCH: Railroads and the Industrial Revolution Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 Search ARES for similar readings	The teacher projects three fully worked, annotated examples — each connected to the passport-photo phenomenon or a related Kenyan context. Example A (LSF → ASF): The passport studio charges by photo area. The small print is 35 mm × 45 mm. The noticeboard print is 105 mm × 135 mm. What is the ratio of their areas? Learners follow along, filling in a structured note-frame with blanks for LSF, LSF², and the area ratio. Example B (ASF → LSF): A githeri pot at a school canteen in Kisumu has a base area of 900 cm². A similar smaller pot has a base area of 100 cm². What is the ratio of their heights? Learners derive $LSF = \sqrt{(ASF)}$. Example C (VSF → LSF → area): Two similar maize storage silos near Eldoret have volumes in the ratio 27:8. Find (i) the LSF, (ii) the ratio of their curved surface areas. Learners work Example C independently for 5 minutes before teacher reveals the solution.	For Example A: narrate each step aloud, pausing at the LSF step. Ask: 'Why do we NOT need to calculate the area directly? What shortcut do our previous lessons give us?' Wait 10 seconds. Cold-call 2 learners. For Example B: write 'ASF = LSF²' on the board and ask: 'If I give you the square, how do I get back to the original number?' Wait 12 seconds. Accept 'square root' responses and annotate: 'LSF = $\sqrt{(ASF)}$'. For Example C: say verbatim: 'I am going to be silent for 5 minutes. Work through this problem in your exercise books step by step — diagnose first, then calculate.' After 5 minutes, cold-call one learner to narrate their solution aloud while the class checks. Prompt: 'Tell us your diagnosis first — which scale factor did the problem give you, and which ones did it ask you to find?'	Structured Note-Frame with Blank-Filling: The partially completed note-frame ensures learners are not passive transcribers but active completers. Gaps are placed at the most conceptually demanding steps (identifying LSF from VSF, deriving LSF from ASF) to focus cognitive effort where misconceptions are most likely. Connecting every example back to the passport photo or Kenyan context keeps the phenomenon as the explanatory anchor.	Observable indicator for Example C: Learners should write $VSF = 27/8 \rightarrow LSF = \sqrt[3]{(27/8)} = 3/2$, then $ASF = (3/2)^2 = 9/4$. A learner who writes $LSF = 27/8 \div 3$ has confused VSF and LSF — this is the most common error at this stage. Collect one exercise book per table group to scan Example C working before Phase 3 begins.

Explain Phase  VIDEO: WATCH: Railroads and the Industrial Revolution Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12  Search ARES for similar readings	Learners work individually on a set of four mixed problems (the 'Kimathi Street Challenge Sheet') before pairing to compare methods. Problem 1: The school noticeboard photo (105 mm × 135 mm) is reprinted as a giant banner for the athletics championship. The banner has an area of 2,835 cm². What is its width in centimetres? Problem 2: A Nairobi architect draws a scale model of a cylindrical water tank for a community project in Kibera. The model tank holds 0.125 litres. The real tank must hold 1,000 litres. What is the LSF of enlargement, and what is the diameter of the real tank if the model's diameter is 4 cm? Problem 3: Two similar triangular warning signs along the Nakuru–Nairobi highway have perimeters in the ratio 2:5. The smaller sign has an area of 48 cm². What is the area of the larger sign? Problem 4: A rectangular farm plot in the Rift Valley and a similar smaller plot have areas of 1,125 m² and 500 m² respectively. The shorter side of the smaller plot is 20 m. Find the shorter side of the larger plot.	Say: 'You have 18 minutes — work in silence for the first 10 minutes, then you may whisper with your partner for 8 minutes. Every problem begins with a written diagnosis line: write the scale factor the problem gives you and the scale factor it is asking for.' Circulate actively. At the 10-minute mark, stop the class and ask: 'Hands up — who has written a diagnosis line for all four problems?' Address any learner who has skipped the diagnosis. Cold-call one learner for Problem 2: 'Read us your diagnosis line only.' After pair discussion, facilitate whole-class debrief for Problems 2 and 4 only — cold-call 3 learners per problem. For Problem 2, ask: 'Where in the passport-photo story would we ever face a volume scale factor problem — what if the photo was printed on a 3D embossed surface?' Probe for transfer of understanding.	Diagnosis-Before-Calculation Routine: Requiring a written 'diagnosis line' before any arithmetic forces metacognitive regulation — learners must consciously categorise the problem type. This mirrors how engineers and designers work: identify what is known and what is needed before computing. Pairing after individual work ensures every learner has committed to a solution path before social influence operates, increasing the quality of peer discussion.	Observable indicators: (1) Correct diagnosis lines for all four problems — signals conceptual fluency. (2) Problem 3: correct answer = 300 cm² (LSF = 5/2, ASF = 25/4, Area = 48 × 25/4). A learner who answers 120 cm² has applied LSF instead of ASF — a persistent misconception. (3) Problem 4: LSF = $\sqrt{(1125/500)}$ = $\sqrt{9/4}$ = 3/2, shorter side = $20 \times 3/2$ = 30 m. Incorrect answer of 56.25 m indicates learner used ASF as if it were LSF². Use these as live data for the debrief.
Driving Question Board (DQB) Creation  VIDEO: WATCH: Railroads and the Industrial Revolution Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML:	Each learner writes two new sticky notes for the class Driving Question Board. Blue note: one thing they can now explain about the passport-photo phenomenon that they could NOT explain at the start of the unit (must reference LSF, ASF, or VSF specifically). Yellow note: one question that the mixed problems raised for them — something still uncertain or a new 'what if?' The teacher facilitates a 5-minute gallery walk where learners read peers' blue notes silently and place a small dot	Say verbatim: 'The blue note is your evidence of growth — it must contain at least one mathematical term from our unit: LSF, ASF, VSF, similar, scale factor, ratio. The yellow note shows scientific curiosity — there are no wrong questions here.' Allow 4 minutes of silent writing. During the gallery walk, circulate and read blue notes. After the walk, select 3 blue notes that represent a progression (basic → intermediate → sophisticated understanding) and read them aloud without naming	DQB Progressive Disclosure: The dot-sticker gallery walk creates a visual 'heat map' of shared understanding across the class — sticky notes with many dots reveal the concepts most widely grasped, while isolated notes with no dots reveal individual insights or persistent uncertainties. Reading three blue notes on a spectrum makes the progression of understanding explicit and models what 'deeper understanding' looks like.	Observable indicator: A blue note demonstrating full understanding will connect all three scale factors in one sentence — e.g., 'The photographer's software uses the LSF to scale every length by the same ratio; the area of the photo changes by LSF² and if the photo were 3D the volume would change by LSF³.' A blue note that mentions only 'making things bigger' without referencing ratios indicates the learner has not yet consolidated the proportional reasoning core of the unit.

Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 Search ARES for similar readings	sticker on any blue note that articulates something they also now understand.	authors. Ask the class: 'Which of these three blue notes most completely answers our driving question — and what would need to be added to make the simplest one as powerful as the most sophisticated one?' Wait 15 seconds. Take 3 responses.		
Model Building Phase VIDEO: WATCH: Railroads and the Industrial Revolution Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 Search ARES for similar readings	Learners open their cumulative model (started in Lesson 1 and revised each lesson) and add a final 'Unified Scale Factor Map' layer. This is a structured diagram with the passport photo at the centre, arrows radiating outward to three boxes labelled LSF (ratio of lengths), ASF = LSF ² (ratio of areas), and VSF = LSF ³ (ratio of volumes). Each box must contain: (1) the formula, (2) one example from the lesson's problems, and (3) one example from the passport-photo phenomenon. Below the diagram, learners write a 'phenomenon explanation paragraph' — a 3–5 sentence explanation of exactly why the photographer's resized photo looks identical but has a different size, referencing all three scale factors.	Say: 'Your model today must do something new — it must show how all three scale factors are one connected system, not three separate rules. Draw the map first, then write the paragraph. The paragraph is your answer to our driving question.' Allow 12 minutes. With 4 minutes remaining, say: 'Swap your model with a partner. Check: does their paragraph mention LSF, ASF, and VSF? Does it explain why the photo looks the same but is a different size? Write one green comment — a specific strength — and one orange comment — one thing to make it more complete.' Facilitate 2-minute share-back: cold-call 2 pairs to share the orange comment they gave and how the author responded to it.	Cumulative Model Revision with Peer Annotation: The Unified Scale Factor Map makes the relational structure of the concept visible — learners are not just recalling three formulas but constructing a connected schema. Writing the phenomenon explanation paragraph requires learners to translate the mathematical relationships back into the real-world context, testing whether their understanding is genuinely explanatory or merely procedural. Peer annotation with colour-coded comments (green = strength, orange = improvement) makes evaluation criteria concrete and actionable.	Observable indicator: A complete, accurate phenomenon explanation paragraph will (1) state that LSF = 3 for the standard-to-noticeboard photo enlargement, (2) explain that every length is multiplied by 3 (so proportions — angles and ratios — are preserved, hence the image looks identical), (3) state that the area increases by 3 ² = 9 (so the noticeboard photo uses 9 times as much paper), and (4) note that if the photo were printed on a 3D embossed surface, the volume would scale by 3 ³ = 27. A paragraph that only addresses lengths and areas, omitting volume, is partially complete — note for Lesson 12 consolidation.

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) Diagnosis Accuracy — during Phase 1 sticky-note posting, what proportion of learners correctly identified volume problem stems? If fewer than 60% correctly flagged the VSF problems, the language of volume problems (capacity, holds, fills) may need explicit vocabulary work in Lesson 12. (2) Persistent Misconception Check — how many learners in Phase 3 applied LSF instead of ASF² for Problem 3? If more than 3 learners per class made this error in the final problem set, plan a targeted 5-minute error analysis opener for Lesson 12 using that specific problem. (3) Model Quality — scan the phenomenon explanation paragraphs collected at the end of Phase 5. Do learners articulate that angles are preserved (not just lengths scaled)? The preservation of angles — and therefore shape — is the heart of the driving question. If this is absent from most paragraphs, the closing lesson must revisit it explicitly. (4) DQB Yellow Notes — read all yellow 'still wondering' notes before Lesson 12. Do any questions point beyond the curriculum (e.g., 'What if the scale factor is less than 1?' or 'Does this work for curved shapes like a football?') — if so, prepare a 2-minute extension hook for Lesson 12 to honour learner curiosity without derailing consolidation objectives.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed three fully worked examples (Kimathi Street passport photo area, Kisumu githeri pot height ratio, Eldoret maize silo surface area) that modelled the complete LSF → ASF → VSF chain, and completed four mixed Kenyan-context problems independently and in pairs.
What did I learn?	Learners consolidated that LSF, ASF ($= \text{LSF}^2$), and VSF ($= \text{LSF}^3$) form a single unified system; that diagnosing which scale factor a problem provides — and which it demands — is the critical first step; and that the passport photographer's software preserves shape because every length is multiplied by the same LSF, keeping all angle ratios and length ratios constant.
How does this explain the phenomenon?	Learners explained in writing why the resized passport photo looks identical but occupies a different area: the LSF scales every length equally (preserving all proportions and angles), the area changes by LSF^2 , and if the photo were three-dimensional the volume would change by LSF^3 — demonstrating that one number, the Linear Scale Factor, governs the entire transformation.

LESSON 12 (80 minutes): Final Explanation — Weaving the Three Relationships: LSF, ASF, and VSF

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate all three scale-factor relationships — Linear (LSF), Area (ASF = LSF^2), and Volume (VSF = LSF^3) — through a structured three-problem set drawn from Kenyan contexts, followed by peer-marking, error analysis, and final DQB completion.
Knowledge	By the end of the lesson the learner should be able to: (a) recall and state the relationships $ASF = LSF^2$ and $VSF = LSF^3$; (b) identify and apply the correct scale-factor relationship for length, area, and volume problems; (c) explain why every dimension in a similar figure must be scaled by the same LSF.
Skills	By the end of the lesson the learner should be able to: (a) solve multi-step similarity problems involving map distances, roofing areas, and water-tank volumes; (b) use peer-marking and error analysis to identify and correct common misconceptions; (c) construct and annotate a final cumulative model linking LSF, ASF, and VSF.
Attitudes	The learner appreciates the power of a single mathematical relationship (LSF) to govern length, area, and volume simultaneously, and values accuracy and intellectual honesty during peer-marking.
Key Inquiry Question	How does the linear scale factor relate to the area and volume of similar figures and solids?
Purpose in Storyline	This is the culminating lesson of the unit. Learners return directly to the passport photo phenomenon: the photographer's software applied one $LSF = 3$ to every pixel dimension, which simultaneously made the area $9\times$ larger — and they now have the tools to prove it. Three structured Kenyan problems (Nairobi–Nakuru map; Mombasa roofing model; Kericho water tank) weave all three relationships into one extended consolidation task. The DQB receives its final entries — $ASF = LSF^2$ and $VSF = LSF^3$ — and learners compare their Day-1 model with their final model to see the full arc of the unit's learning.
Safety Notes	No hazardous materials. Ensure peer-marking is conducted respectfully; remind learners that errors are learning opportunities, not sources of ridicule — model this language explicitly before the peer-marking phase begins.

B. LESSON OVERVIEW

Lesson 12 is the Final Explanation consolidation lesson for Sub-Strand 2.1. Learners open by revisiting the passport photo phenomenon and articulating what they now understand about LSF. They then work independently through a structured three-problem set — (1) a map/linear problem (Nairobi to Nakuru), (2) a roofing area problem (Mombasa architect's model), and (3) a water-tank volume problem (Kericho tea farm) — before participating in a structured peer-marking and error-analysis protocol. The lesson closes with final DQB entries, a side-by-side comparison of Day-1 vs. final cumulative models, and a whole-class Final Explanation of the driving question. All phases are anchored to the passport photo phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Conservation and	Learners receive a single A5 stimulus card showing three labelled diagrams side by side: (A)	Distribute the stimulus card face-down. Say: 'Do NOT turn the card over until I say go.' After 10	Prediction Journaling with Deferred Feedback — by committing to written predictions	Teacher scans written predictions during circulation. Observable indicator: can the learner write

the race to save biodiversity Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 Search ARES for similar readings	two similar rectangles with LSF = 4 labelled; (B) a map strip showing Nairobi → Nakuru with a scale bar; (C) two similar cylindrical water tanks labelled 'Model' and 'Real'. Before any instruction, learners write three silent predictions in their exercise books: (i) 'If LSF = 4, the area ratio will be ____'; (ii) 'If LSF = 4, the volume ratio will be ____'; (iii) 'The one mathematical rule that connects all three diagrams is ____.' Learners are told they will return to these predictions at the end of the lesson to judge their own accuracy.	seconds of silence to build anticipation, say: 'Turn over — read all three diagrams — write your three predictions NOW. You have 3 minutes. No talking.' Set a visible timer. Circulate silently, noting 3–4 interesting or divergent predictions without commenting. After 3 minutes: 'Pens down. Do NOT share yet — we will test your predictions today.' Cold-call 2 learners only to read prediction (i) aloud — do not affirm or correct. Say: 'Hold that thought — the problems today will tell us who is right.'	before instruction, learners activate prior knowledge from Lessons 1–11 and create a personal benchmark against which new evidence will be measured. Deferring feedback sustains cognitive tension throughout the lesson.	'area ratio = 16' ($LSF^2 = 4^2 = 16$) and 'volume ratio = 64' ($LSF^3 = 4^3 = 64$)? Learners who write 'area ratio = 4' reveal the persistent misconception that area scales linearly — flag these for targeted cold-calling during the Explain phase.
Observe Phase VIDEO: Conservation and the race to save biodiversity Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 Search ARES for similar readings	Learners receive the structured three-problem worksheet (one page, three clearly boxed problems). They work INDEPENDENTLY and in SILENCE for 18 minutes, showing all working. PROBLEM 1 — MAP (Linear/LSF): A map of Kenya uses a scale of 1 : 2 500 000. On the map, the distance from Nairobi to Nakuru is 5.8 cm. (a) Calculate the actual distance in kilometres. (b) A second, smaller map of the same region has a scale of 1 : 5 000 000. What is the distance from Nairobi to Nakuru on this smaller map? (c) What is the LSF of the second map relative to the first map? PROBLEM 2 — ROOFING AREA (ASF): An architect in Mombasa builds a scale model of a coastal villa. The model roof has an area of 450 cm². The LSF from model to real building is 1 : 25. (a) Calculate the area of the real roof in cm². (b) Convert this to m². (c) Roofing tiles cost Ksh 850 per m². What is the total cost of tiling the real roof? PROBLEM 3 — WATER TANK VOLUME (VSF): A Kericho tea farm uses two geometrically	Distribute worksheets. Say: 'This is individual silent work — it is your chance to show yourself what you know. You have 18 minutes. Begin.' Set timer visibly. Circulate continuously using a two-column clipboard: left column — learner name, right column — which problem they are on and any error observed (e.g., 'used LSF instead of LSF³ for volume'). Do NOT correct errors during circulation — only note them for use in the Explain phase. At the 15-minute mark say: 'Three minutes remaining — if you have not reached Problem 3, write what relationship you would use and why.' After 18 minutes: 'Pens down. Swap your worksheet with the person directly to your right — you will now peer-mark.'	Three-Context Spiral Evidence Gathering — the three problems deliberately move from linear (map) → area (roof) → volume (tank), mirroring the conceptual hierarchy of the unit. By encountering all three relationships in a single sitting, learners build a unified mental structure rather than treating each relationship as isolated.	Observable indicator: learner correctly identifies and applies $ASF = LSF^2$ in Problem 2 (expected: $450 \times 625 = 281\,250$ cm² = 28.125 m²; cost = $28.125 \times 850 = \text{Ksh } 23\,906.25$) and $VSF = LSF^3$ in Problem 3 (expected: $1\,200 \times 27 = 32\,400$ litres). Teacher clipboard data from circulation identifies the 2–3 most common errors for explicit address in the Explain phase.

	similar cylindrical water tanks. The smaller tank has a volume of 1 200 litres. The LSF from the smaller tank to the larger tank is 1 : 3. (a) Calculate the volume of the larger tank in litres. (b) Water costs Ksh 4.50 per litre to pump. How much more does it cost to fill the larger tank than the smaller tank? (c) A third tank is geometrically similar to the smaller tank with LSF = 1 : 1.5. What is its volume?			
Explain Phase  VIDEO: Conservation and the race to save biodiversity Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12  Search ARES for similar readings	PART A — Structured Peer-Marking (10 min): Learners receive a model-answer strip (teacher-prepared, printed on coloured paper) and mark the swapped worksheet using a three-colour protocol: GREEN tick = fully correct with working; ORANGE tick = correct answer but incomplete working; RED cross = error, with the marker writing one sentence explaining the error. Markers sign the bottom of the marked sheet. Worksheets are returned to owners. PART B — Error Analysis Discussion (10 min): Teacher selects 3 errors captured on the clipboard and writes them anonymously on the board. For each error, learners discuss in pairs (2 minutes) before a whole-class cold-call. PART C — Phenomenon Re-Connection (5 min): Teacher projects the passport photo image from Lesson 1. Learners write a Final Explanation paragraph (4–6 sentences) in their books answering the driving question in full, using the terms LSF, ASF, VSF, similar figures, and angle preservation.	PART A: Distribute model-answer strips. Say: 'You are marking your partner's thinking — be precise and be kind. Write one sentence for every red cross explaining what went wrong.' Give 10 minutes. PART B: Write 3 anonymous errors on the board, e.g., Error 1: 'Area of real roof = $450 \times 25 = 11\,250\text{ cm}^2$' (used LSF, not LSF²). Error 2: 'Volume of large tank = $1\,200 \times 3 = 3\,600$ litres' (used LSF, not LSF³). Error 3: 'LSF of second map = $5\,000\,000 \div 2\,500\,000 = 2$' (correct value but inverted interpretation). For each error say: 'Talk to your partner — what mistake was made and what is the correct approach? 2 minutes.' Allow 15 seconds wait time after pairs stop talking. Cold-call 3 different learners per error (9 cold-calls total). After each: 'Does anyone want to add to or challenge that explanation?' PART C: Project passport photo. Say: 'You have been building to this moment since Lesson 1. Write your Final Explanation now — 4 to 6 sentences. Use ALL five terms on the board: LSF, ASF, VSF, similar figures, angle preservation. You have 5 minutes.' Circulate and read over shoulders — do not correct yet.	Collaborative Error Analysis + Final Explanation Writing — naming and unpacking errors publicly (anonymously) targets the two most durable misconceptions in the unit (area scales as LSF, volume scales as LSF). The Final Explanation paragraph forces learners to synthesise across the entire unit in their own words, anchored to the original phenomenon, which is the hallmark of deep conceptual consolidation.	Observable indicator for peer-marking: marker correctly identifies the error type (linear vs. square vs. cubic scaling) and writes a meaningful correction sentence. Observable indicator for Final Explanation: paragraph uses all five mandated terms correctly and makes explicit reference to the passport photo (e.g., 'The photographer's software applied LSF = 3 to both width and height, which is why the image looks identical but the area is 9× larger — ASF = LSF² = 9').

Driving Question Board (DQB) Creation  VIDEO: Conservation and the race to save biodiversity Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12  Search ARES for similar readings	Each learner receives two sticky notes. On STICKY NOTE 1 (yellow): they write the two final mathematical entries for the DQB — '$ASF = LSF^2$' and '$VSF = LSF^3$' — and place them in the 'What We Now Know' column of the class DQB. On STICKY NOTE 2 (blue): they write one remaining wonder — a question that goes beyond this unit (e.g., 'Does LSF^4 apply to 4D shapes?' or 'How does Google Maps handle scale changes in real time?'). The teacher reads 4–5 blue sticky notes aloud to honour curiosity and point toward future learning. The class then reviews the full DQB together, tracing the arc from the first lesson's questions ('Why doesn't the face distort?') to today's final entries.	Hand out two sticky notes per learner. Say: 'Yellow note — write the two formulas that complete our DQB. Blue note — write one question this unit has opened up for you, not answered.' Give 3 minutes. Invite all learners to the DQB simultaneously to post notes. Then stand at the DQB and say: 'Let's read our journey.' Read aloud 3 questions from Lesson 1's 'What We Wonder' column and the corresponding answers now in 'What We Know.' Narrate: 'In Lesson 1 you asked why the face doesn't distort. Today you can answer that yourself.' Cold-call 2 learners: 'In one sentence — why does the face NOT distort when LSF is applied equally to both dimensions?' Allow 10 seconds wait time. Read 4–5 blue sticky notes aloud and say: 'These questions belong to the next chapter of your mathematical life.'	DQB Arc Review — by explicitly tracing questions from Lesson 1 to Lesson 12 answers, learners experience the unit as a coherent knowledge-building journey rather than a sequence of isolated topics. This metacognitive move strengthens retention and models how scientific/mathematical understanding develops incrementally.	Observable indicator: yellow sticky note contains both correct formulas written symbolically ($ASF = LSF^2$, $VSF = LSF^3$). Blue sticky note contains a genuine question demonstrating curiosity beyond the unit — this signals whether the learner has internalised the concept deeply enough to wonder about its extensions.
Model Building Phase  VIDEO: Conservation and the race to save biodiversity Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12  Search ARES for similar readings	Learners open to the model page in their exercise books where they have been building their cumulative model across the unit. They now draw their FINAL MODEL — a single annotated diagram that shows: (1) the two passport photos (35×45 mm and 105×135 mm) with $LSF = 3$ labelled; (2) a branching structure showing $LSF \rightarrow ASF = LSF^2 = 9$ (with 'all areas multiply by 9' annotated) and $LSF \rightarrow VSF = LSF^3 = 27$ (with 'all volumes multiply by 27' annotated); (3) the condition for similarity: 'All corresponding angles equal AND all corresponding lengths in the same ratio ($= LSF$)'; (4) a note: 'If only width is scaled, angles change $\rightarrow$ distortion $\rightarrow$ NOT similar.' Alongside the final model,	Say: 'Open your model page. You are drawing your FINAL model — this is the full picture you have been building for 12 lessons. You have 8 minutes.' Circulate. After 8 minutes, project the side-by-side Day-1 vs. final model comparison. Say: 'Look at what you knew on Day 1 and what you know now.' Cold-call 3 learners: 'Tell me ONE thing your final model shows that your Day-1 model could not.' Allow 10 seconds wait time per response. Close the lesson by returning to the driving question one final time: 'Why does resizing a passport photo change its size but never its shape — and what is the mathematics that makes this possible?' Cold-call one learner to give the complete answer in 3–4 sentences. Affirm and summarise:	Final Model Comparison (Day-1 vs. Final) — placing the initial and final models side by side makes conceptual growth visible and concrete. This strategy operationalises the NGSS practice of Developing and Using Models and gives learners tangible evidence of their own intellectual development, reinforcing self-efficacy and the value of iterative model revision.	Observable indicator: final model contains all four required elements (two passport photos with $LSF = 3$; branching to ASF and VSF with correct exponents and annotations; similarity condition; distortion note). The one-sentence comparison correctly names something absent from the Day-1 model (e.g., VSF, the LSF^3 relationship, or the distortion consequence of unequal scaling).

	learners write a one-sentence comparison: 'In Lesson 1 my model showed ____ but my final model now shows ____.' The teacher projects a sample Day-1 model (from a willing learner's Lesson 1 work, shared anonymously) and a sample final model side by side for the class to compare.	'The Linear Scale Factor — a single number — governs every length. Angles are preserved. Area scales by LSF^2 . Volume scales by LSF^3 . That is the mathematics behind every perfectly resized image, every architect's model, and every similar figure you will ever meet.'		
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D. TEACHER REFLECTION

After this lesson, reflect on the following: (1) PEER-MARKING QUALITY — did learners write meaningful error-explanation sentences (red crosses), or did they simply mark wrong without explanation? If the latter, introduce a sentence-starter scaffold ('The error here is ____ because ____ . The correct method is ____.') in future consolidation lessons. (2) FINAL EXPLANATION PARAGRAPHS — did learners use all five mandated terms (LSF, ASF, VSF, similar figures, angle preservation) correctly and in context? Collect 5 random books and check tonight. Any learner who omits VSF or angle preservation needs a brief one-to-one follow-up before the next sub-strand begins. (3) DQB ARC — did tracing the unit's question arc generate genuine metacognitive conversation, or did it feel procedural? Consider assigning a short home task: 'Find one real-world object in your home or farm that involves similar figures — describe it using LSF, ASF, or VSF.' (4) MODEL COMPARISON — did the side-by-side Day-1 vs. final model projection produce visible surprise or satisfaction in learners? This emotional moment is pedagogically important — it signals genuine understanding. If learners seemed indifferent, consider making the model-building protocol more deliberate in future units by labelling each lesson's model addition at the time it is made.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners examined three Kenyan-context problems (Nairobi–Nakuru map, Mombasa architect's roof model, Kericho tea-farm water tanks) that each required a different scale-factor relationship — linear, area, or volume — and peer-marked each other's solutions using a three-colour protocol.
What did I learn?	$ASF = LSF^2$ and $VSF = LSF^3$: a single Linear Scale Factor simultaneously determines how lengths, areas, and volumes change between similar figures. All corresponding angles remain equal and all corresponding lengths remain in the same ratio — these are the two conditions for similarity. If only one dimension is scaled (unequal LSF), angles change and the figure is no longer similar (distortion occurs).
How does this explain the phenomenon?	The passport photo photographer's software applied one $LSF = 3$ equally to both width and height of the digital image. This preserved all angles (so the face looks identical) and all length ratios (so proportions are perfect), while making the area exactly $9\times$ larger ($ASF = 3^2 = 9$) and — if the image were a 3D object — the volume would be $27\times$ larger ($VSF = 3^3 = 27$). The mathematics that makes perfect resizing possible is the Linear Scale Factor applied uniformly to every dimension.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.